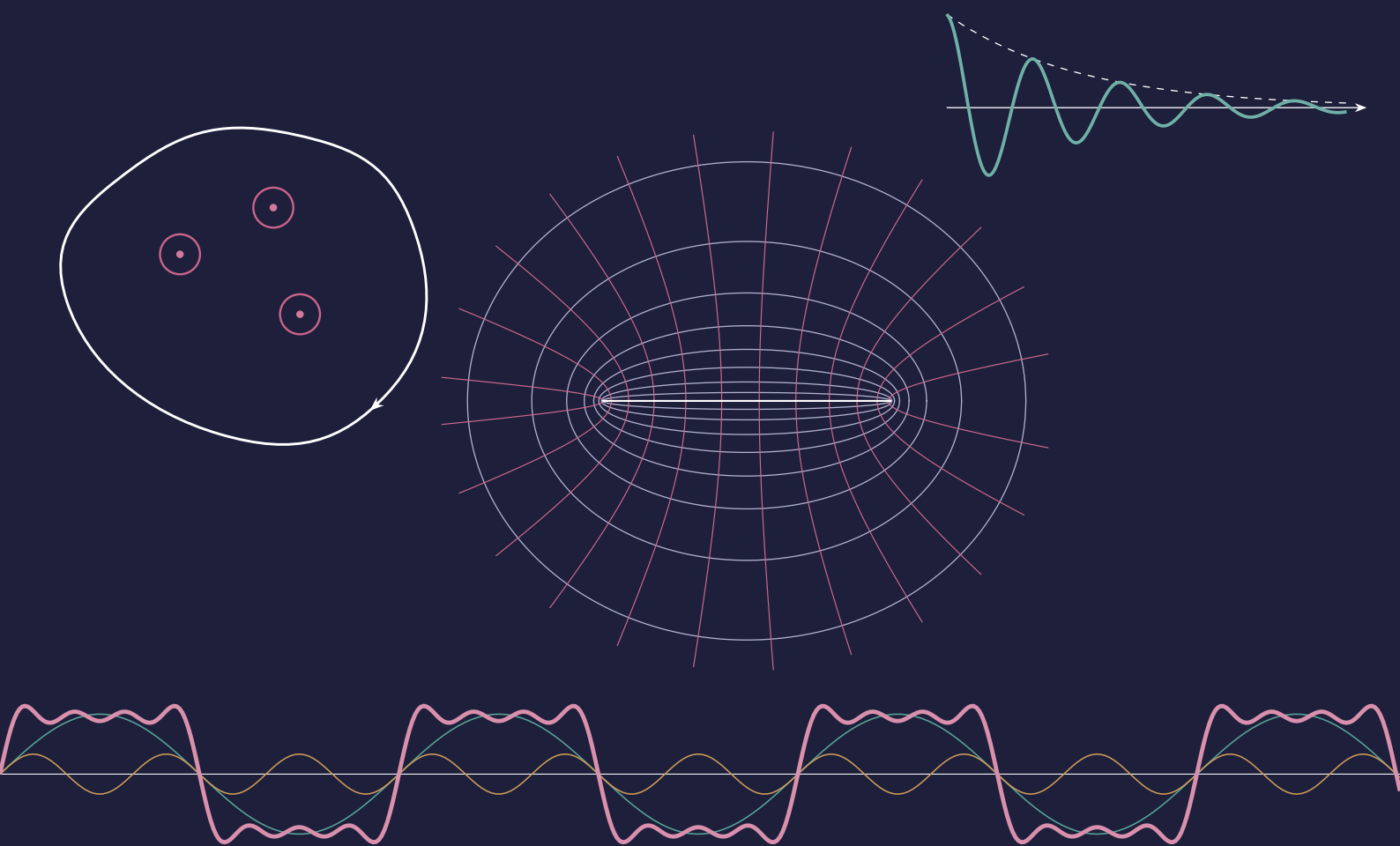




Engineering Mathematics

ریاضی مهندسی

Complex Analysis • Fourier Analysis • Partial Differential Equations

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Compiled and edited by: **Aidin Shekari**

About these notes

These are the typeset lecture notes of the course *Engineering Mathematics*, taught by Abbas Ebrahimi Moghadam. They follow the instructor's slides and keep their order, notation and examples, in three parts: **complex analysis** (chapters 1–7), **Fourier analysis** (chapters 8–10), and **partial differential equations and the Laplace transform** (chapters 11–14).

The notes were compiled and typeset by Aidin Shekari. Every figure is drawn in TikZ or PGFPlots from the underlying mathematical object. Slips in the slides that were corrected while typesetting are listed in the file `REVIEW.md` of the repository.

How to read the notes. Numbered boxes share one counter within each chapter, so "Example 5.12" is easy to find:

Definition 0.1 ■ A concept being defined

Definitions are set in indigo boxes and are numbered within the chapter.

Theorem 0.2 ■ A theorem or a formula to know

Theorems and lemmas are set in rose boxes.

Key result 0.3 ■ A key result

The results worth finding again quickly are framed.

Example 0.4 ■ A worked example

Examples are green; the statement comes first.

Solution. The solution follows inside the same box.

Exercise 0.5 ■ An exercise

Exercises are amber and are left to the reader.

A remark

Remarks and consequences have a thin grey rule and no number.

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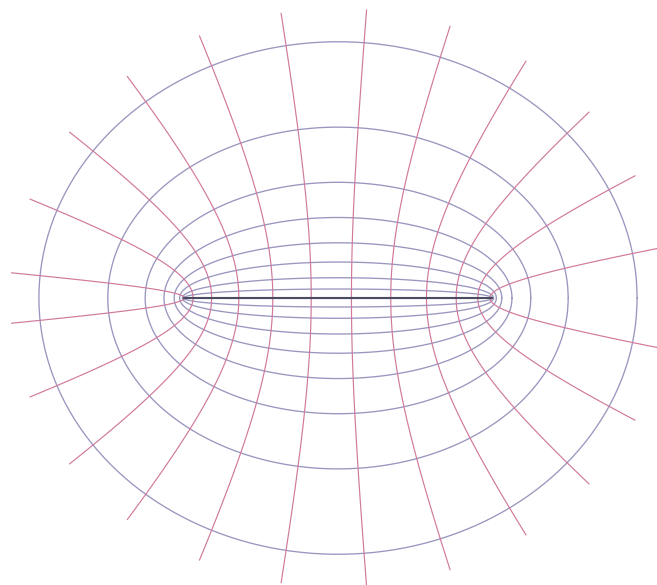
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Part I

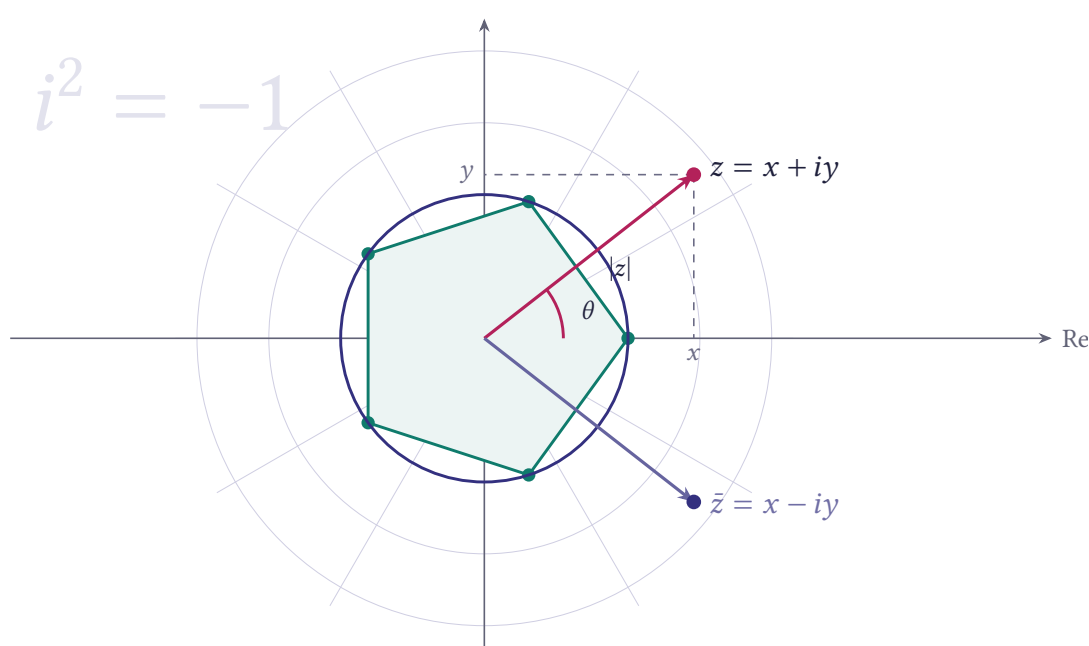
Complex Analysis

آنالیز مختلط

Chapters 1–7

Complex Numbers

اعداد مختلط



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1.1 The Set of Complex Numbers

Definition 1.1 ■ Complex numbers

The set of complex numbers generalizes the real numbers on the real line to a plane of numbers having two components, a real one and an imaginary one.

$$\mathbb{R} \rightarrow \mathbb{C}$$

$$\mathbb{C} = \{(x, y) \mid x, y \in \mathbb{R}\}, \quad z = (x, y), \quad x = \operatorname{Re}\{z\}, \quad y = \operatorname{Im}\{z\}$$

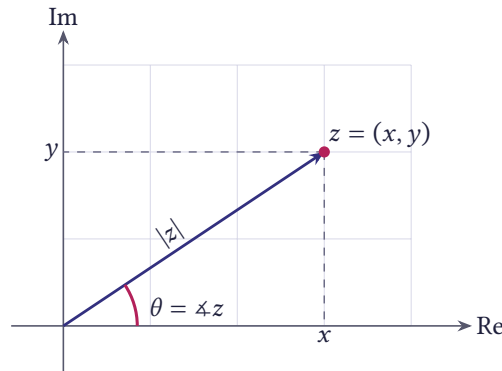


Figure 1.1 The complex number $z = (x, y)$ in the complex plane; its modulus $|z|$ and argument θ

Magnitude or norm of a complex number:

$$|z| = \sqrt{x^2 + y^2}$$

Phase or argument:

$$\theta = \arg z = \tan^{-1}\left(\frac{y}{x}\right), \quad z = |z|e^{j\theta}$$

$$x = |z| \cos \theta, \quad y = |z| \sin \theta$$

$$z = x \times (1, 0) + y \times (0, 1) = x + jy$$

Key result 1.2

$$j^2 = -1$$

1.2 The Four Basic Operations

Let $z_1 = (a_1, b_1) = a_1 + jb_1$ and $z_2 = (a_2, b_2) = a_2 + jb_2$.

(I) Addition:

$$z_1 + z_2 = (a_1 + a_2, b_1 + b_2) = a_1 + a_2 + j(b_1 + b_2)$$

(II) Subtraction:

$$z_1 - z_2 = (a_1 - a_2, b_1 - b_2) = a_1 - a_2 + j(b_1 - b_2)$$

(III) Multiplication:

$$\begin{aligned}
z_1 \times z_2 &= (a_1 + ib_1) \times (a_2 + ib_2) \\
&= a_1 \times a_2 + a_1 \times ib_2 + ib_1 \times a_2 + ib_1 \times ib_2 \\
&= (a_1 a_2 - b_1 b_2) + i(a_1 b_2 + b_1 a_2)
\end{aligned}$$

Multiplication by a real number:

$$\forall \alpha \in \mathbb{R} : \quad \alpha z_1 = \alpha a_1 + i \alpha b_1$$

(IV) Division:

$$\begin{aligned}
z_1 \div z_2 &= (a_1 + ib_1) \div (a_2 + ib_2) = \frac{a_1 + ib_1}{a_2 + ib_2} = \frac{(a_1 + ib_1)(a_2 - ib_2)}{(a_2 + ib_2)(a_2 - ib_2)} \\
&= \frac{(a_1 a_2 + b_1 b_2) + i(b_1 a_2 - a_1 b_2)}{a_2^2 + b_2^2} = \frac{a_1 a_2 + b_1 b_2}{a_2^2 + b_2^2} + i \frac{b_1 a_2 - a_1 b_2}{a_2^2 + b_2^2}
\end{aligned}$$

1.3 Properties of Addition and Multiplication**Theorem 1.3 ■ The complex numbers form a field**

The set of complex numbers, together with addition and multiplication, forms a field that cannot be ordered.

For $z_1, z_2, z_3 \in \mathbb{C}$:

$$z_1 + z_2 \in \mathbb{C}, \quad z_1 \times z_2 \in \mathbb{C}$$

$$z_1 + z_2 = z_2 + z_1, \quad z_1 \times z_2 = z_2 \times z_1$$

$$(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3), \quad (z_1 z_2) z_3 = z_1 (z_2 z_3)$$

$$z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$$

$$z_1 + (0, 0) = z_1 + 0 = z_1, \quad z_1 + (-z_1) = (0, 0) = 0$$

$$-z_1 = -(a_1, b_1) = (-a_1, -b_1) = -a_1 - ib_1$$

$$z_1 \times (1, 0) = z_1 \times 1 = z_1, \quad z_1 \times z_1^{-1} = (1, 0) = 1$$

$$z_1^{-1} = \left(\frac{a_1}{a_1^2 + b_1^2}, \frac{-b_1}{a_1^2 + b_1^2} \right) = \frac{a_1}{a_1^2 + b_1^2} - i \frac{b_1}{a_1^2 + b_1^2}$$

No ordering

An order relation $z_1 < z_2$ cannot be defined on the set of complex numbers.

1.4 The Complex Conjugate

Definition 1.4 ■ Complex conjugate of a number

$$z = x + iy \implies \bar{z} \triangleq x - iy$$

Example 1.5

Plot the conjugate of the complex number $z = 5 + 2i$ in the complex plane.

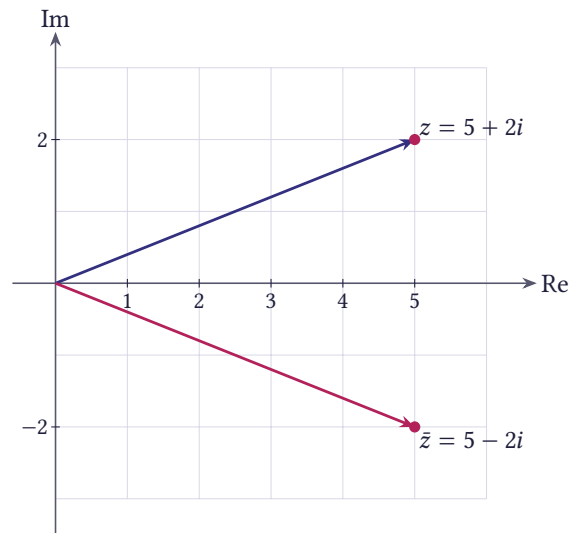


Figure 1.2 The complex number $z = 5 + 2i$ and its conjugate $\bar{z} = 5 - 2i$

Exercise 1.6

Prove the following statements.

$$\operatorname{Re}\{z\} = \frac{1}{2}(z + \bar{z}), \quad \operatorname{Im}\{z\} = \frac{1}{2i}(z - \bar{z}), \quad |z|^2 = z \cdot \bar{z}$$

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2, \quad \overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2, \quad \overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$$

$$|z| = |\bar{z}|, \quad |\operatorname{Re}\{z\}| \leq |z|, \quad |\operatorname{Im}\{z\}| \leq |z|$$

1.5 The Triangle Inequality

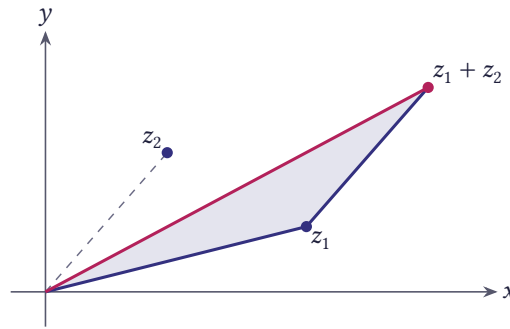


Figure 1.3 The triangle inequality: a triangle with vertices 0, z_1 and $z_1 + z_2$

Exercise 1.7

- Prove the triangle inequality for any two complex numbers: $|z_1 + z_2| \leq |z_1| + |z_2|$.
- Prove the general form of the triangle inequality for n arbitrary complex numbers:

$$|z_1 + z_2 + \dots + z_n| \leq |z_1| + |z_2| + \dots + |z_n|$$
- Prove that $|z_1 + z_2| \geq |z_1| - |z_2|$.

1.6 The Polar Form of Complex Numbers

$$z = x + iy, \quad \begin{cases} x = |z| \cos \theta \\ y = |z| \sin \theta \end{cases} \implies z = |z| (\cos \theta + i \sin \theta)$$

$$\theta = \text{Arg}(z) = \tan^{-1}(y/x), \quad -\pi < \theta \leq \pi$$

$$\arg(z) = \text{Arg}(z) + 2k\pi, \quad k \in \mathbb{Z}$$

1.7 Multiplying Two Complex Numbers in Polar Form

$$z_1 = |z_1|(\cos \theta_1 + i \sin \theta_1), \quad z_2 = |z_2|(\cos \theta_2 + i \sin \theta_2)$$

$$\begin{aligned} z_1 \cdot z_2 &= |z_1||z_2|(\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) \\ &= |z_1||z_2|[(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) \\ &\quad + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)] \end{aligned}$$

Key result 1.8

$$z_1 \cdot z_2 = |z_1||z_2|[\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

Consequence 1

When two complex numbers are multiplied, their magnitudes are multiplied and their phases are added.

Consequence 2

If a complex number is raised to a natural power, its magnitude is raised to that power and its argument (phase) is multiplied by it (prove by induction). In other words:

$$z^k = |z|^k(\cos k\theta + i \sin k\theta)$$

$$z^k = [|z|(\cos \theta + i \sin \theta)]^k = |z|^k(\cos \theta + i \sin \theta)^k = |z|^k(\cos k\theta + i \sin k\theta)$$

Key result 1.9 ■ De Moivre's formula

$$(\cos \theta + i \sin \theta)^k = \cos k\theta + i \sin k\theta$$

Exercise 1.10

Using De Moivre's formula, express $\cos 3\theta$ and $\cos 4\theta$ in terms of $\cos \theta$ and $\sin \theta$.

1.8 The m -th Root of a Complex Number

$$w = \sqrt[m]{z} = z^{1/m}, \quad z = |z|(\cos \theta + i \sin \theta)$$

$$z = w^m, \quad w = |w|(\cos \varphi + i \sin \varphi)$$

$$z = |z|(\cos \theta + i \sin \theta) = |w|^m(\cos m\varphi + i \sin m\varphi)$$

$$|w|^m = |z| \implies |w| = \sqrt[m]{|z|}$$

$$m\varphi = \theta + 2k\pi \implies \varphi = \frac{\theta + 2k\pi}{m}, \quad k = 0, 1, 2, \dots, m-1$$

Key result 1.11

$$w = \sqrt[m]{z} = \sqrt[m]{|z|} \left[\cos\left(\frac{\theta + 2k\pi}{m}\right) + i \sin\left(\frac{\theta + 2k\pi}{m}\right) \right], \quad k = 0, 1, 2, \dots, m-1$$

Example 1.12

What are the roots of the equation $z^n = 1$?

$$z = \sqrt[n]{1}, \quad 1 = 1 \angle 0 \implies z = \cos\left(\frac{2k\pi}{n}\right) + i \sin\left(\frac{2k\pi}{n}\right), \quad k = 0, 1, \dots, n-1$$

Defining $\omega = 1 \angle \frac{2\pi}{n}$, the roots are

$$z = 1, \omega, \omega^2, \dots, \omega^{n-1}$$

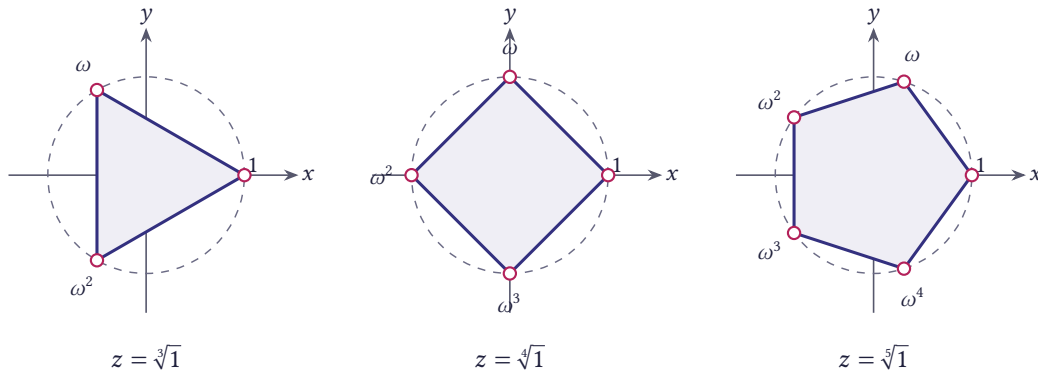


Figure 1.4 The roots $z = \sqrt[3]{1}$, $z = \sqrt[4]{1}$ and $z = \sqrt[5]{1}$ on the unit circle

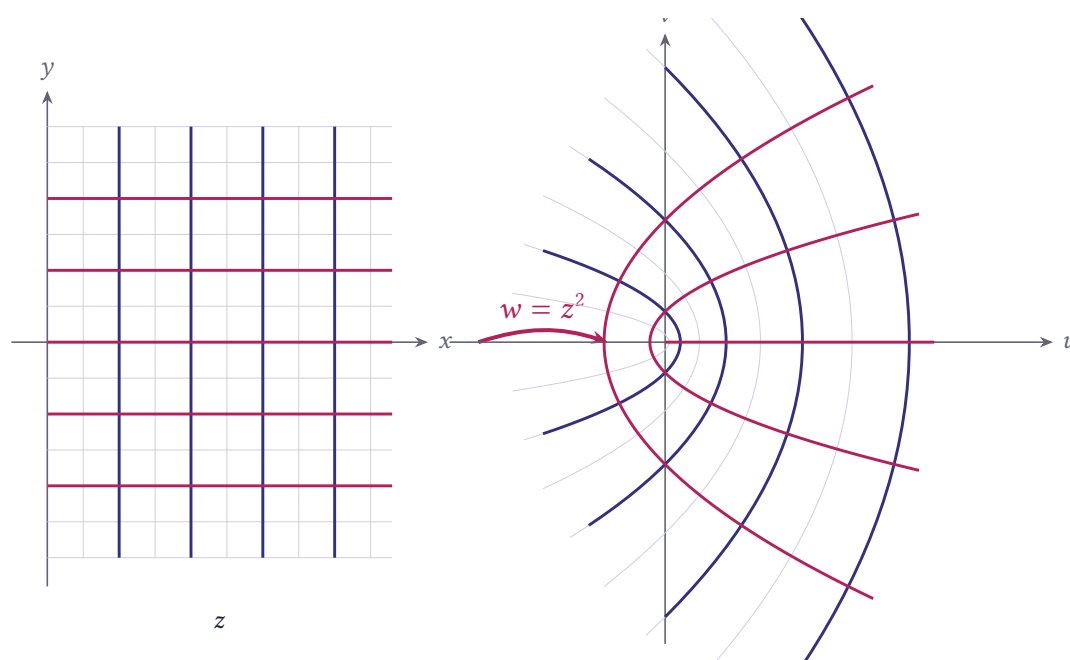
Exercise 1.13

- Find the roots of the equation $z^6 + 6 + 8i = 0$ and plot them in the complex plane.
- Verify the identity $1 + z + z^2 + \dots + z^n = \frac{1 - z^{n+1}}{1 - z}$ and then derive the following identity (Lagrange's identity).

$$1 + \cos \theta + \cos 2\theta + \dots + \cos n\theta = \frac{1}{2} + \frac{\sin\left[\left(n + \frac{1}{2}\right)\theta\right]}{2 \sin(\theta/2)}$$

Complex Functions, the Derivative and the Cauchy–Riemann Conditions

توابع مختلط، مشتق و شرایط کشی--ریمان



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2.1 Complex Functions

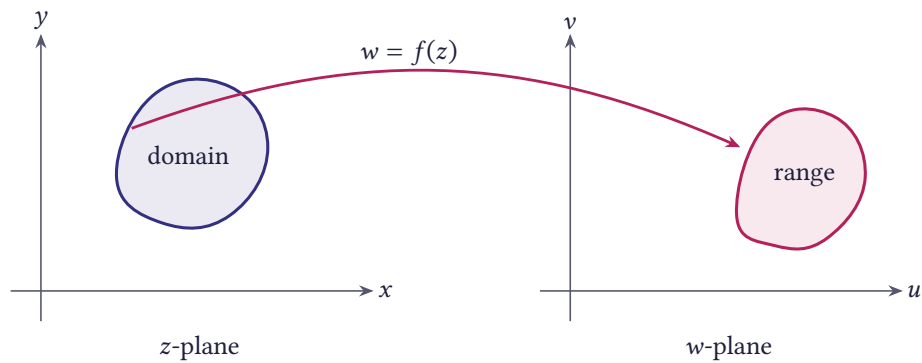


Figure 2.5 A complex function $w = f(z)$ maps points of the domain in the z -plane to points of the range in the w -plane

$$z = x + iy, \quad w = u + iv$$

$$f : \mathbb{C} \rightarrow \mathbb{C}, \quad f : \begin{cases} u = u(x, y) \\ v = v(x, y) \end{cases}$$

Example 2.14

Consider the function $w = z^2$:

$$u + iv = (x + iy)^2 = (x^2 - y^2) + i(2xy) \implies \begin{cases} u = x^2 - y^2 \\ v = 2xy \end{cases}$$

2.2 Some Important Loci

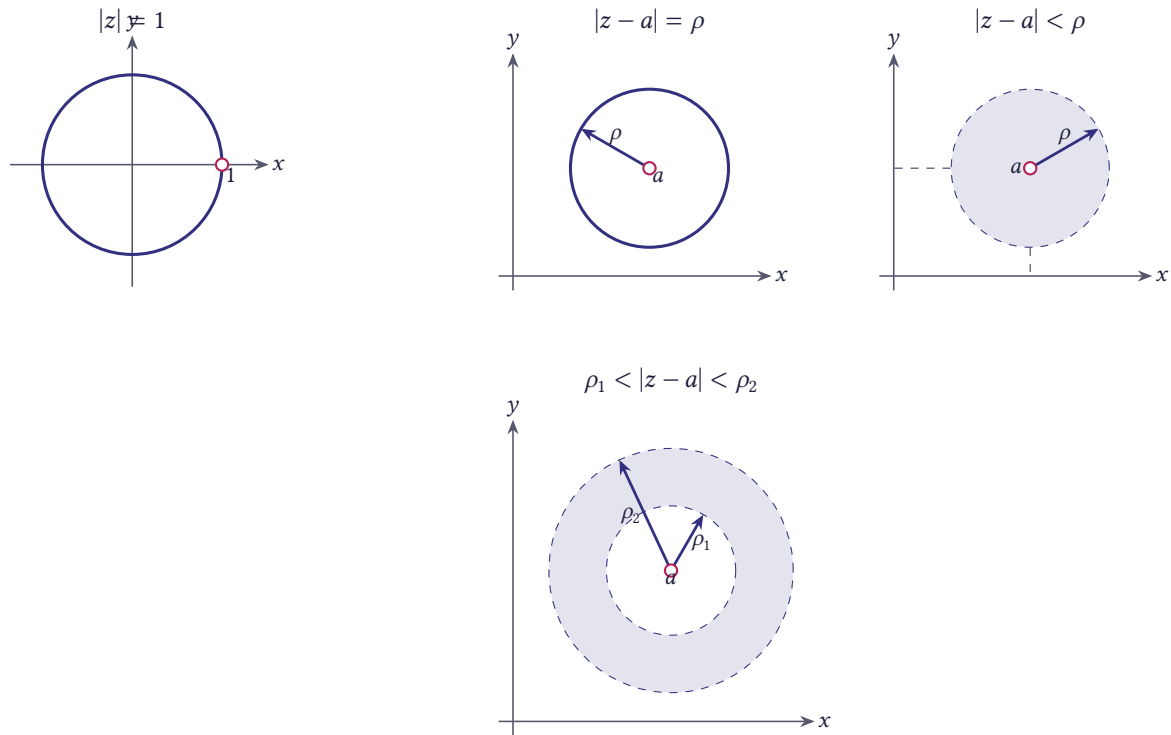


Figure 2.6 Important loci: a circle, the ρ -neighbourhood of a point a , and an annular region

$$|z| = 1, \quad |z - a| = \rho, \quad |z - a| < \rho, \quad \rho_1 < |z - a| < \rho_2$$

The set $|z - a| < \rho$ is called the ρ -neighbourhood of a .

2.3 Limits of Complex Functions

Definition 2.15 ■ Limit

If for every $\varepsilon > 0$ one can find $\delta > 0$ such that $|z - z_0| < \delta$ implies $|f(z) - l| < \varepsilon$, then

$$\lim_{z \rightarrow z_0} f(z) = l$$

Hence for all points inside the circle of centre z_0 and radius δ , $f(z)$ must lie inside the circle of centre l and radius ε . Here ε is given and $\delta(\varepsilon)$ has to be found. Clearly the concept of limit here is broader than that for real functions.

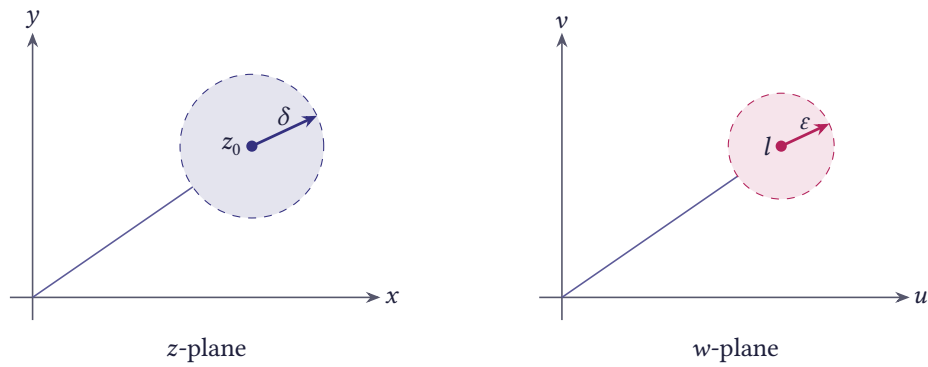


Figure 2.7 The definition of the limit: the δ -neighbourhood of z_0 in the z -plane and the ε -neighbourhood of l in the w -plane

2.4 Continuity of Complex Functions

Definition 2.16 ■ Continuity at a point

The function $w = f(z)$ is continuous at the point $z = z_0$ if the following three conditions hold:

1. $f(z)$ is defined at $z = z_0$.
2. $\lim_{z \rightarrow z_0} f(z)$ exists.
3. $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

Example 2.17

Is the function $f(z) = \sin(1/z)$ continuous at $z = 0$?

Example 2.18

Is the function

$$f(z) = \begin{cases} \sin(1/z) & z \neq 0 \\ 5 + 4i & z = 0 \end{cases}$$

continuous at $z = 0$?

Example 2.19

Is the function

$$f(z) = \begin{cases} 3z^3 + 4 & z \neq 0 \\ 17 & z = 0 \end{cases}$$

continuous at $z = 0$?

■ Properties of Continuous Complex Functions

Theorem 2.20 ■ Theorems

1. If $f(z)$, $g(z)$ are continuous at z_0 , then $f(z) \pm g(z)$ is continuous at z_0 .
2. If $f(z)$, $g(z)$ are continuous at z_0 , then $f(z) \times g(z)$ is continuous at z_0 .
3. If $f(z)$, $g(z)$ are continuous at z_0 and $g(z_0) \neq 0$, then $\frac{f(z)}{g(z)}$ is continuous at z_0 .

Definition 2.21 ■ Continuity on a domain

The function $w = f(z)$ is continuous on the domain D if f is continuous at every point of D .

Domain: all the points inside a closed curve.

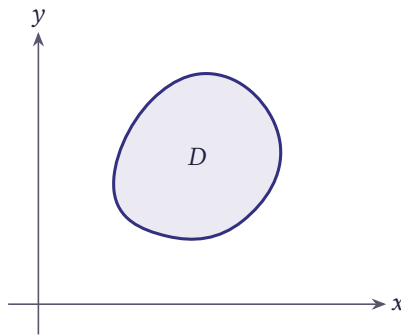


Figure 2.8 A domain D in the complex plane

2.5 The Derivative of Complex Functions

Definition 2.22 ■ Derivative

The function $w = f(z)$ is differentiable at the point $z = z_0$ if the following limit exists:

$$f'(z_0) = \lim_{|\Delta z| \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

The existence of the derivative of a complex function requires stricter conditions than for real functions: since z_0 can be approached from *all* directions in the z -plane, this limit must exist and must not depend on the chosen path towards z_0 .

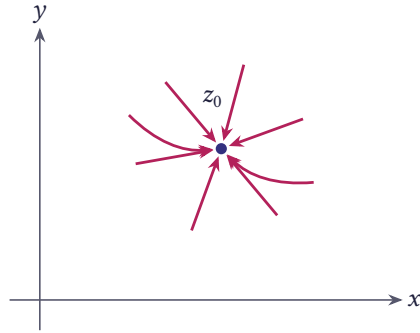


Figure 2.9 Approaching the point z_0 from all directions

2.6 The Cauchy–Riemann Conditions

Theorem 2.23 ■ Necessary condition for differentiability

A necessary condition for the complex function $w = f(z)$ to be differentiable is that the Cauchy–Riemann conditions hold:

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases} \quad w = u + iv = f(x + iy) = f(z)$$

Proof.

$$f'(z) = \lim_{|\Delta z| \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}, \quad z = x + iy, \quad \Delta z = \Delta x + i\Delta y$$

$$\begin{aligned} f(z + \Delta z) - f(z) &= [u(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y)] - [u(x, y) + iv(x, y)] \\ &= [u(x + \Delta x, y + \Delta y) - u(x, y)] + i[v(x + \Delta x, y + \Delta y) - v(x, y)] \end{aligned}$$

$$\frac{f(z + \Delta z) - f(z)}{\Delta z} = \frac{u(x + \Delta x, y + \Delta y) - u(x, y)}{\Delta x + i\Delta y} + i \frac{v(x + \Delta x, y + \Delta y) - v(x, y)}{\Delta x + i\Delta y}$$

A necessary condition for the existence of the derivative of $w = f(z)$ is that the value of the limit defining $f'(z)$ does not change if we approach z from any direction (in particular along the horizontal or the vertical direction).

Path 1: the vertical direction. $\Delta z = \Delta x + i\Delta y$ with $\Delta x = 0$, i.e. $\Delta z = i\Delta y$:

$$\begin{aligned} f'(z) &= \lim_{|\Delta z| \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \\ &= \lim_{\Delta y \rightarrow 0} \left[\frac{u(x, y + \Delta y) - u(x, y)}{i\Delta y} + i \frac{v(x, y + \Delta y) - v(x, y)}{i\Delta y} \right] \\ &= \frac{1}{i} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y} \end{aligned}$$

Path 2: the horizontal direction. $\Delta z = \Delta x + i\Delta y$ with $\Delta y = 0$, i.e. $\Delta z = \Delta x$:

$$\begin{aligned} f'(z) &= \lim_{|\Delta z| \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \\ &= \lim_{\Delta x \rightarrow 0} \left[\frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x} \right] \\ &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \end{aligned}$$

Therefore

$$f'(z) = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

and equating real and imaginary parts:

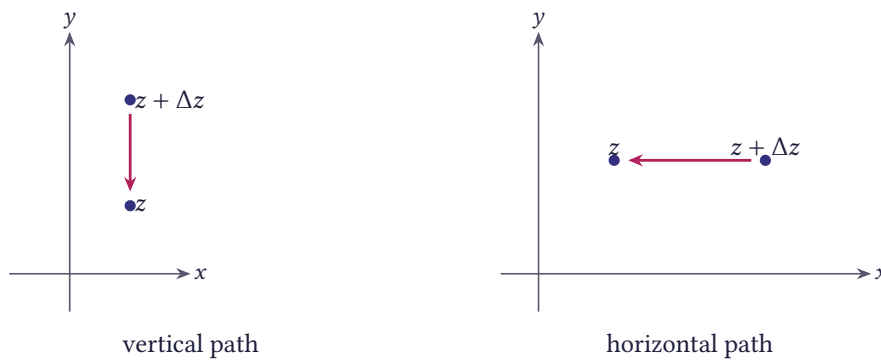


Figure 2.10 Two paths towards z : the vertical direction ($\Delta z = i\Delta y$) and the horizontal direction ($\Delta z = \Delta x$)

Key result 2.24 ■ The Cauchy–Riemann conditions

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Consequence

1. If these conditions do not hold, the derivative does not exist at the point z .
2. The following ways of computing the derivative are suggested:

$$f'(z) = \lim_{|\Delta z| \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} = \frac{\partial v}{\partial y} + i \frac{\partial v}{\partial x}$$

Theorem 2.25 ■ Sufficient condition for differentiability

A sufficient condition for the complex function $w = f(z)$ to be differentiable is that the Cauchy–Riemann conditions hold.

The proof is beyond the scope of this class.

Theorem 2.26 ■ The Cauchy–Riemann theorem

A necessary and sufficient condition for the complex function $w = f(z)$ to be differentiable is that the Cauchy–Riemann conditions hold.

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases} \quad w = u + iv = f(x + iy) = f(z)$$

A Consequence of the Cauchy–Riemann Conditions (Analyticity of a Function)

Differentiating the Cauchy–Riemann conditions with respect to x and y :

$$\left. \begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{\partial^2 v}{\partial x \partial y} \\ \frac{\partial^2 u}{\partial y^2} &= -\frac{\partial^2 v}{\partial x \partial y} \end{aligned} \right\} \xrightarrow{\text{add}} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\left. \begin{aligned} \frac{\partial^2 u}{\partial x \partial y} &= \frac{\partial^2 v}{\partial y^2} \\ \frac{\partial^2 u}{\partial x \partial y} &= -\frac{\partial^2 v}{\partial x^2} \end{aligned} \right\} \xrightarrow{\text{subtract}} \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Consequence

The real and imaginary parts of an analytic function satisfy the two-dimensional Laplace equation.

2.7 Differentiability of Complex Functions: Examples**Example 2.27**

Consider the function $w = z^2$; is it analytic?

$$w = f(z) = z^2 = (x + iy)^2 = (x^2 - y^2) + i 2xy, \quad u(x, y) = x^2 - y^2, \quad v(x, y) = 2xy$$

$$\frac{\partial u}{\partial x} = 2x = \frac{\partial v}{\partial y} \quad \checkmark \quad \frac{\partial u}{\partial y} = -2y = -\frac{\partial v}{\partial x} \quad \checkmark$$

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = 2x + i 2y = 2(x + iy) = 2z$$

$$\nabla^2 u = 2 - 2 = 0, \quad \nabla^2 v = 0 + 0 = 0$$

Example 2.28

Consider the function $w = |z|^2$; is it analytic?

$$w = f(z) = |z|^2 = (x + iy)(x - iy) = x^2 + y^2, \quad u(x, y) = x^2 + y^2, \quad v(x, y) = 0$$

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial y} = 0, \quad \frac{\partial u}{\partial y} = 2y, \quad -\frac{\partial v}{\partial x} = 0$$

$$\nabla^2 u = 2 + 2 = 4, \quad \nabla^2 v = 0 + 0 = 0$$

Analytic nowhere except at the origin.

Exercise 2.29

Determine whether each of the following functions is analytic. When it is, find its derivative.

1. $f(z) = \bar{z}$
2. $f(z) = \operatorname{Re}\{z\}$
3. $f(z) = z^3 + 2z^2 + 3i$

2.8 Analytic Complex Functions**Example 2.30**

$u = x^2 - y^2$ is the real part of an analytic function $f(z)$. Find $f(z)$.

Since $f(z)$ is analytic, its real and imaginary parts satisfy the Cauchy–Riemann conditions, and consequently they must also satisfy the two-dimensional Laplace equation.

- 1) First check whether the real part satisfies the two-dimensional Laplace equation:

$$\nabla^2 u = \frac{\partial^2}{\partial x^2}(x^2 - y^2) + \frac{\partial^2}{\partial y^2}(x^2 - y^2) = 2 - 2 = 0 \quad \checkmark$$

- 2) We use the Cauchy–Riemann relations:

$$\frac{\partial u}{\partial x} = 2x = \frac{\partial v}{\partial y} \implies v(x, y) = 2xy + \phi(x)$$

$$-\frac{\partial v}{\partial x} = -2y - \phi'(x) = \frac{\partial u}{\partial y} = -2y \implies \phi'(x) = 0 \implies \phi(x) = c, \quad c \in \mathbb{R}$$

$$f(z) = (x^2 - y^2) + i(2xy + c) = z^2 + C$$

Example 2.31

Let $u = \cos x \cosh y$. (a) Show that this function is harmonic in the plane ($\nabla^2 u = 0$). (b) Find $v(x, y)$ such that $u + iv$ is analytic.

(a) Check whether the real part satisfies the two-dimensional Laplace equation:

$$\nabla^2 u = \frac{\partial^2}{\partial x^2}(\cos x \cosh y) + \frac{\partial^2}{\partial y^2}(\cos x \cosh y) = -\cos x \cosh y + \cos x \cosh y = 0 \quad \checkmark$$

(b) We use the Cauchy–Riemann relations:

$$\frac{\partial u}{\partial x} = -\sin x \cosh y = \frac{\partial v}{\partial y} \implies v(x, y) = -\sin x \sinh y + \phi(x)$$

$$-\frac{\partial v}{\partial x} = \cos x \sinh y - \phi'(x) = \frac{\partial u}{\partial y} = \cos x \sinh y \implies \phi'(x) = 0$$

$$\phi(x) = c, \quad c \in \mathbb{R} \implies v(x, y) = -\sin x \sinh y + c$$

2.9 The Cauchy–Riemann Conditions in Polar Coordinates

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \quad \begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \tan^{-1} \frac{y}{x} \end{cases} \quad r_x = \frac{x}{r}, \quad \theta_x = \frac{-y}{r^2}, \quad r_y = \frac{y}{r}, \quad \theta_y = \frac{x}{r^2}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x} = \frac{x}{r} \frac{\partial u}{\partial r} - \frac{y}{r^2} \frac{\partial u}{\partial \theta} = \cos \theta \frac{\partial u}{\partial r} - \frac{1}{r} \sin \theta \frac{\partial u}{\partial \theta}$$

$$\frac{\partial v}{\partial y} = \frac{\partial v}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial v}{\partial \theta} \frac{\partial \theta}{\partial y} = \frac{y}{r} \frac{\partial v}{\partial r} + \frac{x}{r^2} \frac{\partial v}{\partial \theta} = \sin \theta \frac{\partial v}{\partial r} + \frac{1}{r} \cos \theta \frac{\partial v}{\partial \theta}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \implies \cos \theta \frac{\partial u}{\partial r} - \frac{1}{r} \sin \theta \frac{\partial u}{\partial \theta} = \sin \theta \frac{\partial v}{\partial r} + \frac{1}{r} \cos \theta \frac{\partial v}{\partial \theta}$$

Similarly

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \implies \sin \theta \frac{\partial u}{\partial r} + \frac{1}{r} \cos \theta \frac{\partial u}{\partial \theta} = -\cos \theta \frac{\partial v}{\partial r} + \frac{1}{r} \sin \theta \frac{\partial v}{\partial \theta}$$

Key result 2.32 ■ The Cauchy–Riemann conditions in polar form

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad -\frac{1}{r} \frac{\partial u}{\partial \theta} = \frac{\partial v}{\partial r}$$

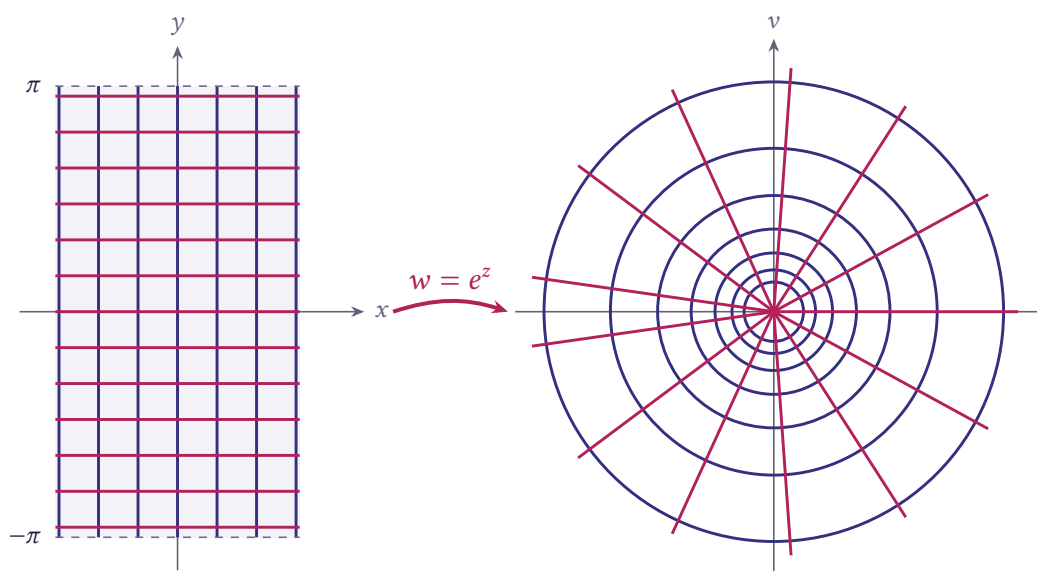
Exercise 2.33

Using the Cauchy–Riemann conditions in polar form, prove that u and v satisfy Laplace's equation in polar form, i.e.

$$\nabla^2 u = u_{rr} + \frac{1}{r^2} u_{\theta\theta} + \frac{1}{r} u_r = 0, \quad \nabla^2 v = v_{rr} + \frac{1}{r^2} v_{\theta\theta} + \frac{1}{r} v_r = 0$$

Elementary Complex Functions

توابع مختلط مقدماتی



In this chapter

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3.2	Trigonometric Functions	22
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3.4	The Logarithm Function	25

The elementary complex functions generalize the elementary real functions.

3.1 The Exponential Function

The real exponential function $f(x) = e^x$ has the following properties:

$$\begin{aligned} 1) \quad & f(x_1 + x_2) = f(x_1)f(x_2) & e^{x_1}e^{x_2} &= e^{x_1+x_2} \\ 2) \quad & \frac{d}{dx}f(x) = f(x) & \frac{d}{dx}e^x &= e^x \\ 3) \quad & e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \cdots \end{aligned}$$

We define the complex exponential function inspired by the third property, and then verify the other properties for it.

Definition 3.34 ■ The complex exponential function

$$w = e^z \triangleq 1 + z + \frac{z^2}{2!} + \cdots + \frac{z^n}{n!} + \cdots$$

If in the definition of the complex exponential we substitute $z = iy$:

$$\begin{aligned} e^{iy} &= 1 + iy + \frac{1}{2!}(iy)^2 + \frac{1}{3!}(iy)^3 + \cdots \\ &= 1 + iy - \frac{1}{2!}y^2 - \frac{i}{3!}y^3 + \frac{1}{4!}y^4 + \frac{i}{5!}y^5 + \cdots \\ &= \underbrace{\left(1 - \frac{1}{2!}y^2 + \frac{1}{4!}y^4 + \cdots\right)}_{\cos y} + i \underbrace{\left(y - \frac{1}{3!}y^3 + \frac{1}{5!}y^5 + \cdots\right)}_{\sin y} = \cos y + i \sin y \end{aligned}$$

Key result 3.35

$$w = e^z = e^{x+iy} = e^x e^{iy} = e^x (\cos y + i \sin y)$$

Example 3.36

Is the function $w = e^z$ analytic?

$$\begin{aligned} u(x, y) &= e^x \cos y, & v(x, y) &= e^x \sin y \\ \frac{\partial u}{\partial x} &= e^x \cos y = \frac{\partial v}{\partial y} \quad \checkmark & \frac{\partial u}{\partial y} &= -e^x \sin y = -\frac{\partial v}{\partial x} \quad \checkmark \end{aligned}$$

Since the Cauchy–Riemann conditions hold in the whole complex plane, the function $w = e^z$ is differentiable in the whole complex plane.

Example 3.37

Find the derivative of $w = e^z$.

$$\frac{d}{dz}e^z = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = e^x \cos y + ie^x \sin y = e^x(\cos y + i \sin y) = e^z$$

Key result 3.38 ■ Important remark

From now on we can write every complex number in exponential form:

$$z = x + iy = |z|(\cos \theta + i \sin \theta) = |z|e^{i\theta}$$

Observe that

$$e^{z+i2k\pi} = e^z e^{i2k\pi} = e^z [\cos(2k\pi) + i \sin(2k\pi)] = e^z$$

so $w = e^z$ is a periodic function in the complex plane with period $i2\pi$.

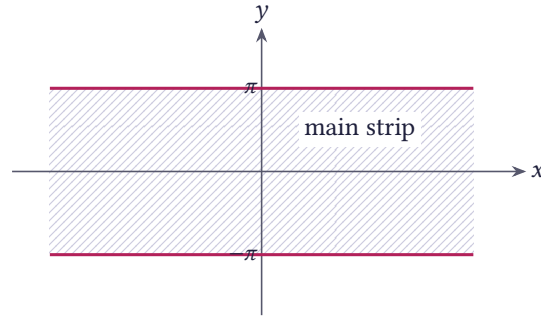


Figure 3.11 The fundamental strip of the exponential function: $-\pi < y \leq \pi$

3.2 Trigonometric Functions

$$\begin{aligned} e^{i\theta} &= \cos \theta + i \sin \theta \quad \implies \quad e^{-i\theta} = \cos \theta - i \sin \theta \\ \implies \quad \cos \theta &= \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \end{aligned}$$

Definition 3.39 ■ Complex cosine and sine

$$\cos z \triangleq \frac{e^{iz} + e^{-iz}}{2}, \quad \sin z \triangleq \frac{e^{iz} - e^{-iz}}{2i}$$

$$\begin{aligned} \cos z &= \frac{1}{2} [e^{i(x+iy)} + e^{-i(x+iy)}] = \frac{1}{2} [e^{ix}e^{-y} + e^{-ix}e^y] \\ &= \frac{1}{2} [e^{-y}(\cos x + i \sin x) + e^y(\cos x - i \sin x)] \\ &= \cos x \left(\frac{e^y + e^{-y}}{2} \right) - i \sin x \left(\frac{e^y - e^{-y}}{2} \right) \\ &= \cos x \cosh y - i \sin x \sinh y \end{aligned}$$

$$\begin{aligned}
\sin z &= \frac{1}{2i} \left[e^{i(x+iy)} - e^{-i(x+iy)} \right] = \frac{1}{2i} \left[e^{ix} e^{-y} - e^{-ix} e^y \right] \\
&= \frac{1}{2i} \left[e^{-y} (\cos x + i \sin x) - e^y (\cos x - i \sin x) \right] \\
&= \sin x \left(\frac{e^y + e^{-y}}{2} \right) + i \cos x \left(\frac{e^y - e^{-y}}{2} \right) \\
&= \sin x \cosh y + i \cos x \sinh y
\end{aligned}$$

■ Analyticity and Differentiability of $\cos z$

$$\cos z = u(x, y) + iv(x, y) = \cos x \cosh y - i \sin x \sinh y, \quad u = \cos x \cosh y, \quad v = -\sin x \sinh y$$

$$\frac{\partial u}{\partial x} = -\sin x \cosh y = \frac{\partial v}{\partial y} \quad \checkmark \quad \frac{\partial u}{\partial y} = \cos x \sinh y = -\frac{\partial v}{\partial x} \quad \checkmark$$

So this function is analytic in the whole complex plane. We obtain its derivative with one of the methods described earlier:

$$f'(z) = \frac{d}{dz} \cos z = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = -\sin x \cosh y - i \cos x \sinh y = -\sin z$$

■ Analyticity and Differentiability of $\sin z$

$$\sin z = u(x, y) + iv(x, y) = \sin x \cosh y + i \cos x \sinh y, \quad u = \sin x \cosh y, \quad v = \cos x \sinh y$$

$$\frac{\partial u}{\partial x} = \cos x \cosh y = \frac{\partial v}{\partial y} \quad \checkmark \quad \frac{\partial u}{\partial y} = \sin x \sinh y = \frac{\partial v}{\partial x} \quad \checkmark$$

So this function is analytic in the whole complex plane and

$$f'(z) = \frac{d}{dz} \sin z = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \cos x \cosh y + i \sin x \sinh y = \cos z$$

Key result 3.40

$$(\sin z)' = \cos z, \quad (\cos z)' = -\sin z$$

■ Unboundedness of $\sin z$ and $\cos z$

$$|\cos z| = \sqrt{\cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y} = \sqrt{\cos^2 x + \sinh^2 y}, \quad (\cosh^2 y = 1 + \sinh^2 y, \quad \sin^2 x = 1 - \cos^2 x)$$

$$|\sin z| = \sqrt{\sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y} = \sqrt{\sin^2 x + \sinh^2 y}, \quad (\cosh^2 y = 1 + \sinh^2 y, \quad \cos^2 x = 1 - \sin^2 x)$$

Since $\sinh y$ is an unbounded function, the functions $\cos z$ and $\sin z$ are **unbounded**.

■ Periodicity

$$\begin{aligned}\cos(z + 2\pi) &= \cos(x + iy + 2\pi) = \cos(x + 2\pi) \cosh y - i \sin(x + 2\pi) \sinh y \\ &= \cos x \cosh y - i \sin x \sinh y = \cos z\end{aligned}$$

In the same way $\sin(z + 2\pi) = \sin z$.

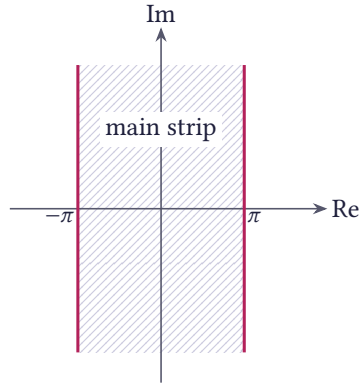


Figure 3.12 The fundamental strip of $\sin z$ and $\cos z$: $-\pi < x \leq \pi$

■ Remark

$$\cos(iy) = \cosh y, \quad \sin(iy) = i \sinh y$$

$$\cos(x + iy) = \cos x \cos(iy) - \sin x \sin(iy) = \cos x \cosh y - i \sin x \sinh y$$

$$\sin(x + iy) = \sin x \cos(iy) + \cos x \sin(iy) = \sin x \cosh y + i \cos x \sinh y$$

The other trigonometric functions:

$$\tan z \triangleq \frac{\sin z}{\cos z}, \quad \cot z \triangleq \frac{\cos z}{\sin z}, \quad \sec z \triangleq \frac{1}{\cos z}, \quad \csc z \triangleq \frac{1}{\sin z}$$

$$(\tan z)' = \sec^2 z, \quad \dots$$

3.3 Hyperbolic Functions

$$\cosh z \triangleq \frac{e^z + e^{-z}}{2}, \quad \sinh z \triangleq \frac{e^z - e^{-z}}{2}, \quad e^z = e^x(\cos y + i \sin y)$$

Exercise 3.41

Verify that

$$\cosh z = \cosh x \cos y + i \sinh x \sin y, \quad \sinh z = \sinh x \cos y + i \cosh x \sin y$$

Exercise 3.42

Verify that $\cosh z$ and $\sinh z$ are analytic in the whole complex plane and that their derivatives are given by

$$(\cosh z)' = \sinh z, \quad (\sinh z)' = \cosh z$$

$$\tanh z = \frac{\sinh z}{\cosh z}, \quad \coth z = \frac{\cosh z}{\sinh z}, \quad \operatorname{sech} z = \frac{1}{\cosh z}, \quad \operatorname{csch} z = \frac{1}{\sinh z}$$

$$\cosh(z + i2\pi) = \cosh(z), \quad \sinh(z + i2\pi) = \sinh(z)$$

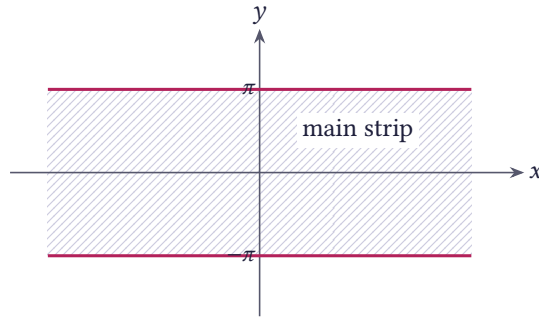


Figure 3.13 The fundamental strip of the hyperbolic functions: $-\pi < y \leq \pi$

3.4 The Logarithm Function

$$w = \ln z, \quad z = |z|e^{i \arg z} = e^w = e^{\ln |z|} e^{i \arg z}$$

$$w = \ln z = \ln |z| + i \arg z = \ln(x^2 + y^2)^{1/2} + i \tan^{-1} \frac{y}{x}$$

$$u(x, y) = \frac{1}{2} \ln(x^2 + y^2), \quad v(x, y) = \tan^{-1} \frac{y}{x}$$

$$\frac{\partial u}{\partial x} = \frac{x}{x^2 + y^2} = \frac{\partial v}{\partial y} \quad \checkmark \quad \frac{\partial u}{\partial y} = \frac{y}{x^2 + y^2} = -\frac{\partial v}{\partial x} \quad \checkmark$$

Hence the logarithm function is analytic in the whole complex plane except at the origin.

$$\frac{d}{dz} \ln z = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{x}{x^2 + y^2} + i \frac{-y}{x^2 + y^2} = \frac{\bar{z}}{z\bar{z}} = \frac{1}{z}$$

$$z = re^{i\theta} = re^{i(\theta + 2k\pi)}$$

So that the definition of a function is not violated, the range of this function is the region shown in the figure.

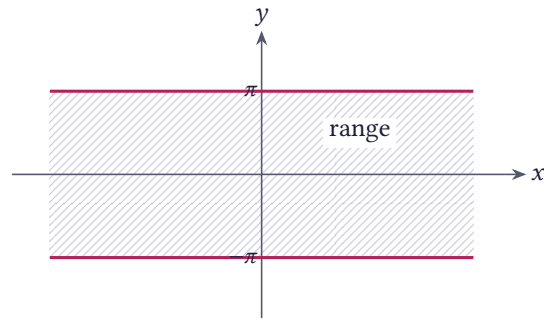


Figure 3.14 The range of the logarithm function: the strip $-\pi < \text{Im } w \leq \pi$

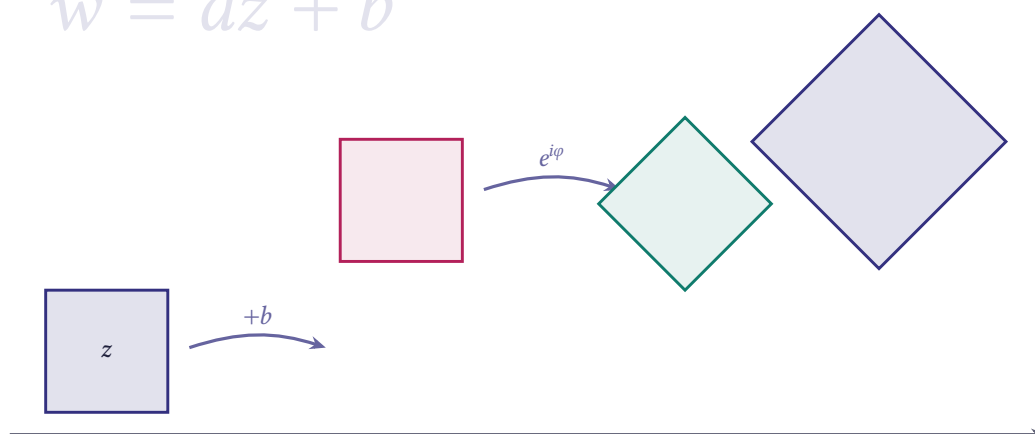
The following properties hold for the logarithm function:

$$\ln(z_1 \times z_2) = \ln z_1 + \ln z_2, \quad \ln\left(\frac{z_1}{z_2}\right) = \ln z_1 - \ln z_2, \quad \ln(z^c) = c \ln z$$

Linear Mappings

نگاشت‌های خطی

$$w = az + b$$



In this chapter

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This chapter examines some simple mappings from the domain plane (z) to the range plane (w).

4.1 Translation

$$\begin{aligned} w = z + b &= (x + iy) + \operatorname{Re}\{b\} + i\operatorname{Im}\{b\} \\ &= (x + \operatorname{Re}\{b\}) + i(y + \operatorname{Im}\{b\}) \end{aligned}$$

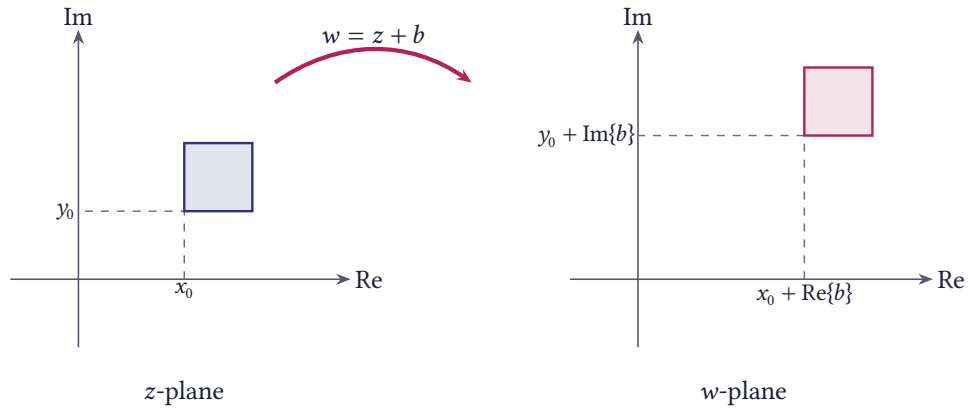


Figure 4.15 Translation $w = z + b$: a region of the z -plane is shifted by b without changing its shape

4.2 Rotation

$$\begin{aligned} w &= az, \quad |a| = 1, \quad a = e^{i\varphi}, \quad \varphi = \arg\{a\} \\ z &= |z|e^{i\theta} \implies w = |z|e^{i(\theta+\varphi)} \end{aligned}$$

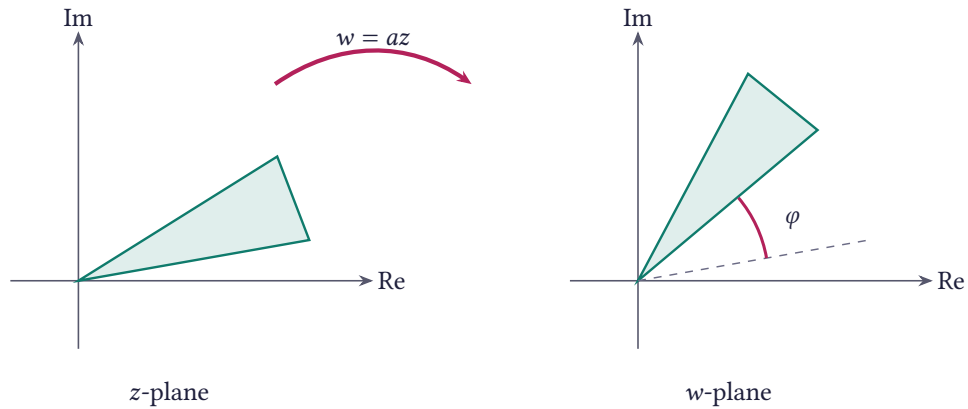


Figure 4.16 Rotation $w = az$ with $|a| = 1$: a region of the z -plane is rotated about the origin by $\varphi = \arg a$

4.3 Dilation or Contraction

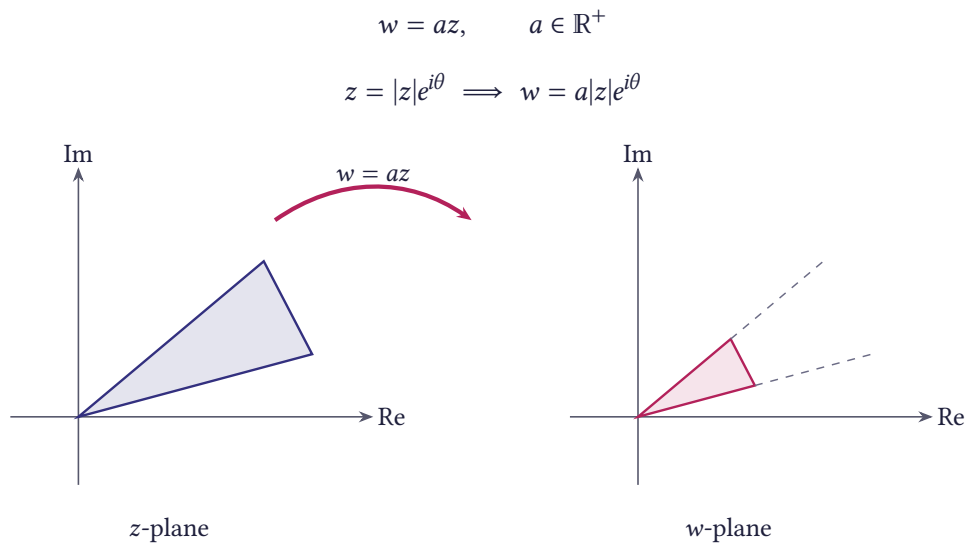


Figure 4.17 Contraction $w = az$ with $0 < a < 1$; for $a > 1$ the mapping is a dilation

4.4 The General Linear Mapping

$$w = az + b, \quad a, b \in \mathbb{C}$$

- (a) a rotation by $\arg a$
- (b) a dilation or contraction by $|a|$
- (c) a translation by b

Example 4.43

For $w = (1 + i)z + (1 - i)$, find the image of the unit square.

$$w(0) = 1 - i, \quad w(1) = 2, \quad w(i) = 0, \quad w(1 + i) = 1 + i$$

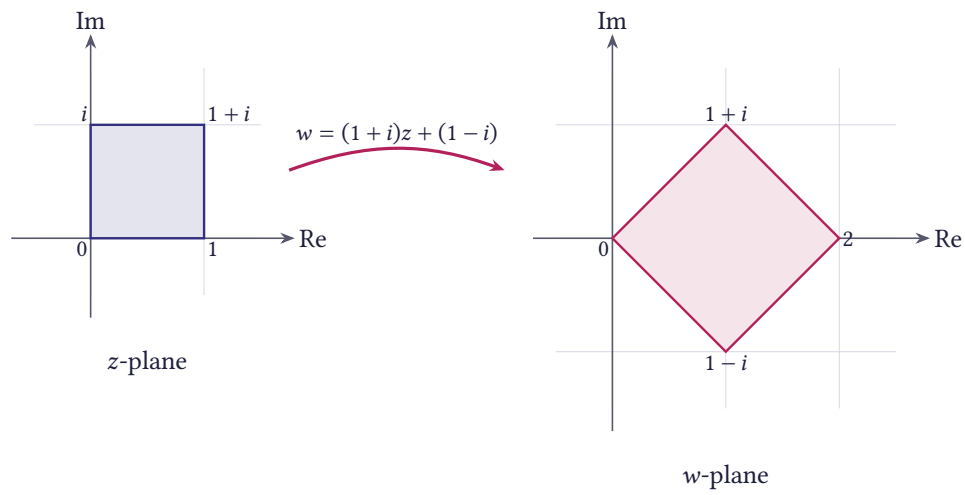
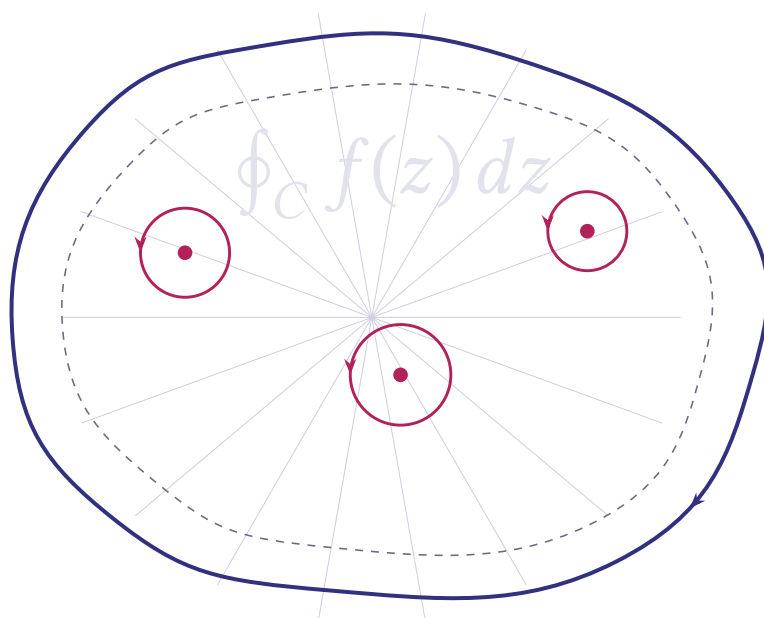


Figure 4.18 The image of the unit square under the mapping $w = (1+i)z + (1-i)$

Integration of Complex Functions

انتگرال توابع مختلط



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5.1 The Definite Integral and the Path of Integration

The integral of complex functions, like that of real functions, appears in two forms:

- the indefinite integral, which amounts to finding an antiderivative;
- the definite integral, known as the integral along a path.

In the setting of complex functions the definite integral is also called the path integral or *line integral*, and is written

$$\int_C f(z) dz$$

C is called the path of integration.

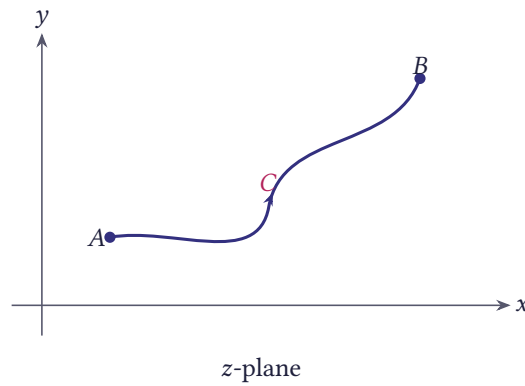


Figure 5.19 A path of integration C from the point A to the point B in the z -plane

■ The Path of Integration

A path of integration in the complex plane consists of a starting point, an intermediate route and an end point. In many cases the best way to describe a path is to use an auxiliary variable which, as it increases, traverses the path from start to end in the desired direction.

$$z(t) = x(t) + iy(t), \quad a \leq t \leq b, \quad t, a, b \in \mathbb{R}$$

$$z(a) = A, \quad z(b) = B, \quad z, A, B \in \mathbb{C}$$

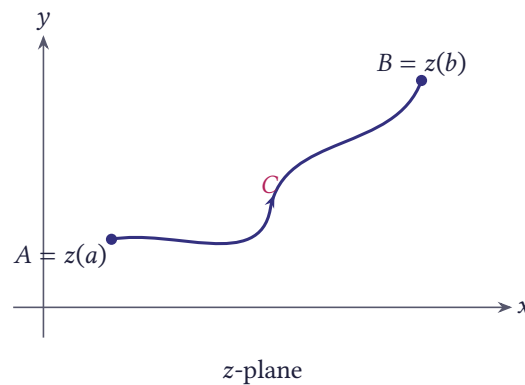


Figure 5.20 The path C , directed from $A = z(a)$ to $B = z(b)$

Example 5.44

$$z(t) = t + i2t, \quad 1 \leq t \leq 2$$

This path is the line segment from the point $1 + 2i$ to the point $2 + 4i$.

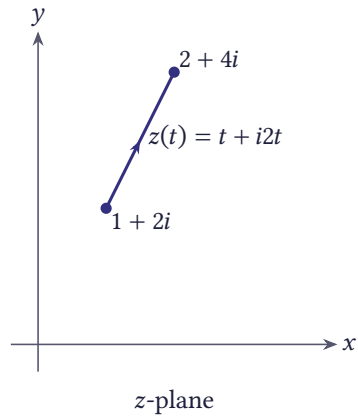


Figure 5.21 The path $z(t) = t + i2t$, $1 \leq t \leq 2$

Example 5.45

$$z(t) = \cos t + i \sin t, \quad 0 \leq t \leq 2\pi$$

$$z(t) = e^{it}, \quad 0 \leq t \leq 2\pi$$

This path is the unit circle, traversed counterclockwise.

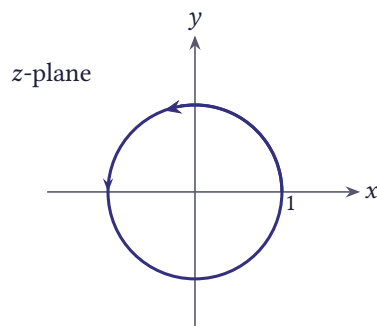


Figure 5.22 The path $z(t) = e^{it}$, $0 \leq t \leq 2\pi$ (the unit circle)

■ The Derivative of the Path

$$\dot{z}(t) = \frac{d}{dt}z(t) = \lim_{\Delta t \rightarrow 0} \frac{z(t + \Delta t) - z(t)}{\Delta t} = \frac{d}{dt}x(t) + i \frac{d}{dt}y(t) = \dot{x}(t) + i\dot{y}(t)$$

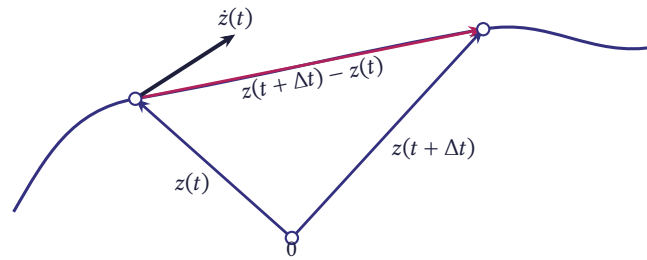


Figure 5.23 The derivative of the path $\dot{z}(t)$ as the limit of $\frac{z(t+\Delta t)-z(t)}{\Delta t}$

5.2 The Definite Integral of Complex Functions

$$\begin{aligned} \int_C f(z) dz &= \int_C (u + iv)(dx + i dy) = \int_C (u dx - v dy) + i \int_C (v dx + u dy) \\ \int_C f(z) dz &= \int_t f(z(t)) \dot{z}(t) dt \end{aligned}$$

Definition 5.46 ■ Steps for computing a complex definite integral along a path

1. Describe the path parametrically: $z(t) = x(t) + iy(t)$, $a \leq t \leq b$.
2. Compute the derivative of $z(t)$ with respect to t : $\dot{z}(t) = \dot{x}(t) + i\dot{y}(t)$.
3. Substitute $z(t)$ into $f(z)$.
4. Integrate $f(z(t))\dot{z}(t)$ with respect to t for $a \leq t \leq b$.

Example 5.47

Compute the integral of the function $f(z) = \bar{z}$ along the path shown.

$$z(t) = t + it^2, \quad -1 \leq t \leq 1, \quad \dot{z}(t) = 1 + i2t$$

$$f(z) = \bar{z} = x - iy, \quad f(z(t)) = \bar{z}(t) = t - it^2$$

$$\begin{aligned} \int_{-1+i}^{1+i} \bar{z} dz &= \int_{t=-1}^1 (t - it^2)(1 + i2t) dt \\ &= \int_{t=-1}^1 [(t + 2t^3) + i(2t^2 - t^2)] dt = \int_{t=-1}^1 [(t + 2t^3) + it^2] dt \\ &= \left(\frac{1}{2}t^2 + \frac{2}{4}t^4 \right) + i \times \frac{1}{3}t^3 \Big|_{-1}^1 = 0 + i \frac{2}{3} \end{aligned}$$

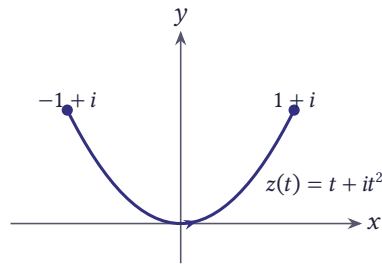


Figure 5.24 The path $z(t) = t + it^2$, $-1 \leq t \leq 1$, from $-1 + i$ to $1 + i$

Example 5.48

Compute the integral of the function $f(z) = \bar{z}$ along the path shown.

$$z(t) = t + i, \quad -1 \leq t \leq 1, \quad \dot{z}(t) = 1$$

$$f(z) = \bar{z} = x - iy, \quad f(z(t)) = \bar{z}(t) = t - i$$

$$\int_{-1+i}^{1+i} \bar{z} dz = \int_{t=-1}^1 (t - i)(1) dt = \left(\frac{t^2}{2} - it \right) \Big|_{-1}^1 = 0 - i2$$

So for this complex function the value of the integral depends on the chosen path.

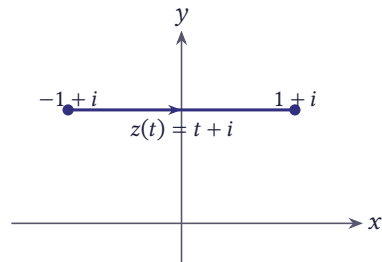


Figure 5.25 The path $z(t) = t + i$, $-1 \leq t \leq 1$, from $-1 + i$ to $1 + i$

■ Some Properties of the Definite Integral

$$\int_C [k_1 f_1(z) + k_2 f_2(z)] dz = k_1 \int_C f_1(z) dz + k_2 \int_C f_2(z) dz \quad (\text{linearity})$$

$$\int_{z_0}^Z f(z) dz = - \int_Z^{z_0} f(z) dz \quad (\text{reversing the direction of integration})$$

$$\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz \quad (\text{splitting the path of integration})$$

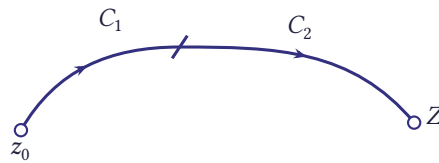


Figure 5.26 A path C from z_0 to Z split into two pieces C_1 and C_2

5.3 Simple Closed Curves and Simply Connected Domains

Definition 5.49 ■ Simple closed curve

A simple closed curve is a closed curve that does not intersect itself.

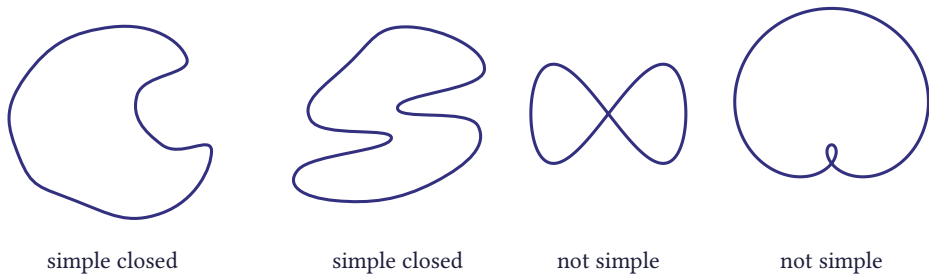


Figure 5.27 Simple closed curves and closed curves that are not simple

Definition 5.50 ■ Simply connected domain

A domain D is simply connected if, whenever a simple closed curve lies entirely in it, all the points inside the curve belong to this domain.

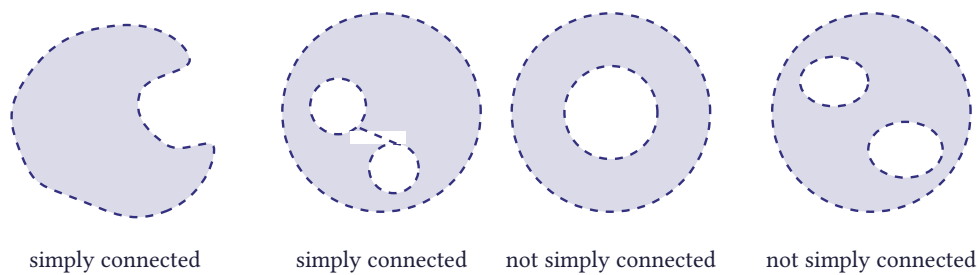


Figure 5.28 Simply connected and not simply connected domains

5.4 Integration by Antiderivative

Theorem 5.51

Let the complex function $f(z)$ be analytic in the simply connected domain D . If $F(z)$ is an antiderivative of $f(z)$, i.e. $F'(z) = f(z)$, then

$$\int_{z_0}^{z_1} f(z) dz = F(z_1) - F(z_0)$$

Example 5.52

$$\int_0^{1+i} z^2 dz = \frac{1}{3} z^3 \Big|_0^{1+i} = \frac{1}{3} (1+i)^3 = \frac{2}{3} (-1+i)$$

Example 5.53

$$\int_{-\pi i}^{\pi i} \cos z dz = \sin z \Big|_{-\pi i}^{\pi i} = 2 \sin(\pi i) = 2i \sinh \pi = 23.097 i$$

Example 5.54

$$\int_{8+\pi i}^{8-3\pi i} e^{z/2} dz = 2e^{z/2} \Big|_{8+\pi i}^{8-3\pi i} = 2(e^{4-3\pi i/2} - e^{4+\pi i/2}) = 0$$

Example 5.55

$$\int_{-i}^i \frac{dz}{z} = \ln i - \ln(-i) = \frac{i\pi}{2} - \left(-\frac{i\pi}{2}\right) = i\pi$$

5.5 Integrals of Complex Functions Along Closed Paths

The integral along a simple closed path C , written

$$\oint_C f(z) dz$$

is of great importance.

Theorem 5.56 ■ Cauchy's integral theorem

If the function $f(z)$ is analytic in the simply connected domain D , then for every simple closed path C in this domain,

$$\oint_C f(z) dz = 0$$

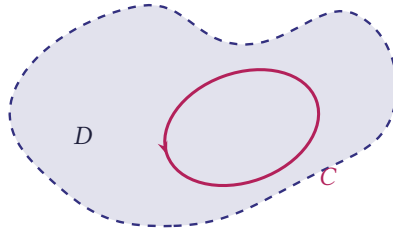


Figure 5.29 A simple closed path C in the simply connected domain D

Proof.

As we saw earlier, since the complex function is analytic, we have

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\oint_C f(z) dz = \oint_C (u + iv)(dx + i dy) = \oint_C (u dx - v dy) + i \oint_C (v dx + u dy)$$

By Green's theorem (which you learned in Mathematics II),

$$\oint_C (L dx + M dy) = \iint_R \left(\frac{\partial M}{\partial x} - \frac{\partial L}{\partial y} \right) dx dy$$

Hence

$$\oint_C (u dx - v dy) + i \oint_C (v dx + u dy) = \iint_R \underbrace{\left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)}_0 dx dy + i \iint_R \underbrace{\left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right)}_0 dx dy = 0$$

where the first term vanishes because of the second Cauchy–Riemann condition and the second because of the first. ■

Example 5.57

If the function is analytic in the whole complex plane, the integral along any closed path is zero:

$$\oint_C \cos z dz = 0, \quad \oint_C e^z dz = 0, \quad \oint_C z^n dz = 0, \quad n = 1, 2, 3, \dots$$

Example 5.58

If the function has non-analytic points that lie neither on the path nor inside C , the integral is still zero:

$$\oint_{|z|=1} \sec z \, dz, \quad \oint_{|z|=1} \frac{dz}{z^2 + 4}$$

(the path is the unit circle.)

Example 5.59

If the function is not analytic, the integral along a closed path **may** be nonzero:

$$\oint_{|z|=1} \bar{z} \, dz, \quad z(t) = \cos t + i \sin t = e^{it}, \quad \dot{z}(t) = ie^{it}$$

$$\oint_{|z|=1} \bar{z} \, dz = \int_0^{2\pi} e^{-it} ie^{it} dt = 2\pi i$$

Example 5.60

If the function is not analytic on the path or inside it, the integral along a closed path **may** be zero:

$$\oint_C \frac{1}{z^2} dz = 0$$

Exercise 5.61

Prove that $\oint_C \frac{1}{z^2} dz = 0$.

Consequence 1

The connectedness of the domain is essential in Cauchy's integral theorem.

Consequence 2

If $f(z)$ is analytic in the simply connected domain D , the definite integral does not depend on the choice of path.

$$\int_{C_1} f(z) dz + \int_{C_2^*} f(z) dz = 0 \implies \int_{C_1} f(z) dz = - \int_{C_2^*} f(z) dz \implies \int_{C_1} f(z) dz = \int_{C_2} f(z) dz$$

Hence the integral from the starting point to the end point does not depend on the chosen path.

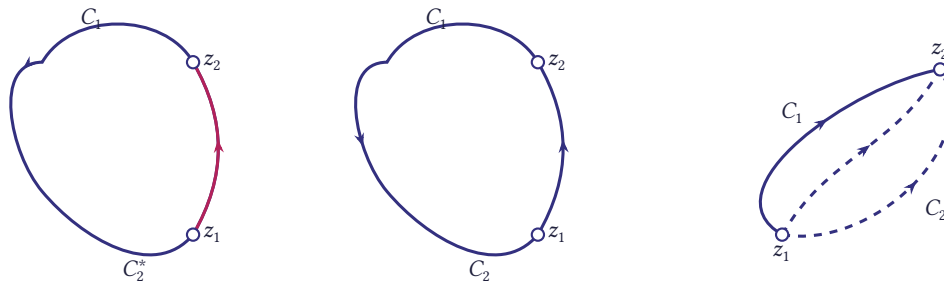


Figure 5.30 The paths C_1 , C_2 and C_2^* (C_2 with reversed direction) between z_1 and z_2

■ A Very Important Example

$$\oint_C \frac{dz}{z - z_0} = ? \quad C : z(t) = z_0 + re^{it}$$

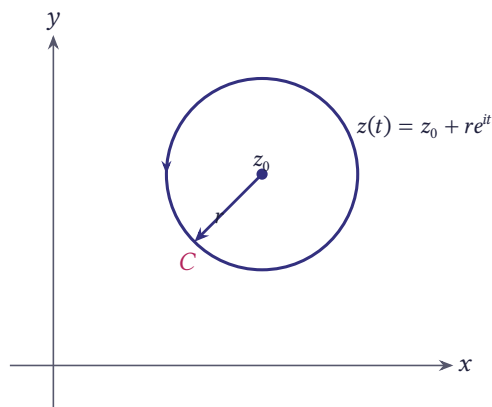


Figure 5.31 A circle C with centre z_0 and radius r

$$z(t) = z_0 + re^{it}, \quad \dot{z}(t) = rie^{it}$$

$$\oint_C \frac{dz}{z - z_0} = \int_0^{2\pi} \frac{rie^{it}}{re^{it}} dt = \int_0^{2\pi} i dt = 2\pi i$$

Key result 5.62

$$\oint_C \frac{dz}{z - z_0} = 2\pi i$$

5.6 Cauchy's Integral Formula

Theorem 5.63 ■ Cauchy's integral formula

If the function $f(z)$ is analytic in the simply connected domain D and z_0 is a point inside C , then

$$\frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz = f(z_0)$$

Proof.

As we saw earlier, since the function $\frac{f(z)}{z - z_0}$ is analytic inside the connected domain Γ , we have

$$\begin{aligned} \oint_{\Gamma} \frac{f(z)}{z - z_0} dz &= 0 = \int_{ACA'} \frac{f(z)}{z - z_0} dz + \int_{A'B'} \frac{f(z)}{z - z_0} dz + \int_{B'C'B} \frac{f(z)}{z - z_0} dz + \int_{BA} \frac{f(z)}{z - z_0} dz \\ \lim_{\substack{A \rightarrow A' \\ B \rightarrow B'}} \int_{BA} \frac{f(z)}{z - z_0} dz &= - \int_{A'B'} \frac{f(z)}{z - z_0} dz \end{aligned}$$

Consequently (the two integrals along the connecting segments cancel)

$$\lim_{\substack{A \rightarrow A' \\ B \rightarrow B'}} \int_{ACA'} \frac{f(z)}{z - z_0} dz + \int_{B'C'B} \frac{f(z)}{z - z_0} dz = 0 \implies \oint_C \frac{f(z)}{z - z_0} dz = \oint_{C'} \frac{f(z)}{z - z_0} dz$$

(here C' is traversed counterclockwise). Continuing:

$$\begin{aligned} \oint_C \frac{f(z)}{z - z_0} dz &= \oint_{C'} \frac{f(z)}{z - z_0} dz = \oint_{C'} \frac{f(z) - f(z_0) + f(z_0)}{z - z_0} dz \\ &= \oint_{C'} \frac{f(z) - f(z_0)}{z - z_0} dz + \oint_{C'} \frac{f(z_0)}{z - z_0} dz \\ &= \oint_{C'} g(z) dz + f(z_0) \oint_{C'} \frac{1}{z - z_0} dz = 0 + 2\pi i f(z_0) = 2\pi i f(z_0) \end{aligned}$$

where

$$g(z) \triangleq \begin{cases} \frac{f(z) - f(z_0)}{z - z_0} & z \neq z_0 \\ f'(z_0) & z = z_0 \end{cases}$$

is analytic inside C' . Therefore

$$\frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz = f(z_0)$$

■

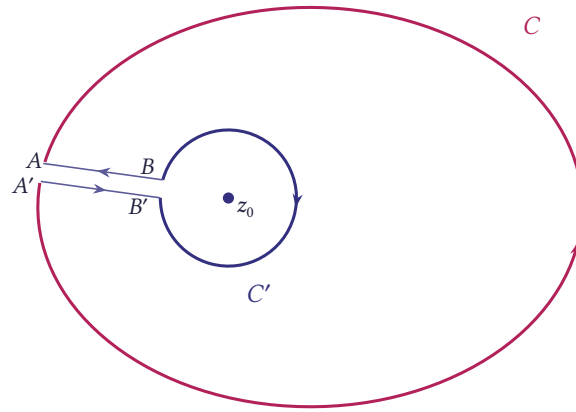


Figure 5.32 The path Γ in the proof of Cauchy's integral formula: the curve C , the small circle C' centred at z_0 , and the two connecting segments AA' and BB'

5.7 Application: Evaluating Real Integrals

Example 5.64

Evaluate the integral $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$ using Cauchy's integral formula.

Consider the function $g(z) = \frac{1}{1+z^2}$. Using the path shown in the figure we find the integral along the closed path.

$$\oint_C \frac{dz}{1+z^2} = \oint_C \left[\frac{1}{z-i} \right] dz = 2\pi i \frac{1}{i+i} = \pi, \quad f(z) = \frac{1}{z+i}, \quad z_0 = i$$

On the other hand

$$\oint_C \frac{dz}{1+z^2} = \int_{-R}^R \frac{dx}{1+x^2} + \int_0^\pi \frac{iRe^{it} dt}{1+R^2e^{i2t}}$$

In the expression above, the first term as $R \rightarrow \infty$ is the required value.

$$\oint_C \frac{dz}{1+z^2} = \pi = \int_{-\infty}^{\infty} \frac{dx}{1+x^2} + I, \quad I = \lim_{R \rightarrow \infty} \int_0^\pi \frac{iRe^{it} dt}{1+R^2e^{i2t}} = 0$$

Therefore

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi$$

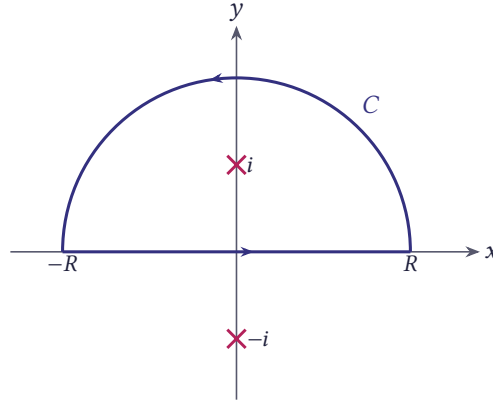


Figure 5.33 The semicircular path in the upper half-plane; the poles are at $z = \pm i$

Example 5.65

Evaluate the integral $\int_{-\infty}^{\infty} \frac{\cos \omega x}{1+x^2} dx$ using Cauchy's integral formula ($\omega > 0$).

Consider the function $g(z) = \frac{e^{i\omega z}}{1+z^2}$. Using the path shown in the figure (the same semicircle as before) we find the integral along the closed path.

$$\oint_C \frac{e^{i\omega z}}{1+z^2} dz = \oint_C \frac{\left[\frac{e^{i\omega z}}{z-i} \right]}{z+i} dz = 2\pi i \frac{e^{i\omega i}}{i+i} = \pi e^{-\omega}, \quad f(z) = \frac{e^{i\omega z}}{z+i}, \quad z_0 = i$$

On the other hand

$$\oint_C \frac{e^{i\omega z}}{1+z^2} dz = \int_{-R}^R \frac{e^{i\omega x}}{1+x^2} dx + \int_0^\pi \frac{e^{i\omega R e^{it}} i R e^{it} dt}{1+R^2 e^{i2t}}$$

But:

$$I_1 = \lim_{R \rightarrow \infty} \int_0^\pi \frac{e^{i\omega R e^{it}} i R e^{it} dt}{1+R^2 e^{i2t}} \rightarrow 0$$

Proof:

$$e^{i\omega R e^{it}} = e^{i\omega R(\cos t + i \sin t)} = e^{i\omega R \cos t} e^{-\omega R \sin t}$$

$$\lim_{R \rightarrow \infty} |e^{i\omega R e^{it}}| = \lim_{R \rightarrow \infty} |e^{i\omega R \cos t}| \times |e^{-\omega R \sin t}| = 1 \times 0 = 0, \quad t \in [0, \pi]$$

On the other hand, in the integral I_1 the remaining factor of the integrand tends to zero as R tends to infinity. So the limit of I_1 as $R \rightarrow \infty$ is zero.

Therefore

$$\pi e^{-\omega} = \oint_C \frac{e^{i\omega z}}{1+z^2} dz = \int_{-R}^R \frac{e^{i\omega x}}{1+x^2} dx = \int_{-R}^R \frac{(\cos x + i \sin x) dx}{1+x^2} = \int_{-R}^R \frac{\cos x dx}{1+x^2} + i \int_{-R}^R \frac{\sin x dx}{1+x^2}$$

The second integral is zero. Why?

$$\int_{-\infty}^{\infty} \frac{\cos \omega x}{1+x^2} dx = \pi e^{-\omega}$$

Example 5.66

Evaluate the integral $I = \int_{-\infty}^{\infty} \frac{dx}{(x^2 + 1)(x^2 + 4)}$ using Cauchy's integral formula.

$$\frac{1}{(x^2 + 1)(x^2 + 4)} = \frac{A}{x^2 + 1} + \frac{B}{x^2 + 4} = \frac{1/3}{x^2 + 1} + \frac{-1/3}{x^2 + 4}$$

$$\int_{-\infty}^{\infty} \frac{dx}{(x^2 + 1)(x^2 + 4)} = \frac{1}{3} \int_{-\infty}^{\infty} \frac{dx}{x^2 + 1} - \frac{1}{3} \int_{-\infty}^{\infty} \frac{dx}{x^2 + 4}$$

First integral: $I_1 = \int_{-\infty}^{\infty} \frac{dx}{x^2 + 1}$ and $g_1(z) = \frac{1}{z^2 + 1}$:

$$\oint_C g_1(z) dz = \oint_C \frac{1}{z^2 + 1} dz = \oint_C \left[\frac{1}{z + i} \right] dz = 2\pi i \frac{1}{i + i} = \pi$$

$$\oint_C \frac{1}{z^2 + 1} dz = \pi = \int_{-R}^R \frac{dx}{1 + x^2} + \int_0^\pi \frac{iRe^{it} dt}{1 + R^2 e^{i2t}}$$

$$\lim_{R \rightarrow \infty} \oint_C \frac{1}{z^2 + 1} dz = \int_{-R}^R \frac{dx}{1 + x^2} + 0 = \pi \implies I_1 = \pi$$

Second integral: $I_2 = \int_{-\infty}^{\infty} \frac{dx}{x^2 + 4}$ and $g_2(z) = \frac{1}{z^2 + 4}$:

$$\oint_C g_2(z) dz = \oint_C \frac{1}{z^2 + 4} dz = \oint_C \left[\frac{1}{z + 2i} \right] dz = 2\pi i \frac{1}{2i + 2i} = \frac{\pi}{2}$$

$$\oint_C \frac{1}{z^2 + 4} dz = \frac{\pi}{2} = \int_{-R}^R \frac{dx}{x^2 + 4} + \int_0^\pi \frac{iRe^{it} dt}{R^2 e^{i2t} + 4}$$

$$\lim_{R \rightarrow \infty} \oint_C \frac{1}{z^2 + 4} dz = \int_{-R}^R \frac{dx}{x^2 + 4} + 0 = \frac{\pi}{2} \implies I_2 = \frac{\pi}{2}$$

As a result

$$I = \frac{1}{3}I_1 - \frac{1}{3}I_2 = \frac{1}{3}\pi - \frac{1}{3} \times \frac{\pi}{2} = \frac{\pi}{6}$$

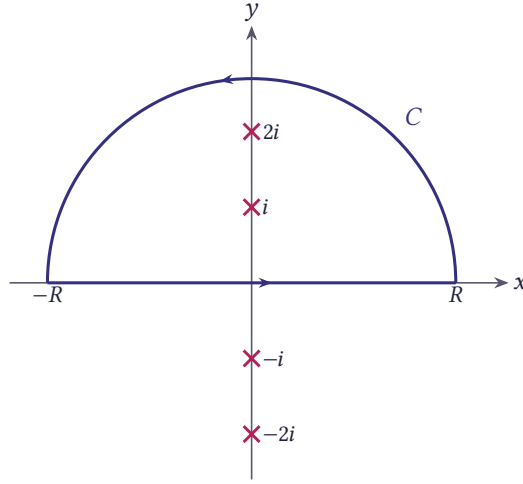


Figure 5.34 The semicircular path; the poles are at $z = \pm i$ and $z = \pm 2i$

Example 5.67

Evaluate the integral $I = \int_0^{\infty} \frac{dx}{x^4 + 1}$ using Cauchy's integral formula.

$$g(z) = \frac{1}{z^4 + 1}, \quad z^4 + 1 = 0 \implies z_1 = e^{i\pi/4}, \quad z_2 = e^{i3\pi/4}, \quad z_3 = e^{i5\pi/4}, \quad z_4 = e^{i7\pi/4}$$

Only the two poles z_1 and z_2 lie inside the path C :

$$\begin{aligned} \oint_C g(z) dz &= \oint_C \frac{1}{z^4 + 1} dz = \oint_{C_1} \frac{1}{z^4 + 1} dz + \oint_{C_2} \frac{1}{z^4 + 1} dz \\ &= \oint_{C_1} \frac{1}{(z - z_2)(z - z_3)(z - z_4)} dz + \oint_{C_2} \frac{1}{(z - z_1)(z - z_3)(z - z_4)} dz \\ &= 2\pi i \frac{1}{(z - z_2)(z - z_3)(z - z_4)} \Big|_{z=z_1} + 2\pi i \frac{1}{(z - z_1)(z - z_3)(z - z_4)} \Big|_{z=z_2} \\ &= 2\pi i \frac{1}{\underbrace{(e^{i\pi/4} - e^{i3\pi/4})}_{\sqrt{2}} \underbrace{(e^{i\pi/4} - e^{i5\pi/4})}_{\sqrt{2}+i\sqrt{2}} \underbrace{(e^{i\pi/4} - e^{i7\pi/4})}_{i\sqrt{2}}} \\ &\quad + 2\pi i \frac{1}{\underbrace{(e^{i3\pi/4} - e^{i\pi/4})}_{-\sqrt{2}} \underbrace{(e^{i3\pi/4} - e^{i5\pi/4})}_{i\sqrt{2}} \underbrace{(e^{i3\pi/4} - e^{i7\pi/4})}_{-\sqrt{2}+i\sqrt{2}}} = \frac{\pi}{\sqrt{2}} \end{aligned}$$

On the other hand

$$\begin{aligned} \oint_C \frac{1}{z^4 + 1} dz &= \frac{\pi}{\sqrt{2}} = \int_{-R}^R \frac{dx}{x^4 + 1} + \int_0^{\pi} \frac{iRe^{it} dt}{R^4 e^{i4t} + 1} \\ \lim_{R \rightarrow \infty} \oint_C \frac{1}{z^4 + 1} dz &= \int_{-\infty}^{\infty} \frac{dx}{x^4 + 1} + \underbrace{\lim_{R \rightarrow \infty} \int_0^{\pi} \frac{iRe^{it} dt}{R^4 e^{i4t} + 1}}_0 = \frac{\pi}{\sqrt{2}} \\ \int_{-\infty}^{\infty} \frac{dx}{x^4 + 1} &= \frac{\pi}{\sqrt{2}} \implies \int_0^{\infty} \frac{dx}{x^4 + 1} = \frac{\pi}{2\sqrt{2}} \end{aligned}$$

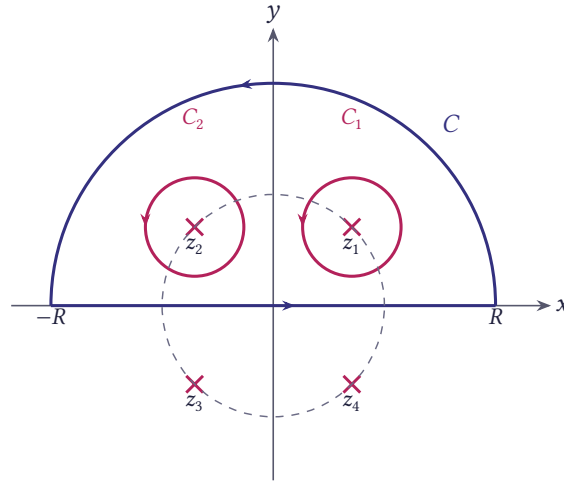


Figure 5.35 The semicircular path and the circles C_1 and C_2 around the poles z_1 and z_2 ; the poles z_3 and z_4 lie in the lower half-plane

5.8 Cauchy's Theorem with Several Non-Analytic Points

Theorem 5.68

If the function $f(z)$ is analytic at all points of the domain D except the points $z_1, z_2, z_3, \dots, z_n$ inside C , then

$$\oint_C f(z) dz = \oint_{C_1} f(z) dz + \oint_{C_2} f(z) dz + \dots + \oint_{C_n} f(z) dz$$

where C_k is a circle centred at z_k that lies inside C and does not contain any of the other points $z_1, \dots, z_n$.

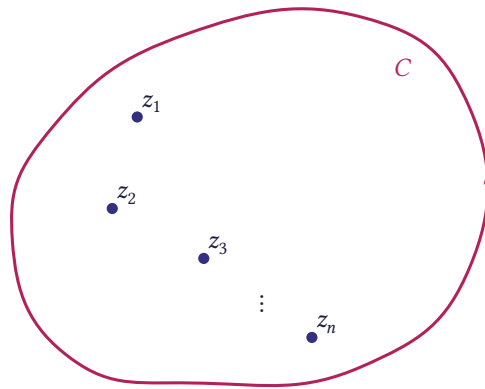


Figure 5.36 A path C and the non-analytic points $z_1, \dots, z_n$ inside it

Proof.

Build the path Γ from C (counterclockwise), the small circles $C_1, \dots, C_n$ (clockwise) and the segments connecting them to C ; the integral along each connecting segment is traversed twice in opposite directions and cancels. The function is analytic inside Γ , so

$$\oint_{\Gamma} f(z) dz = 0 \implies \oint_C f(z) dz + \oint_{C_1^c} f(z) dz + \dots + \oint_{C_n^c} f(z) dz = 0$$

$$\oint_C f(z) dz = -\oint_{C_1^\cup} f(z) dz - \dots - \oint_{C_n^\cup} f(z) dz = \oint_{C_1} f(z) dz + \dots + \oint_{C_n} f(z) dz$$

where $C_k^\cup$ is the circle C_k traversed clockwise and C_k the same circle traversed counterclockwise.

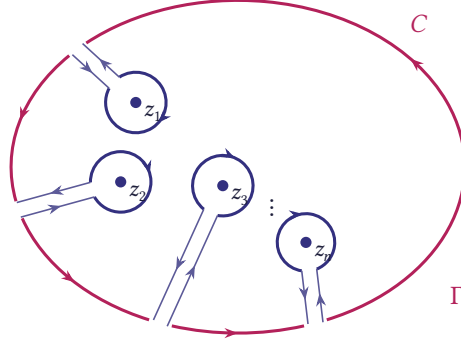


Figure 5.37 The path Γ for the proof: C , the circles $C_1, \dots, C_n$ and the connecting segments

5.9 Derivatives of an Analytic Function

Theorem 5.69 ■ Cauchy's integral formula for the derivative

If the function $f(z)$ is analytic in the simply connected domain D and z_0 is a point inside C , then

$$\frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz = f'(z_0)$$

Proof.

$$f'(z_0) = \lim_{\Delta t \rightarrow 0} \frac{f(z_0 + \Delta t) - f(z_0)}{\Delta t}, \quad f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz, \quad f(z_0 + \Delta t)$$

$$= \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0 - \Delta t} dz$$

$$\begin{aligned} \frac{f(z_0 + \Delta t) - f(z_0)}{\Delta t} &= \frac{1}{2\pi i \Delta t} \left\{ \oint_C \frac{f(z)}{z - z_0 - \Delta t} dz - \oint_C \frac{f(z)}{z - z_0} dz \right\} \\ &= \frac{1}{2\pi i \Delta t} \oint_C f(z) \left[\frac{1}{z - z_0 - \Delta t} - \frac{1}{z - z_0} \right] dz \\ &= \frac{1}{2\pi i \Delta t} \oint_C f(z) \left[\frac{\Delta t}{(z - z_0 - \Delta t)(z - z_0)} \right] dz \\ &= \frac{1}{2\pi i} \oint_C f(z) \left[\frac{1}{(z - z_0 - \Delta t)(z - z_0)} \right] dz \end{aligned}$$

$$\begin{aligned} f'(z_0) &= \lim_{\Delta t \rightarrow 0} \frac{f(z_0 + \Delta t) - f(z_0)}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{1}{2\pi i} \oint_C f(z) \left[\frac{1}{(z - z_0 - \Delta t)(z - z_0)} \right] dz \\ &= \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz \end{aligned}$$

Theorem 5.70 ■ Generalization of Cauchy's integral formula

If the function $f(z)$ is analytic in the simply connected domain D and z_0 is a point inside C , then

$$\frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz = f^{(n)}(z_0)$$

Exercise 5.71

Prove it by mathematical induction.

Example 5.72

Evaluate the following integral. (C is the unit circle.)

$$I = \oint_C \frac{z^2}{(2z - 1)^2} dz$$

$$I = \oint_C \frac{z^2}{4(z - 1/2)^2} dz = \frac{1}{4} \oint_C \frac{z^2}{(z - 1/2)^2} dz = \frac{1}{4} \times 2\pi i \times \left. \frac{d}{dz}(z^2) \right|_{z=1/2} = \frac{1}{4} \times 2\pi i \times (2z) \Big|_{z=1/2}$$

$$= \frac{\pi}{2} i$$

($z = 1/2$ lies inside the unit circle.)

Example 5.73

Evaluate the following integral. (C is the unit circle.)

$$I = \oint_C \frac{e^{-z} \sin z}{z^2} dz$$

$$f(z) = e^{-z} \sin z, \quad z_0 = 0, \quad f'(z) = -e^{-z} \sin z + e^{-z} \cos z, \quad f'(z_0) = f'(0) = 1$$

$$I = 2\pi i f'(z_0) = 2\pi i$$

Exercise 5.74 ■ Problem

If $f(z)$ is analytic in the whole z -plane and bounded in the whole plane, i.e. $|f(z)| < M$, prove that $f(z) = k$, $k \in \mathbb{C}$.

Choose the path C to be a circle of radius R centred at z_0 , and let R tend to infinity.

$$f'(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz$$

$$|f'(z_0)| = \left| \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz \right| \leq \frac{1}{2\pi} \oint_C \frac{|f(z)|}{|z - z_0|^2} |dz| \leq \frac{1}{2\pi} \frac{M}{R^2} \oint_C |dz| = \frac{1}{2\pi} \frac{M}{R^2} \cdot 2\pi R$$

$$= \frac{M}{R} \rightarrow 0 \quad (R \rightarrow \infty)$$

$$\Rightarrow |f'(z_0)| = 0 \Rightarrow f'(z) = 0 \Rightarrow f(z) = k$$

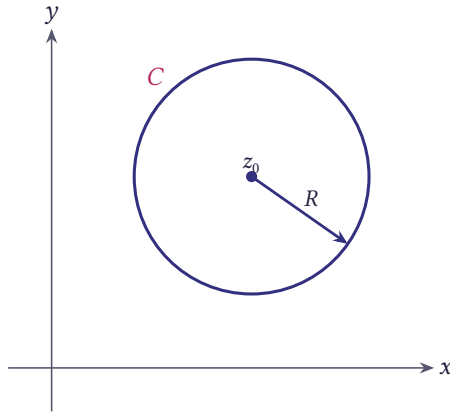


Figure 5.38 A circle C with centre z_0 and radius R

Exercise 5.75

Evaluate the following integrals:

1. $\oint_C \frac{z^2}{(2z - 1)^4} dz$, C : the unit circle
2. $\int_{1+i}^{1+3i} e^z \sin z \, dz$
3. $\int_{1+i}^{2+2i} \operatorname{Re}\{z^2\} \, dz$ along a straight-line path
4. $\oint_C \frac{e^{\cos z} \sin z}{1 + z^6} dz$ along the path below

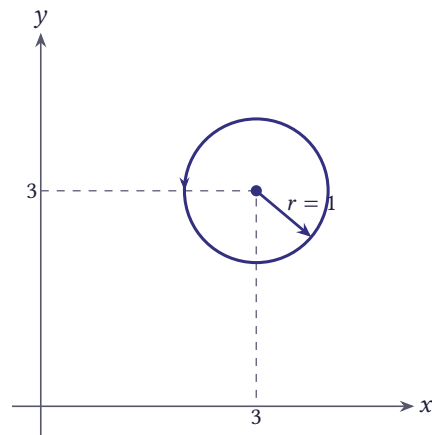
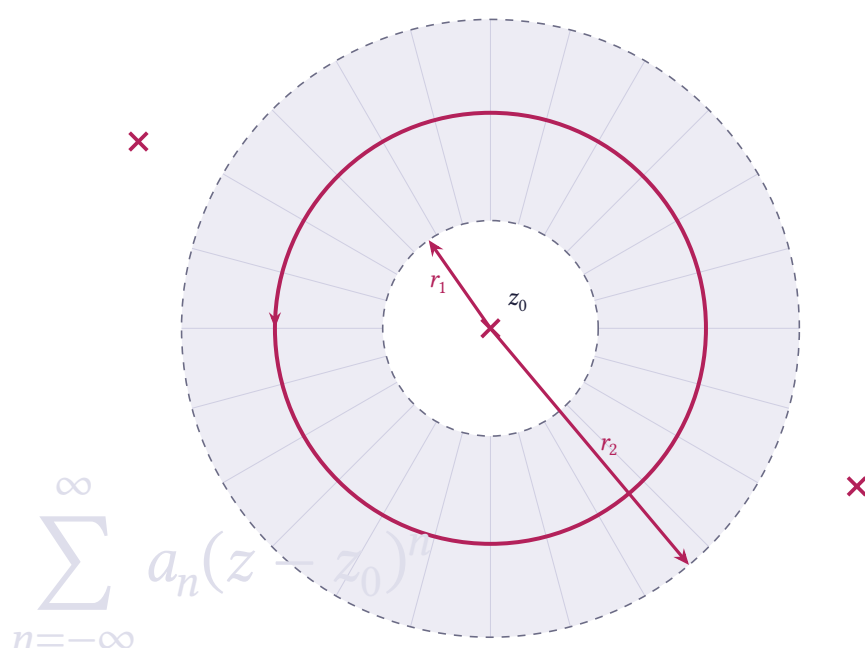


Figure 5.39 The path of exercise 4: a circle with centre $3 + 3i$ and radius $r = 1$

Laurent Series, Singularities and Residues

سری لورنت، نقاط تکین و مانده‌ها



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6.1 The Laurent Series

Theorem 6.76 ■ Laurent series

If the function $f(z)$ is analytic in the domain D , its Laurent series about the point z_0 is written

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n, \quad a_n = \frac{1}{2\pi i} \oint_C \frac{f(v)}{(v - z_0)^{n+1}} dv$$

where C is a circle centred at z_0 and the function is analytic at all points inside and on this circle, except possibly at z_0 .

The radius of convergence is the distance from z_0 to the nearest non-analytic point of $f(z)$.

For non-negative n we have

$$a_n = \frac{1}{2\pi i} \oint_C \frac{f(v)}{(v - z_0)^{n+1}} dv = \frac{f^{(n)}(z_0)}{n!}, \quad n = 0, 1, 2, \dots$$

Therefore

$$\begin{aligned} f(z) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n + \sum_{n=-\infty}^{-1} a_n (z - z_0)^n \\ &= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n + \sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n} \\ &= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n + \sum_{n=1}^{\infty} \frac{A_n}{(z - z_0)^n}, \quad A_n = a_{-n} \end{aligned}$$

$$A_n = a_{-n} = \frac{1}{2\pi i} \oint_C \frac{f(v)}{(v - z_0)^{-n+1}} dv = \frac{1}{2\pi i} \oint_C (v - z_0)^{n-1} f(v) dv, \quad n = 1, 2, \dots$$

Key result 6.77 ■ The Laurent series: Taylor part and principal part

$$f(z) = \underbrace{\sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n}_{\text{Taylor series part}} + \underbrace{\sum_{n=1}^{\infty} \frac{A_n}{(z - z_0)^n}}_{\text{principal part}}, \quad A_n = \frac{1}{2\pi i} \oint_C (v - z_0)^{n-1} f(v) dv$$

If z_0 is an irregular (singular) point, in other words if the function is not analytic there, the Taylor part alone is not sufficient to describe the function.

■ The Different Cases for the Point z_0

1. If all the A_n are zero, the point z_0 is called a **regular** point. In this case the Laurent series is just the Taylor series.
2. If $A_p \neq 0$ but $(A_n = 0 \ \forall n > p)$, then z_0 is called a **pole of order p** .
3. If z_0 is not a regular point and is also not a pole (in other words, no p can be found such that $A_p \neq 0$ and $(A_n = 0 \ \forall n > p)$), then z_0 is called an **essential singularity**.

6.2 The Residue

Definition 6.78 ■ Residue

The residue of $f(z)$ at the point z_0 is A_1 .

$$\text{Res}\{f(z)\}\Big|_{z=z_0} = A_1 = \frac{1}{2\pi i} \oint_C f(v) dv$$

Theorem 6.79

If the function $f(z)$ has a pole of order p at z_0 , then

$$A_k = \frac{1}{(p-k)!} \frac{d^{p-k}}{dz^{p-k}} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0}, \quad k = 1, 2, \dots, p$$

Proof.

$$f(z) = \sum_{n=1}^{\infty} \frac{A_n}{(z-z_0)^n} + \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^n$$

If $f(z)$ has a pole of order p at z_0 we have

$$f(z) = \frac{A_p}{(z-z_0)^p} + \frac{A_{p-1}}{(z-z_0)^{p-1}} + \dots + \frac{A_2}{(z-z_0)^2} + \frac{A_1}{z-z_0} + \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^n$$

Multiplying both sides by $(z-z_0)^p$ gives

$$(z-z_0)^p f(z) = A_p + A_{p-1}(z-z_0) + \dots + A_2(z-z_0)^{p-2} + A_1(z-z_0)^{p-1} + \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^{n+p}$$

Now if we set $z = z_0$ we get

$$(z-z_0)^p f(z) \Big|_{z=z_0} = A_p$$

If we differentiate $(z-z_0)^p f(z)$ once and then set $z = z_0$:

$$\begin{aligned} \frac{d}{dz} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0} &= \left\{ A_{p-1} + \dots + A_2(p-2)(z-z_0)^{p-3} + A_1(p-1)(z-z_0)^{p-2} \right. \\ &\quad \left. + \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^{n+p-1} (n+p) \right\} \Big|_{z=z_0} = A_{p-1} \end{aligned}$$

If we differentiate $(z-z_0)^p f(z)$ twice and then set $z = z_0$:

$$\begin{aligned} \frac{d^2}{dz^2} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0} &= \left\{ 2A_{p-2} + \dots + A_2(p-2)(p-3)(z-z_0)^{p-4} + A_1(p-1)(p-2)(z-z_0)^{p-3} \right. \\ &\quad \left. + \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z-z_0)^{n+p-2} (n+p)(n+p-1) \right\} \Big|_{z=z_0} = 2A_{p-2} \\ \implies A_{p-2} &= \frac{1}{2!} \frac{d^2}{dz^2} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0} \end{aligned}$$

Exercise 6.80

Continue the differentiation in the same way and complete the proof of the theorem.

Consequence

In particular, when $f(z)$ has a pole of order p at z_0 , the residue is given by

$$\text{Res}\{f(z)\} = A_1 = \frac{1}{(p-1)!} \frac{d^{p-1}}{dz^{p-1}} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0}$$

6.3 The Residue Theorem**Theorem 6.81 ■ The residue theorem**

If the function $f(z)$ is analytic everywhere inside the path C except at the points $z_1, z_2, \dots, z_n$, and the residues of the function at these points are $A_1^1, A_1^2, \dots, A_1^n$ respectively, then

$$\oint_C f(z) dz = 2\pi i \sum_{k=1}^n A_1^k$$

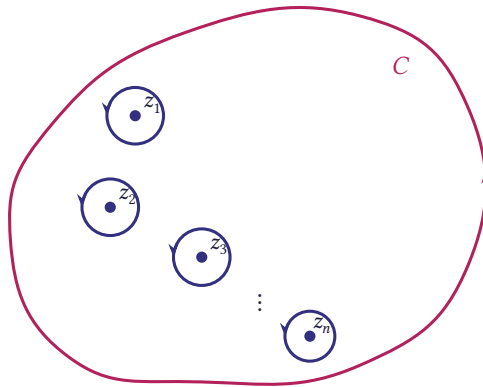


Figure 6.40 A path C and the singular points $z_1, \dots, z_n$ inside it; a small circle is drawn around each point

The proof is left to the reader as an exercise.

6.4 Examples of Taylor and Laurent Expansions

Example 6.82

Find the Laurent series of $f(z) = \frac{1}{z^4 + 1}$ in a neighbourhood of $z_0 = 0$.

We see that $f(z)$ is analytic at $z_0 = 0$, so this is a regular point and the Laurent series is the Taylor (Maclaurin) series. Hence

$$f(z) = f(0) + \frac{1}{1!}f^{(1)}(0)z + \frac{1}{2!}f^{(2)}(0)z^2 + \frac{1}{3!}f^{(3)}(0)z^3 + \dots$$

Geometric series:

$$1 + q + q^2 + q^3 + q^4 + \dots = \frac{1}{1 - q}, \quad |q| < 1$$

$$\frac{1}{1 + z^4} = 1 - z^4 + z^8 - z^{12} + \dots, \quad |z^4| < 1 \implies |z| < 1$$

(Taylor/Maclaurin series; region of convergence of radius 1.)

$$z^4 + 1 = 0 \implies z_1 = e^{i\pi/4}, \quad z_2 = e^{i3\pi/4}, \quad z_3 = e^{i5\pi/4}, \quad z_4 = e^{i7\pi/4}$$

We see that the distance from $z_0 = 0$ to the nearest non-analytic point of the function (the radius of convergence) is 1.

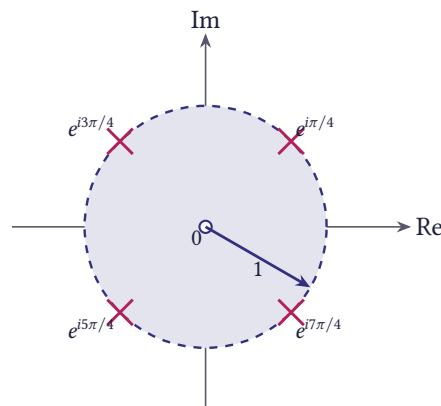


Figure 6.41 The four poles of $z^4 + 1 = 0$ on the unit circle; the radius of convergence of the series about the origin is 1

Example 6.83

Find the Laurent series of $f(z) = \frac{4z^2 + 30z + 68}{(z+4)^2(z-2)}$ in a neighbourhood of $z_0 = 0$.

We see that $f(z)$ is analytic at $z_0 = 0$, so this is a regular point and the Laurent series is the Taylor (Maclaurin) series. Hence

$$f(z) = f(0) + \frac{1}{1!}f^{(1)}(0)z + \frac{1}{2!}f^{(2)}(0)z^2 + \frac{1}{3!}f^{(3)}(0)z^3 + \dots$$

$$\frac{1}{z-2} = \frac{-1}{2-z} = \frac{-1}{2} \frac{1}{1-z/2} = \frac{-1}{2} \left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots \right), \quad \left| \frac{z}{2} \right| < 1 \implies |z| < 2$$

$$\frac{1}{4+z} = \frac{1}{4} \frac{1}{1+z/4} = \frac{1}{4} \left(1 - \frac{z}{4} + \frac{z^2}{16} - \frac{z^3}{64} + \dots \right), \quad \left| -\frac{z}{4} \right| < 1 \implies |z| < 4$$

$$\frac{d}{dz} \left(\frac{1}{4+z} \right) = \frac{-1}{(4+z)^2} = \frac{1}{4} \left(-\frac{1}{4} + \frac{2z}{16} - \frac{3z^2}{64} + \dots \right) \implies \frac{1}{(4+z)^2} = \frac{1}{4} \left(\frac{1}{4} - \frac{2z}{16} + \frac{3z^2}{64} - \dots \right)$$

$$f(z) = (4z^2 + 30z + 68) \times \frac{1}{(z+4)^2} \times \frac{1}{z-2}$$

$$f(z) = (4z^2 + 30z + 68) \times \left(\frac{-1}{2} \right) \left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots \right) \times \frac{1}{4} \left(\frac{1}{4} - \frac{z}{8} + \frac{3z^2}{64} - \dots \right)$$

$$f(z) = \frac{-1}{16} \left(34 + 15z + \frac{67}{8}z^2 + \dots \right), \quad |z| < 2$$

(region of convergence of radius 2.)

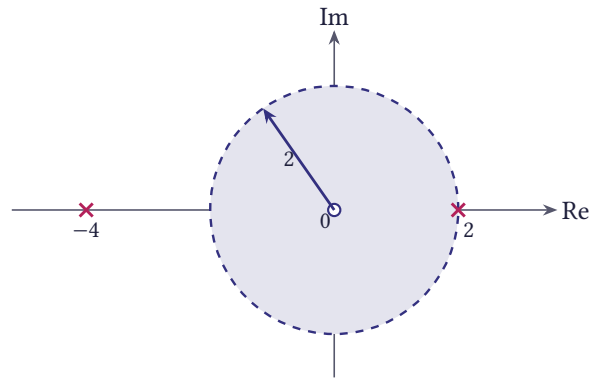


Figure 6.42 The poles $z = -4$ and $z = 2$; the circle of convergence about the origin has radius 2

Example 6.84

Find the Laurent series of $f(z) = \cos z^2$ in a neighbourhood of $z_0 = 0$.

From the definition of the cosine function,

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots$$

$$f(z) = \cos z^2 = 1 - \frac{z^4}{2!} + \frac{z^8}{4!} - \frac{z^{12}}{6!} + \dots$$

The region of convergence is the whole complex plane, with infinite radius of convergence.

Example 6.85

Find the Laurent series of $f(z) = \frac{\cos z}{1 - z^2}$ in a neighbourhood of $z_0 = 0$.

$$\begin{cases} \frac{1}{1 - z^2} = 1 + z^2 + z^4 + z^6 + \dots & |z^2| < 1 \quad |z| < 1 \\ \cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots & |z| < \infty \end{cases}$$

$$\begin{aligned} f(z) = \frac{\cos z}{1 - z^2} &= (1 + z^2 + z^4 + z^6 + \dots) \left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots \right) \\ &= 1 + \left(1 - \frac{1}{2!} \right) z^2 + \left(1 + \frac{1}{4!} - \frac{1}{2!} \right) z^4 + \dots, \quad |z| < 1 \end{aligned}$$

Example 6.86

Find the Laurent series of $f(z) = \frac{1}{z^4}$ in a neighbourhood of $z_0 = 1$.

$$g(z) = \frac{1}{z} = \frac{1}{z - 1 + 1} = \frac{1}{1 + (z - 1)} = 1 - (z - 1) + (z - 1)^2 - (z - 1)^3 + \dots, \quad |z - 1| < 1$$

$$\frac{d}{dz} g(z) = \frac{-1}{z^2} = -1 + 2(z - 1) - 3(z - 1)^2 + 4(z - 1)^3 - \dots, \quad |z - 1| < 1$$

$$\frac{d^2}{dz^2} g(z) = \frac{2}{z^3} = 2 - 6(z - 1) + 12(z - 1)^2 - 20(z - 1)^3 + \dots, \quad |z - 1| < 1$$

$$\frac{d^3}{dz^3} g(z) = \frac{-6}{z^4} = -6 + 24(z - 1) - 60(z - 1)^2 + \dots, \quad |z - 1| < 1$$

$$f(z) = \frac{1}{z^4} = 1 - 4(z - 1) + 10(z - 1)^2 - \dots$$

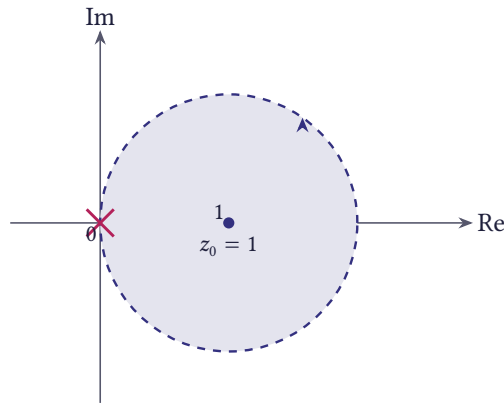


Figure 6.43 The region of convergence of the series about $z_0 = 1$ is a disc of radius 1 (up to the pole $z = 0$)

Exercise 6.87

Find the Laurent expansions of the following functions in a neighbourhood of $z_0 = 0$:

1. $f_1(z) = \frac{1}{\sqrt{1-z^2}}$
2. $f_2(z) = \sin^{-1} z$
3. $f_3(z) = \frac{e^z}{z^2(z^2+1)}$
4. Expand $\sinh z$ in powers of $(z - \pi i)$ and prove that $\lim_{z \rightarrow \pi i} \frac{\sinh z}{z - \pi i} = -1$
5. Prove that $z \cosh z^2 = z + \sum_{n=1}^{\infty} \frac{1}{(2n)!} z^{4n+1}$

6.5 Examples of Singular Points and Residues

Example 6.88

Find the Laurent series of $f(z) = \frac{z}{(z+1)(z+4)^3}$ in the neighbourhoods of its non-analytic points, and find the residue at those points.

The function is not analytic at $z = -1$ and $z = -4$.

(a) In a neighbourhood of $z = -1$, a circle centred at this point with radius less than 3 ($|z+1| < 3$) does not contain the isolated singular point $z = -4$.

$$\begin{aligned} \frac{z}{(z+4)^3} &= \frac{z+4-4}{(z+4)^3} = \frac{z+4}{(z+4)^3} + \frac{-4}{(z+4)^3} = \frac{1}{(z+4)^2} - \frac{4}{(z+4)^3} \\ \frac{1}{z+4} &= \frac{1}{(z+1)+3} = \frac{1}{3} \frac{1}{1+\frac{z+1}{3}} = \frac{1}{3} \left[1 - \frac{z+1}{3} + \left(\frac{z+1}{3} \right)^2 - \left(\frac{z+1}{3} \right)^3 + \dots \right] \end{aligned}$$

$$\begin{aligned}
\frac{-1}{(z+4)^2} &= \frac{1}{3} \left[-\frac{1}{3} + \frac{2}{3^2}(z+1) - \frac{3}{3^3}(z+1)^2 + \dots \right] \Rightarrow \frac{1}{(z+4)^2} \\
&= \frac{1}{9} - \frac{2}{27}(z+1) + \frac{1}{27}(z+1)^2 - \dots \\
\frac{1}{(z+4)^3} &= \frac{1}{6} \left[\frac{2}{3^2} - \frac{2 \times 3}{3^3}(z+1) + \frac{3 \times 4}{3^4}(z+1)^2 - \frac{4 \times 5}{3^5}(z+1)^3 + \dots \right] \\
&= \frac{1}{27} - \frac{1}{27}(z+1) + \frac{2}{81}(z+1)^2 - \dots \\
\frac{z}{(z+4)^3} &= \frac{1}{(z+4)^2} - \frac{4}{(z+4)^3} = -\frac{1}{27} + \frac{2}{27}(z+1) - \frac{5}{81}(z+1)^2 + \dots \\
f(z) &= \frac{1}{z+1} \frac{z}{(z+4)^3} = -\frac{1}{27} \frac{1}{z+1} + \frac{2}{27} - \frac{5}{81}(z+1) + \dots
\end{aligned}$$

Type of singular point: a simple pole (of order 1). Residue: $\text{Res}\{f(z)\}|_{z=-1} = -\frac{1}{27}$.

(b) In a neighbourhood of $z = -4$, a circle centred at this point with radius less than 3 ($|z+4| < 3$) does not contain the isolated singular point $z = -1$.

$$\begin{aligned}
\frac{z}{z+1} &= \frac{z+1-1}{z+1} = \frac{z+1}{z+1} - \frac{1}{z+1} = 1 - \frac{1}{z+1} \\
\frac{1}{z+1} &= \frac{1}{z+4-3} = \frac{1}{-3+(z+4)} = \frac{-1}{3} \frac{1}{1-\frac{z+4}{3}} = \frac{-1}{3} \left[1 + \frac{z+4}{3} + \frac{(z+4)^2}{3^2} + \dots \right], \\
|z+4| &< 3
\end{aligned}$$

$$\begin{aligned}
\frac{z}{z+1} &= 1 + \frac{1}{3} + \frac{1}{3^2}(z+4) + \frac{1}{3^3}(z+4)^2 + \frac{1}{3^4}(z+4)^3 + \dots \\
f(z) &= \frac{z}{(z+1)(z+4)^3} = \frac{1}{(z+4)^3} \frac{z}{z+1} \\
&= \frac{4/3}{(z+4)^3} + \frac{1}{3^2} \frac{1}{(z+4)^2} + \frac{1}{3^3} \frac{1}{z+4} + \frac{1}{3^4} + \frac{1}{3^5}(z+4) + \frac{1}{3^6}(z+4)^2 + \dots
\end{aligned}$$

Type of singular point: a pole of order 3. Residue: $\text{Res}\{f(z)\}|_{z=-4} = \frac{1}{27}$.

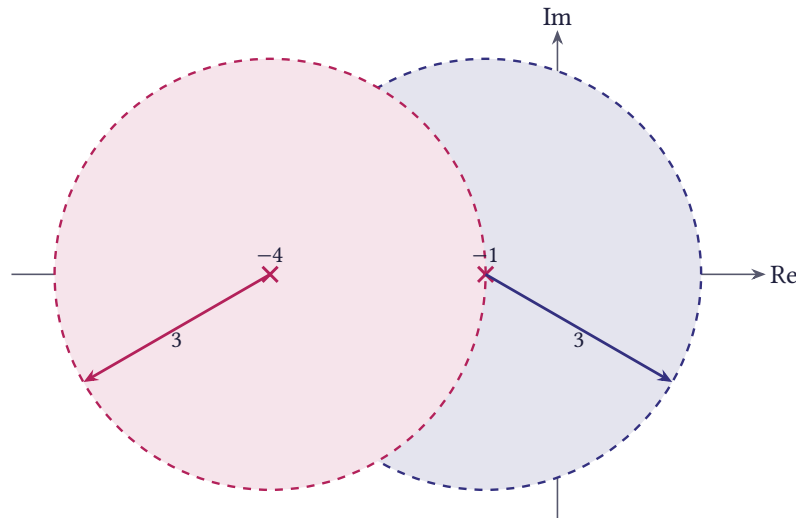


Figure 6.44 The poles $z = -1$ and $z = -4$ of $f(z) = \frac{z}{(z+1)(z+4)^3}$

Example 6.89

Find the Laurent series of $f(z) = e^{1/z}$ in a neighbourhood of its non-analytic point, and find the residue at that point.

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots \implies e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \dots$$

Type of singular point: an essential singularity (infinitely many negative powers). Residue: $\text{Res}\{f(z)\}_{z=0} = 1$.

Example 6.90

Find the Laurent series of $f(z) = \cos\left(\frac{z}{z-1}\right)$ in a neighbourhood of its non-analytic point, and find the residue at that point.

$$\frac{z}{z-1} = 1 + \frac{1}{z-1} \implies \cos\left(\frac{z}{z-1}\right) = \cos\left(1 + \frac{1}{z-1}\right) = \cos 1 \cos\left(\frac{1}{z-1}\right) - \sin 1 \sin\left(\frac{1}{z-1}\right)$$

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots, \quad \sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots$$

$$\begin{aligned} \cos\left(\frac{z}{z-1}\right) &= \cos 1 \left(1 - \frac{1}{2!} \frac{1}{(z-1)^2} + \frac{1}{4!} \frac{1}{(z-1)^4} - \dots\right) \\ &\quad - \sin 1 \left(\frac{1}{z-1} - \frac{1}{3!} \frac{1}{(z-1)^3} + \frac{1}{5!} \frac{1}{(z-1)^5} - \dots\right) \\ &= \cos 1 - \sin 1 \frac{1}{z-1} - \cos 1 \frac{1}{2!} \frac{1}{(z-1)^2} + \sin 1 \frac{1}{3!} \frac{1}{(z-1)^3} + \dots \end{aligned}$$

Type of singular point: an essential singularity. Residue: $\text{Res}\{f(z)\}_{z=1} = -\sin 1$.

Theorem 6.91 ■ Residue at a simple pole

Prove that if $f(z) = q(z)/g(z)$ and $g(z)$ has a simple root at z_0 , then

$$\text{Res}\{f(z)\}\Big|_{z=z_0} = \frac{q(z_0)}{g'(z_0)}$$

Proof.

Since z_0 is a pole of order 1 we have ($p = 1$):

$$\begin{aligned} \text{Res}\{f(z)\}\Big|_{z=z_0} &= A_1 = \frac{1}{(p-1)!} \frac{d^{p-1}}{dz^{p-1}} \left[(z-z_0)^p f(z) \right] \Big|_{z=z_0} = (z-z_0) \frac{q(z)}{g(z)} \Big|_{z=z_0} \\ &= (z-z_0) \frac{q(z)}{g(z) - g(z_0)} \Big|_{z=z_0} = \frac{q(z)}{\frac{g(z) - g(z_0)}{z-z_0}} \Big|_{z=z_0} = \frac{q(z_0)}{g'(z_0)} \end{aligned}$$

■

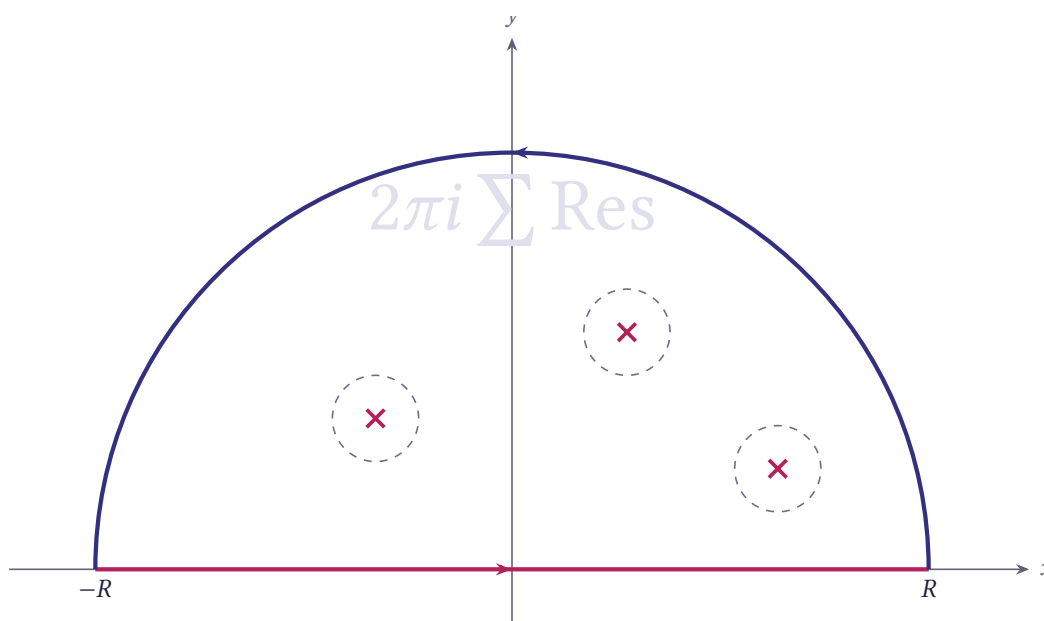
Exercise 6.92

Find the Laurent series of the following functions about $z = 0$ and determine the type of this point.

$$f_1(z) = \frac{z}{e^z - 1}, \quad f_2(z) = \frac{e^{-z}}{z^3}, \quad f_3(z) = \frac{1}{z^2(1 - z^2)}, \quad f_4(z) = \frac{\sin z^2}{z^3}$$

Evaluating Definite Integrals by Residues

محاسبهٔ انتگرال‌های معین به روش مانده‌ها



In this chapter

7.1 Integrals of the First Class	63
7.2 Integrals of the Second Class	64

Theorem 7.93 ■ The residue theorem

If the function $f(z)$ is analytic everywhere inside the path C except at the points $z_1, z_2, \dots, z_n$, and the residues of the function at these points are $A_1^1, A_1^2, \dots, A_1^n$ respectively, then

$$\oint_C f(z) dz = 2\pi i \sum_{k=1}^n A_1^k$$

Many different kinds of integrals can be solved by the method of residues. Two classes of them are mentioned here:

Definition 7.94 ■ Integrals of the first class

In these integrals the denominator of the integrand is a polynomial, and the numerator is a polynomial, an exponential, or a sine or cosine function (which can also be expressed exponentially).

Definition 7.95 ■ Integrals of the second class

In these integrals the integrand is a function of sine or cosine.

7.1 Integrals of the First Class**Example 7.96**

Evaluate the integral $\int_0^\infty \frac{\cos \omega x}{(1+x^2)^2} dx$ ($\omega > 0$).

Consider the function $f(z) = e^{i\omega z} / (1+z^2)^2$. It is not analytic at $z = i$ and $z = -i$. The path C is the upper semicircle of radius R .

$$\begin{aligned} \oint_C \frac{e^{i\omega z}}{(1+z^2)^2} dz &= 2\pi i \times \text{Res} \left\{ \frac{e^{i\omega z}}{(1+z^2)^2} \right\} \Big|_{z=i} \\ &= 2\pi i \times \left[\frac{d}{dz} \left\{ (z-i)^2 \frac{e^{i\omega z}}{(z-i)^2(z+i)^2} \right\} \right]_{z=i} \\ &= 2\pi i \times \left[\frac{i\omega e^{i\omega z}(z+i) - 2e^{i\omega z}}{(z+i)^3} \right]_{z=i} \\ &= 2\pi i \times e^{-\omega} \left(\frac{i\omega(2i) - 2}{(2i)^3} \right) \\ &= 2\pi i \times e^{-\omega} \frac{\omega + 1}{4i} = \frac{\pi}{2}(\omega + 1)e^{-\omega} \end{aligned}$$

On the other hand

$$\oint_C \frac{e^{i\omega z}}{(1+z^2)^2} dz = \int_{-R}^R \frac{e^{i\omega x}}{(1+x^2)^2} dx + \int_0^\pi \frac{e^{i\omega R e^{it}} R i e^{it}}{(1+R^2 e^{i2t})^2} dt$$

and as $R \rightarrow \infty$ the integral along the semicircle vanishes, while the imaginary part of the integral

along the real axis (the odd function $\sin \omega x / (1 + x^2)^2$) is zero:

$$\oint_C \frac{e^{i\omega z}}{(1 + z^2)^2} dz \xrightarrow{R \rightarrow \infty} 2 \int_0^\infty \frac{\cos \omega x}{(1 + x^2)^2} dx$$

$$\int_0^\infty \frac{\cos \omega x}{(1 + x^2)^2} dx = \frac{\pi}{4}(\omega + 1)e^{-\omega}$$

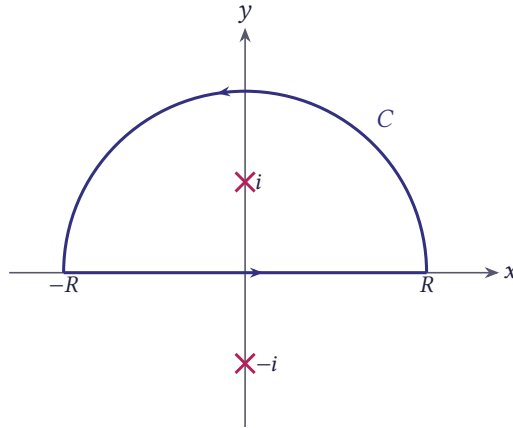


Figure 7.45 The upper semicircular path; the double pole $z = i$ lies inside the path and $z = -i$ outside it

7.2 Integrals of the Second Class

$$\int_0^{2\pi} f(a \cos t, b \sin t) dt$$

To solve such integrals we put $z = e^{it}$. Then

$$dz = ie^{it} dt \implies dt = \frac{dz}{ie^{it}} = \frac{dz}{iz}$$

$$\sin t = \frac{e^{it} - e^{-it}}{2i} = \frac{z - z^{-1}}{2i}, \quad \cos t = \frac{e^{it} + e^{-it}}{2} = \frac{z + z^{-1}}{2}$$

Key result 7.97

$$\int_0^{2\pi} f(a \cos t, b \sin t) dt = \oint_C f\left(a \frac{z + z^{-1}}{2}, b \frac{z - z^{-1}}{2i}\right) \frac{dz}{iz}$$

Here C is the unit circle, and the integral is easily solved with the method of residues.

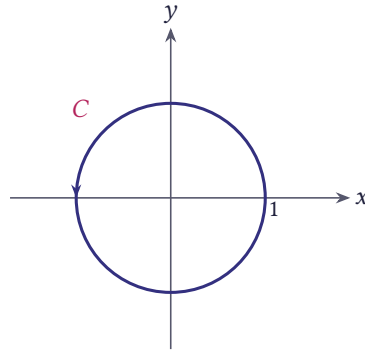


Figure 7.46 The unit circle $z = e^{it}$, $0 \leq t \leq 2\pi$

Example 7.98

Evaluate the integral $\int_0^{2\pi} \frac{1}{2 + \cos x} dx$.

$$z = e^{ix}, \quad dz = ie^{ix} dx, \quad dx = \frac{dz}{iz}, \quad \cos x = \frac{z + z^{-1}}{2} = \frac{z^2 + 1}{2z}$$

$$2 + \cos x = 2 + \frac{z^2 + 1}{2z} = \frac{z^2 + 4z + 1}{2z}$$

$$\begin{aligned} \int_0^{2\pi} \frac{1}{2 + \cos x} dx &= \oint_C \frac{2z}{z^2 + 4z + 1} \frac{dz}{iz} = \frac{2}{i} \oint_C \frac{1}{z^2 + 4z + 1} dz \\ &= \frac{2}{i} \times 2\pi i \times \sum_k \operatorname{Res} \left\{ \frac{1}{z^2 + 4z + 1} \right\} \Big|_{z=z_k} \end{aligned}$$

$$z^2 + 4z + 1 = 0 \implies z_1 = -2 + \sqrt{3} \approx -0.27, \quad z_2 = -2 - \sqrt{3} \approx -3.73$$

Only z_1 lies inside the unit circle.

$$\begin{aligned} \int_0^{2\pi} \frac{1}{2 + \cos x} dx &= 4\pi \times \operatorname{Res} \left\{ \frac{1}{z^2 + 4z + 1} \right\} \Big|_{z=z_1} \\ &= 4\pi \left[(z + 2 - \sqrt{3}) \frac{1}{(z + 2 - \sqrt{3})(z + 2 + \sqrt{3})} \right]_{z_1 = -2 + \sqrt{3}} = \frac{2\pi}{\sqrt{3}} \end{aligned}$$

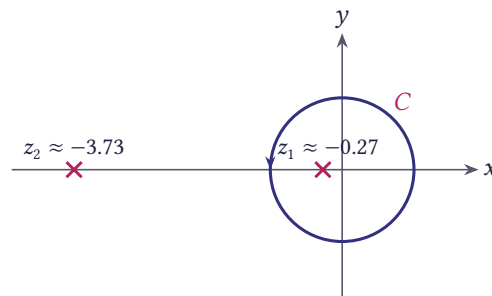


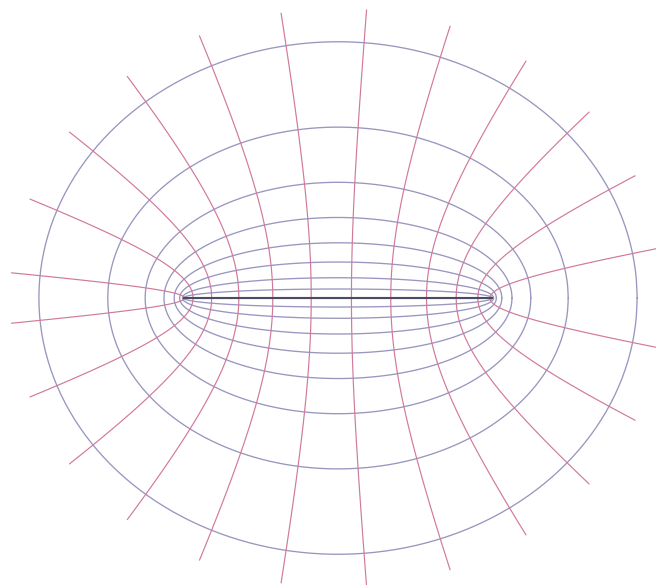
Figure 7.47 The unit circle and the poles $z_1 \approx -0.27$ (inside) and $z_2 \approx -3.73$ (outside)

Exercise 7.99 ■ Applying the residue theorem to integrals

Using the residue theorem, evaluate the following definite integrals:

$$I_1 = \int_0^{2\pi} \frac{dt}{1 - 2p \cos t + p^2}, \quad I_2 = \int_{-\infty}^{\infty} \frac{\sin 2x}{x^2 + x + 1} dx$$

$$I_3 = \int_{-\infty}^{\infty} \frac{x}{(x^2 - 2x + 2)^2} dx, \quad I_4 = \int_{-\infty}^{\infty} \frac{\cos^4 x}{(x^2 + 1)(x^2 + 4)} dx$$



Part II

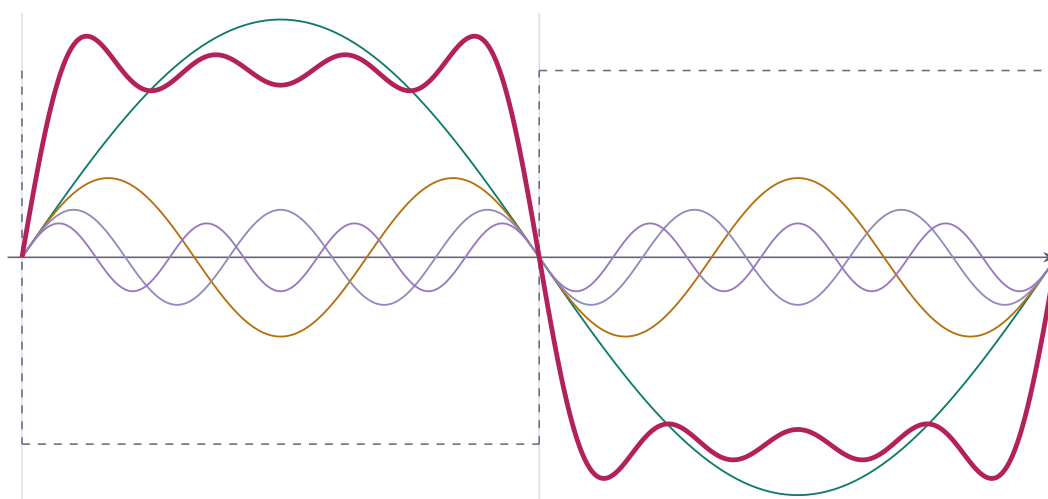
Fourier Analysis

آنالیز فوریه

Chapters 8–10

Fourier Series

سری فوریه



In this chapter

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8.1 Periodic Functions

$$f(x + p) = f(x) \implies f(x + np) = f(x)$$

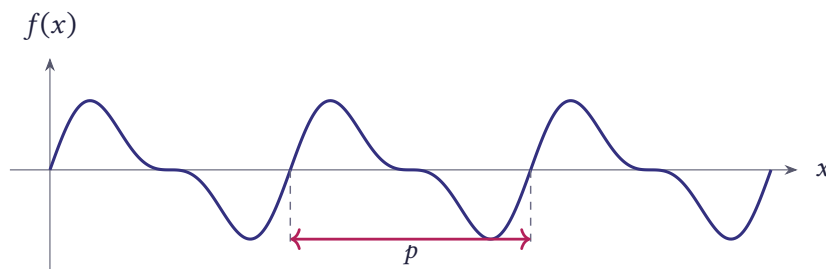


Figure 8.48 A periodic function with period p

- The smallest positive value of p for which the function is periodic is called the fundamental period of the function.
- If a function is periodic with period p , it is also periodic with period np , where n is a natural number.

■ The Periodic Functions Sine and Cosine

Example 8.100

The function $\sin x$ is periodic with period 2π , and $\sin 2x$ is periodic with fundamental period π , although 2π is also a period of it. Likewise $\sin 3x$ is periodic with period $2\pi/3$, although 2π is also a period of it.

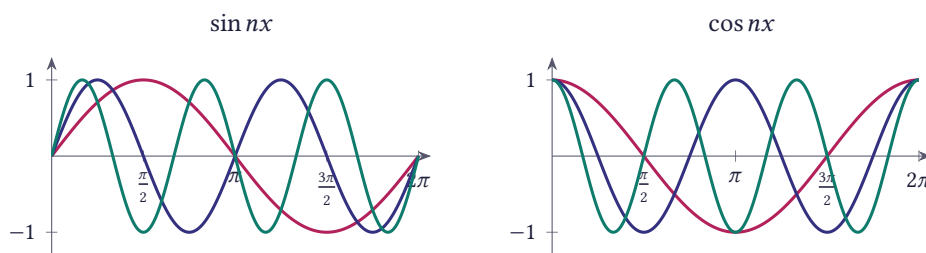


Figure 8.49 The functions $\sin nx$ and $\cos nx$ for $n = 1, 2, 3$ on $[0, 2\pi]$

8.2 Review of Trigonometry

$$\sin^2 \alpha + \cos^2 \alpha = 1$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1 = 1 - 2 \sin^2 \alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\sin^2 \alpha = \frac{1 - \cos 2\alpha}{2} \quad \cos^2 \alpha = \frac{1 + \cos 2\alpha}{2}$$

$$\sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

$$\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$$

$$\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha - \beta) + \sin(\alpha + \beta)]$$

8.3 Review: Integrals and Orthogonality

$$\int_{-\pi}^{\pi} \sin nt \, dt = 0, \quad n \in \mathbb{Z}$$

$$\int_{-\pi}^{\pi} \cos nt \, dt = 0, \quad n \in \mathbb{Z} \setminus \{0\}$$

Proof.

$$\int_{-\pi}^{\pi} \sin nt \, dt = -\frac{1}{n} \cos nt \Big|_{-\pi}^{\pi} = 0, \quad \int_{-\pi}^{\pi} \cos nt \, dt = \frac{1}{n} \sin nt \Big|_{-\pi}^{\pi} = 0 \quad (n \neq 0)$$

■

■ Orthogonality

$$\int_{-\pi}^{\pi} \sin nt \sin mt \, dt = 0, \quad n, m \in \mathbb{Z}, n \neq m$$

$$\int_{-\pi}^{\pi} \cos nt \cos mt \, dt = 0, \quad n, m \in \mathbb{Z}, n \neq m$$

$$\int_{-\pi}^{\pi} \sin nt \cos mt \, dt = 0, \quad n, m \in \mathbb{Z} \quad (n \neq m \text{ or } n = m)$$

Proof.

$$\int_{-\pi}^{\pi} \sin nx \sin mx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} \cos(n-m)x \, dx - \frac{1}{2} \int_{-\pi}^{\pi} \cos(n+m)x \, dx$$

$$\int_{-\pi}^{\pi} \cos nx \cos mx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} \cos(n+m)x \, dx + \frac{1}{2} \int_{-\pi}^{\pi} \cos(n-m)x \, dx$$

$$\int_{-\pi}^{\pi} \sin nx \cos mx \, dx = \frac{1}{2} \int_{-\pi}^{\pi} \sin(n+m)x \, dx + \frac{1}{2} \int_{-\pi}^{\pi} \sin(n-m)x \, dx$$

■

$$\int_{-\pi}^{\pi} \sin^2 nt \, dt = \pi, \quad \int_{-\pi}^{\pi} \cos^2 nt \, dt = \pi, \quad n \in \mathbb{Z} \setminus \{0\}$$

Proof.

$$\int_{-\pi}^{\pi} \sin^2 nt \, dt = \int_{-\pi}^{\pi} \frac{1 - \cos 2nt}{2} \, dt = \pi, \quad \int_{-\pi}^{\pi} \cos^2 nt \, dt = \int_{-\pi}^{\pi} \frac{1 + \cos 2nt}{2} \, dt = \pi$$

■

8.4 The Fourier Series

History

Joseph Fourier (1768–1830) put forward the idea of expressing periodic functions as a series of sinusoidal functions. When this revolutionary idea was presented, some mathematicians, Lagrange among them, objected to its generality, until Dirichlet stated the conditions for the convergence of the Fourier series.

Theorem 8.101 ■ The Fourier series

If the function $f(x)$ is periodic with period 2π , it can be expressed as a trigonometric series as follows:

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

where

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \, dx, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx$$

Proof.

Suppose the periodic function has been written as the trigonometric series above; it suffices to find the coefficients a_0 , a_n and b_n .

Computing a_0 . Integrating both sides over an interval of length 2π gives

$$\begin{aligned} \int_{-\pi}^{\pi} f(x) \, dx &= \int_{-\pi}^{\pi} \left[a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \right] dx \\ &= a_0 \int_{-\pi}^{\pi} dx + \sum_{n=1}^{\infty} \left(\underbrace{a_n \int_{-\pi}^{\pi} \cos nx \, dx}_0 + \underbrace{b_n \int_{-\pi}^{\pi} \sin nx \, dx}_0 \right) = 2\pi a_0 \\ \implies a_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \, dx \end{aligned}$$

a_0 is the mean value or DC level of the function.

Computing a_n . Multiplying both sides by $\cos mx$ and integrating over an interval of length 2π :

$$\begin{aligned} \int_{-\pi}^{\pi} f(x) \cos mx \, dx &= \int_{-\pi}^{\pi} a_0 \cos mx \, dx + \sum_{n=1}^{\infty} \int_{-\pi}^{\pi} a_n \cos nx \cos mx \, dx + \sum_{n=1}^{\infty} \int_{-\pi}^{\pi} b_n \sin nx \cos mx \, dx \\ &= \pi a_m \end{aligned}$$

$$\Rightarrow a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx$$

Computing b_n . Multiplying both sides by $\sin mx$ and integrating over an interval of length 2π :

$$\begin{aligned} \int_{-\pi}^{\pi} f(x) \sin mx \, dx &= \int_{-\pi}^{\pi} a_0 \sin mx \, dx + \sum_{n=1}^{\infty} \int_{-\pi}^{\pi} a_n \cos nx \sin mx \, dx + \sum_{n=1}^{\infty} \int_{-\pi}^{\pi} b_n \sin nx \sin mx \, dx \\ &= \pi b_m \\ \Rightarrow b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx \end{aligned}$$

■

Key result 8.102 ■ The Fourier series and Euler's formulas

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx), \quad f(x + 2\pi) = f(x)$$

$$a_0 = \frac{1}{2\pi} \int_{2\pi} f(x) \, dx, \quad a_n = \frac{1}{\pi} \int_{2\pi} f(x) \cos nx \, dx, \quad b_n = \frac{1}{\pi} \int_{2\pi} f(x) \sin nx \, dx$$

(the integral is taken over one period of length 2π .)

At points of discontinuity of the function, the value of the Fourier series equals the average of the upper and lower limits at that point.

8.5 Review: The Definite Integral of a Product of Two Functions

By integration by parts:

$$\int_{\alpha}^{\beta} f(x)g(x) \, dx = \left[f(x)G_1(x) - f^{(1)}(x)G_2(x) + f^{(2)}(x)G_3(x) - + \dots \right]_{\alpha}^{\beta}$$

where $f^{(k)}$ is the k -th derivative of f and G_k is the k -th integral of g :

derivatives of f		integrals of g
$f(x)$	$\searrow$	$g(x)$
$f^{(1)}(x)$	$\searrow$	$G_1(x)$
$f^{(2)}(x)$	$\searrow$	$G_2(x)$
$f^{(3)}(x)$	$\searrow$	$G_3(x)$
$\vdots$		$\vdots$

8.6 Examples of Fourier Series

Example 8.103

Find the Fourier series of the following function:

$$f(x) = \begin{cases} -k & -\pi < x < 0 \\ k & 0 < x < \pi \end{cases} \quad f(x + 2\pi) = f(x)$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = 0$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \left[\int_{-\pi}^0 (-k) \cos nx dx + \int_0^{\pi} k \cos nx dx \right] \\ &= \frac{1}{\pi} \left[-k \frac{\sin nx}{n} \Big|_{-\pi}^0 + k \frac{\sin nx}{n} \Big|_0^{\pi} \right] = 0 \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \left[\int_{-\pi}^0 (-k) \sin nx dx + \int_0^{\pi} k \sin nx dx \right] \\ &= \frac{1}{\pi} \left[k \frac{\cos nx}{n} \Big|_{-\pi}^0 - k \frac{\cos nx}{n} \Big|_0^{\pi} \right] = \frac{k}{n\pi} [\cos 0 - \cos(-n\pi) - \cos n\pi + \cos 0] = \frac{2k}{n\pi} (1 - \cos n\pi) \end{aligned}$$

$$\cos n\pi = \begin{cases} -1 & n \text{ odd} \\ 1 & n \text{ even} \end{cases} \implies 1 - \cos n\pi = \begin{cases} 2 & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

$$b_1 = \frac{4k}{\pi}, \quad b_2 = 0, \quad b_3 = \frac{4k}{3\pi}, \quad b_4 = 0, \quad b_5 = \frac{4k}{5\pi}, \dots$$

$$f(x) = \frac{4k}{\pi} \left(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right)$$

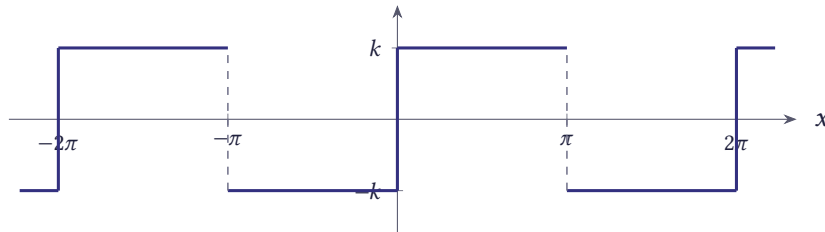


Figure 8.50 A square wave $f(x)$ with period 2π and values $\pm k$

The partial sums of the series:

$$S_1 = \frac{4k}{\pi} \sin x, \quad S_2 = \frac{4k}{\pi} \left(\sin x + \frac{1}{3} \sin 3x \right), \quad S_3 = \frac{4k}{\pi} \left(\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x \right)$$

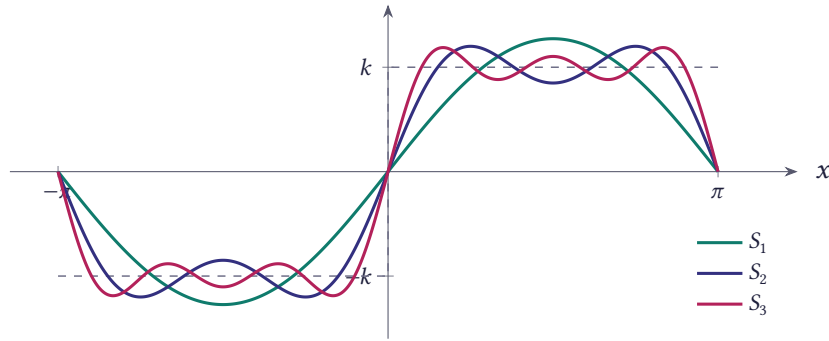


Figure 8.51 The partial sums S_1 , S_2 and S_3 of the Fourier series of the square wave (the dashed line is the square wave itself)

Setting $x = \pi/2$:

$$f(\pi/2) = \frac{4k}{\pi} \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right) = k \implies \pi = 4 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right)$$

Example 8.104

Find the Fourier series of the following function:

$$f(x) = x, \quad -\pi \leq x \leq \pi, \quad f(x + 2\pi) = f(x)$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = 0$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos nx dx = \frac{1}{\pi} \left[x \left(\frac{1}{n} \sin nx \right) - (1) \left(-\frac{1}{n^2} \cos nx \right) \right]_{-\pi}^{\pi} \\ &= \frac{1}{\pi} \left[\frac{1}{n^2} (\cos n\pi - \cos(-n\pi)) \right] = 0 \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin nx dx = \frac{2}{\pi} \int_0^{\pi} x \sin nx dx = \frac{2}{\pi} \left[x \left(-\frac{1}{n} \cos nx \right) - (1) \left(-\frac{1}{n^2} \sin nx \right) \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[-\frac{\pi}{n} \cos n\pi \right] = \frac{2}{n} (-1)^{n+1} \end{aligned}$$

$$f(x) = \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin nx = 2 \left(\sin x - \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x - \dots \right)$$

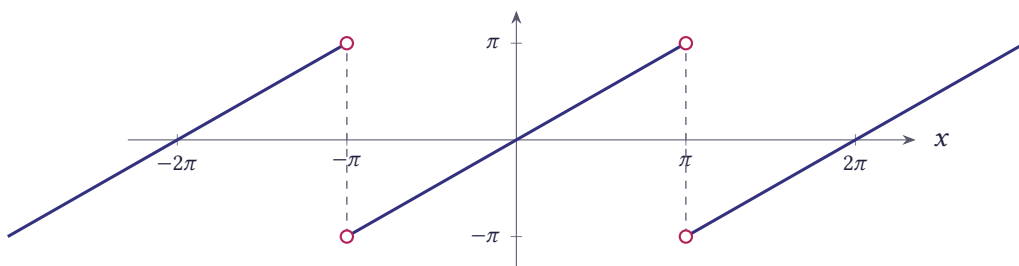


Figure 8.52 The sawtooth wave $f(x) = x$ on $[-\pi, \pi]$ with its periodic extension

Setting $x = \pi/2$:

$$\begin{aligned} f\left(\frac{\pi}{2}\right) &= \frac{\pi}{2} = \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin\left(n\frac{\pi}{2}\right) = 2\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots\right) = \sum_{n=1}^{\infty} \frac{2}{2n-1} (-1)^{n+1} \Rightarrow \pi \\ &= \sum_{n=1}^{\infty} \frac{4}{2n-1} (-1)^{n+1} \end{aligned}$$

The value of the series at the point of discontinuity $x = \pi$:

$$f(\pi) = ? \quad \Rightarrow \quad \sum_{n=1}^{\infty} \frac{2}{n} (-1)^{n+1} \sin(n\pi) = 0$$

Key result 8.105 ■ Important remarks

1. If a periodic function is odd, all the a coefficients are zero.
2. At a point of discontinuity, the value of the Fourier series equals the average of the upper and lower limits at that point.

Example 8.106

Find the Fourier series of the following function:

$$f(x) = x^2, \quad -\pi \leq x \leq \pi, \quad f(x+2\pi) = f(x)$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{2}{2\pi} \int_0^{\pi} x^2 dx = \frac{1}{\pi} \left[\frac{1}{3} x^3 \right]_0^{\pi} = \frac{\pi^2}{3}$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos nx dx = \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx dx \\ &= \frac{2}{\pi} \left[x^2 \left(\frac{1}{n} \sin nx \right) - 2x \left(\frac{-1}{n^2} \cos nx \right) + 2 \left(\frac{-1}{n^3} \sin nx \right) \right]_0^{\pi} = \frac{2}{\pi} \left[\frac{2\pi}{n^2} \cos n\pi \right] = \frac{4}{n^2} (-1)^n \end{aligned}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \sin nx dx = 0$$

$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^n \cos nx$$

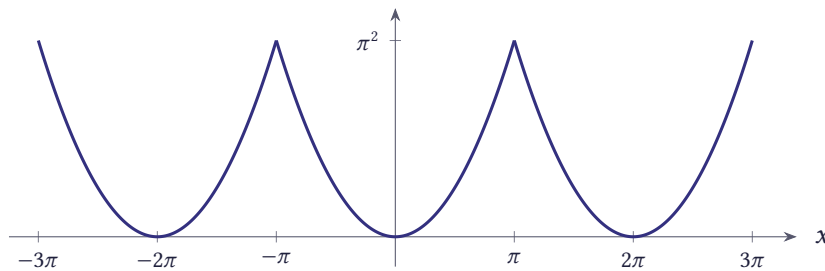


Figure 8.53 The function $f(x) = x^2$ on $[-\pi, \pi]$ with its periodic extension

Key result 8.107 ■ Important remark

If a periodic function is even, all the b coefficients are zero.

Setting $x = 0$:

$$f(0) = 0 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^n \cos(n \cdot 0) \implies \frac{\pi^2}{3} = \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^{n+1}$$

$$\pi^2 = \sum_{n=1}^{\infty} \frac{12}{n^2} (-1)^{n+1} = 12 \left(1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \dots \right)$$

Example 8.108

Find the Fourier series of the following function:

$$f(x) = x^2, \quad 0 \leq x \leq 2\pi, \quad f(x + 2\pi) = f(x)$$

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} x^2 dx = \frac{1}{2\pi} \left[\frac{1}{3} x^3 \right]_0^{2\pi} = \frac{1}{2\pi} \cdot \frac{1}{3} 8\pi^3 = \frac{4\pi^2}{3}$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_0^{2\pi} x^2 \cos nx dx = \frac{1}{\pi} \left[x^2 \left(\frac{1}{n} \sin nx \right) - 2x \left(\frac{-1}{n^2} \cos nx \right) + 2 \left(\frac{-1}{n^3} \sin nx \right) \right]_0^{2\pi} \\ &= \frac{1}{\pi} (2 \times 2\pi) \left(\frac{1}{n^2} \cos 2\pi n \right) = \frac{4}{n^2} \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_0^{2\pi} x^2 \sin nx dx = \frac{1}{\pi} \left[x^2 \left(\frac{-1}{n} \cos nx \right) - 2x \left(\frac{-1}{n^2} \sin nx \right) + 2 \left(\frac{1}{n^3} \cos nx \right) \right]_0^{2\pi} \\ &= \frac{1}{\pi} \left[(4\pi^2) \left(\frac{-1}{n} \right) + 2 \cdot \frac{1}{n^3} - 2 \cdot \frac{1}{n^3} \right] = -\frac{4\pi}{n} \end{aligned}$$

$$f(x) = \frac{4\pi^2}{3} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} \cos nx - \frac{4\pi}{n} \sin nx \right)$$

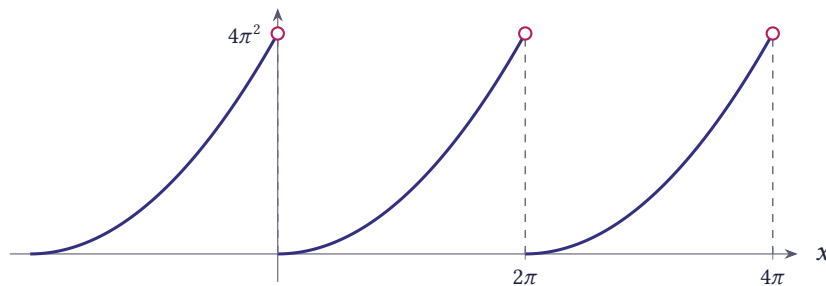


Figure 8.54 The function $f(x) = x^2$ on $[0, 2\pi]$ with its periodic extension

8.7 Fourier Series of Functions with an Arbitrary Period

$$f(x + T) = f(x)$$

Using the auxiliary variable $t = \frac{2\pi}{T}x$ ($x : 0 \rightarrow T$ and $t : 0 \rightarrow 2\pi$):

$$f\left(\frac{T}{2\pi}t\right) = a_0 + \sum_{n=1}^{\infty} a_n \cos nt + b_n \sin nt$$

$$a_0 = \frac{1}{2\pi} \int_{2\pi} f\left(\frac{T}{2\pi}t\right) dt, \quad a_n = \frac{1}{\pi} \int_{2\pi} f\left(\frac{T}{2\pi}t\right) \cos nt dt, \quad b_n = \frac{1}{\pi} \int_{2\pi} f\left(\frac{T}{2\pi}t\right) \sin nt dt$$

Returning to $x = \frac{T}{2\pi}t$:

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi}{T}nx + b_n \sin \frac{2\pi}{T}nx$$

$$a_0 = \frac{1}{2\pi} \int_T f(x) \frac{2\pi}{T} dx = \frac{1}{T} \int_T f(x) dx$$

$$a_n = \frac{1}{\pi} \int_T f(x) \cos\left(\frac{2\pi}{T}nx\right) \frac{2\pi}{T} dx = \frac{2}{T} \int_T f(x) \cos\left(\frac{2\pi}{T}nx\right) dx$$

$$b_n = \frac{1}{\pi} \int_T f(x) \sin\left(\frac{2\pi}{T}nx\right) \frac{2\pi}{T} dx = \frac{2}{T} \int_T f(x) \sin\left(\frac{2\pi}{T}nx\right) dx$$

Theorem 8.109 ■ The Fourier series (period T)

If the function $f(x)$ is periodic with period T , it can be expressed as a trigonometric series as follows:

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi}{T}nx + b_n \sin \frac{2\pi}{T}nx, \quad f(x + T) = f(x)$$

$$a_0 = \frac{1}{T} \int_T f(x) dx, \quad a_n = \frac{2}{T} \int_T f(x) \cos \frac{2\pi}{T}nx dx, \quad b_n = \frac{2}{T} \int_T f(x) \sin \frac{2\pi}{T}nx dx$$

At points of discontinuity of the function, the value of the Fourier series equals the average of the upper and lower limits there; the value of the Fourier series at the point x is

$$\frac{f(x^+) + f(x^-)}{2}$$

8.8 Dirichlet's Theorem

Theorem 8.110 ■ Dirichlet's theorem

If the function $f(x)$ is periodic and **bounded**, has a **finite number of maxima and minima**, and also a **finite number of points of discontinuity** in each period, then this function has a Fourier series, and the value of the Fourier series at points of discontinuity equals the average of the right and left limits at those points.

Examples of functions that do not satisfy Dirichlet's conditions:

$$f(x) = 1/x, \quad 0 < x \leq 1, \quad f(x+1) = f(x)$$

$$f(x) = \sin(1/x), \quad 0 < x \leq 1, \quad f(x+1) = f(x)$$

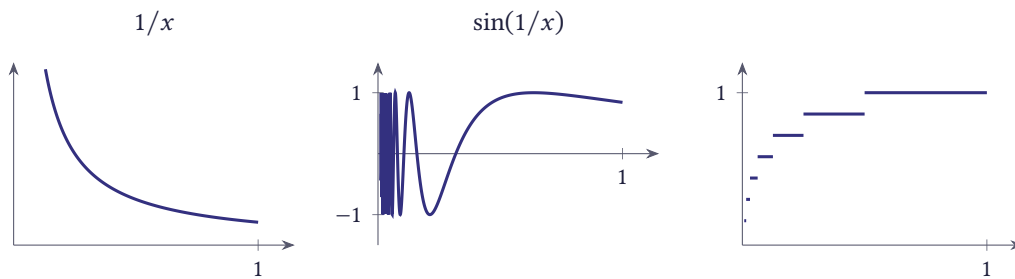


Figure 8.55 Functions that violate Dirichlet's conditions: $1/x$ (unbounded), $\sin(1/x)$ (infinitely many maxima and minima) and a staircase function with infinitely many steps

8.9 Solving Some Problems

Example 8.111 ■ Problem 1

Find the Fourier series of the following function:

$$f(x) = \sin \alpha x, \quad \alpha \in \mathbb{R}, \quad x \in [-\pi, \pi), \quad f(x + 2\pi) = f(x)$$

Since this function is odd, $a_0 = 0$ and $a_n = 0$.

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} \sin \alpha x \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} \sin \alpha x \sin nx \, dx = \frac{1}{\pi} \int_0^{\pi} [\cos(\alpha - n)x - \cos(\alpha + n)x] \, dx \\ &= \frac{1}{\pi} \left[\frac{\sin(\alpha - n)x}{\alpha - n} - \frac{\sin(\alpha + n)x}{\alpha + n} \right]_0^{\pi} = \dots = \frac{2n(-1)^n \sin \alpha \pi}{\pi(\alpha^2 - n^2)} \end{aligned}$$

$$f(x) = \sum_{n=1}^{\infty} \frac{2n(-1)^n \sin \alpha \pi}{\pi(\alpha^2 - n^2)} \sin nx$$

We examine the series when $\alpha = 1$:

$$\lim_{\alpha \rightarrow 1} b_1 = \lim_{\alpha \rightarrow 1} \frac{2(-1)^1 \sin \alpha \pi}{\pi(\alpha^2 - 1^2)} = \lim_{\alpha \rightarrow 1} \frac{-2(\pi \cos \alpha \pi)}{\pi(2\alpha)} = 1, \quad b_n \Big|_{n \neq 1} = 0$$

Hence the Fourier series is simply $f(x) = \sin x$.

Exercise 8.112

Examine the series when $\alpha = m$, $m \in \mathbb{N}$.

Example 8.113 ■ Problem 2

Find the Fourier series of the following function and deduce $\csc \alpha\pi$ from it.

$$f(x) = \cos \alpha x, \quad \alpha \in \mathbb{R}, \quad x \in [-\pi, \pi), \quad f(x + 2\pi) = f(x)$$

Since this function is even, $b_n = 0$ and $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$.

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos \alpha x \, dx = \frac{1}{\pi} \int_0^{\pi} \cos \alpha x \, dx = \frac{1}{\pi} \left[\frac{\sin \alpha x}{\alpha} \right]_0^{\pi} = \frac{1}{\alpha\pi} \sin \alpha\pi$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} \cos \alpha x \cos nx \, dx = \frac{1}{\pi} \int_0^{\pi} [\cos(\alpha - n)x + \cos(\alpha + n)x] \, dx \\ &= \frac{1}{\pi} \left[\frac{\sin(\alpha - n)x}{\alpha - n} + \frac{\sin(\alpha + n)x}{\alpha + n} \right]_0^{\pi} = \dots = \frac{(-1)^n 2\alpha \sin \alpha\pi}{\pi(\alpha^2 - n^2)} \end{aligned}$$

$$f(x) = \frac{1}{\alpha\pi} \sin \alpha\pi + \frac{2\alpha \sin \alpha\pi}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(\alpha^2 - n^2)} \cos nx$$

The periodic function is continuous at $x = 0$, so

$$\begin{aligned} f(0) = 1 &= \frac{1}{\alpha\pi} \sin \alpha\pi + \frac{2\alpha \sin \alpha\pi}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(\alpha^2 - n^2)} \implies \frac{1}{\sin \alpha\pi} = \csc \alpha\pi \\ &= \frac{1}{\alpha\pi} + \frac{2\alpha}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{(\alpha^2 - n^2)} \end{aligned}$$

Exercise 8.114

Prove that if $\alpha = m$, $m \in \mathbb{N}$, then

$$a_0 = 0, \quad a_n = \begin{cases} 1 & n = m \\ 0 & n \neq m \end{cases}$$

Example 8.115 ■ Problem 3

Find the Fourier series of the following function and deduce $\coth \alpha\pi$ from it.

$$f(x) = e^{-\alpha x}, \quad \alpha \in \mathbb{R}, \quad x \in [-\pi, \pi), \quad f(x + 2\pi) = f(x)$$

Lemma 1:

$$\int e^{\alpha x} \cos nx \, dx = \frac{e^{\alpha x}}{\alpha^2 + n^2} (\alpha \cos nx + n \sin nx)$$

Proof: integrating by parts twice, with $I = \int e^{\alpha x} \cos nx \, dx$:

$$\begin{aligned} I &= \frac{1}{\alpha} e^{\alpha x} \cos nx - \int \frac{1}{\alpha} e^{\alpha x} (-n \sin nx) \, dx = \frac{1}{\alpha} e^{\alpha x} \cos nx + \frac{n}{\alpha} \int e^{\alpha x} \sin nx \, dx \\ &= \frac{1}{\alpha} e^{\alpha x} \cos nx + \frac{n}{\alpha} \left[\frac{1}{\alpha} e^{\alpha x} \sin nx - \int \frac{1}{\alpha} e^{\alpha x} (n \cos nx) \, dx \right] = \frac{1}{\alpha} e^{\alpha x} \cos nx + \frac{n}{\alpha^2} e^{\alpha x} \sin nx - \frac{n^2}{\alpha^2} I \end{aligned}$$

$$\Rightarrow I = \frac{e^{\alpha x}}{\alpha^2 + n^2} (\alpha \cos nx + n \sin nx)$$

Lemma 2:

$$\int e^{\alpha x} \sin nx \, dx = \frac{e^{\alpha x}}{\alpha^2 + n^2} (\alpha \sin nx - n \cos nx)$$

(proof left as an exercise.)

The function is neither even nor odd, so all the Fourier coefficients must be computed.

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-\alpha x} \, dx = \frac{1}{-2\pi\alpha} \left[e^{-\alpha x} \right]_{-\pi}^{\pi} = \frac{1}{2\pi\alpha} [e^{\alpha\pi} - e^{-\alpha\pi}] = \frac{\sinh \alpha\pi}{\alpha\pi}$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-\alpha x} \cos nx \, dx = \left\{ \frac{e^{-\alpha x}}{\pi(\alpha^2 + n^2)} (-\alpha \cos nx + n \sin nx) \right\}_{-\pi}^{\pi} = \frac{\alpha(-1)^n}{\pi(\alpha^2 + n^2)} [e^{\alpha\pi} - e^{-\alpha\pi}] \\ &= \frac{2\alpha(-1)^n}{\pi(\alpha^2 + n^2)} \sinh \alpha\pi \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-\alpha x} \sin nx \, dx = \left\{ \frac{e^{-\alpha x}}{\pi(\alpha^2 + n^2)} (-\alpha \sin nx - n \cos nx) \right\}_{-\pi}^{\pi} = \frac{n(-1)^n}{\pi(\alpha^2 + n^2)} [e^{\alpha\pi} - e^{-\alpha\pi}] \\ &= \frac{2n(-1)^n}{\pi(\alpha^2 + n^2)} \sinh \alpha\pi \end{aligned}$$

$$\begin{aligned} f(x) &= \frac{\sinh \alpha\pi}{\alpha\pi} + \sum_{n=1}^{\infty} \left(\frac{2\alpha(-1)^n}{\pi(\alpha^2 + n^2)} \sinh \alpha\pi \cos nx + \frac{2n(-1)^n}{\pi(\alpha^2 + n^2)} \sinh \alpha\pi \sin nx \right) \\ &= \sinh \alpha\pi \left(\frac{1}{\alpha\pi} + \sum_{n=1}^{\infty} \left(\frac{2\alpha(-1)^n}{\pi(\alpha^2 + n^2)} \cos nx + \frac{2n(-1)^n}{\pi(\alpha^2 + n^2)} \sin nx \right) \right) \end{aligned}$$

Since $x = \pi$ is a point of discontinuity, the value of the Fourier series there equals the average of the upper and lower limits of the function at that point, i.e.

$$\frac{f(\pi^+) + f(\pi^-)}{2} = \frac{e^{\alpha\pi} + e^{-\alpha\pi}}{2} = \cosh \alpha\pi$$

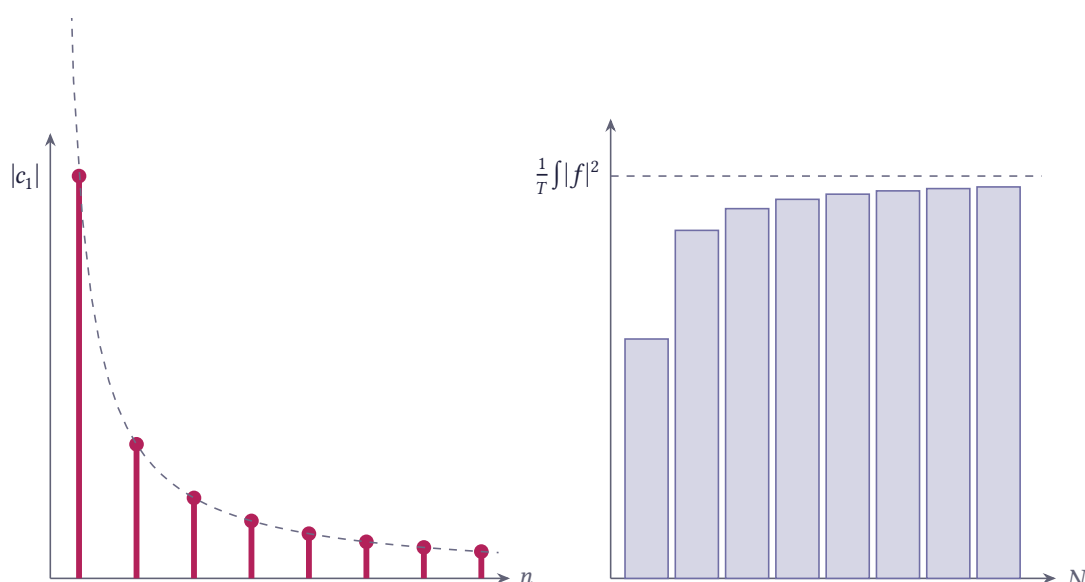
$$\cosh \alpha\pi = \sinh \alpha\pi \left(\frac{1}{\alpha\pi} + \sum_{n=1}^{\infty} \left(\frac{2\alpha(-1)^n}{\pi(\alpha^2 + n^2)} \cos n\pi + \frac{2n(-1)^n}{\pi(\alpha^2 + n^2)} \sin n\pi \right) \right)$$

$$= \sinh \alpha\pi \left(\frac{1}{\alpha\pi} + \sum_{n=1}^{\infty} \frac{2\alpha}{\pi(\alpha^2 + n^2)} \right)$$

$$\coth \alpha\pi = \frac{\cosh \alpha\pi}{\sinh \alpha\pi} = \frac{1}{\alpha\pi} + \sum_{n=1}^{\infty} \frac{2\alpha}{\pi(\alpha^2 + n^2)}$$

Properties of Fourier Series, Parseval's Theorem and the Complex Form

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9.1 Approximating a Periodic Function by a Finite Trigonometric Series

Theorem 9.116

For an N -term approximation of a periodic function $f(x)$ by a series of sines and cosines, the best coefficients (in the MMSE sense) are the Fourier series coefficients.

Consider a periodic function with period 2π :

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx, \quad f(x) = f(x + 2\pi)$$

The N -term trigonometric approximation:

$$f(x) \approx a_0 + \sum_{n=1}^N a_n \cos nx + b_n \sin nx$$

Is the following trigonometric approximation not **better**?

$$F(x) \triangleq \alpha_0 + \sum_{n=1}^N \alpha_n \cos nx + \beta_n \sin nx$$

What is the **criterion** for an approximation being better? One criterion: the error function:

$$E(x) = |f(x) - F(x)| = \left| f(x) - \alpha_0 - \sum_{n=1}^N (\alpha_n \cos nx + \beta_n \sin nx) \right|$$

A good criterion: the mean squared error:

$$D \triangleq \int_{-\pi}^{\pi} E^2(x) dx = \int_{-\pi}^{\pi} \left| f(x) - \alpha_0 - \sum_{n=1}^N (\alpha_n \cos nx + \beta_n \sin nx) \right|^2 dx$$

Proof.

For the best approximation we minimize the mean squared error. This is done by differentiating D with respect to the coefficients of the trigonometric series and setting these derivatives equal to zero. In other words:

$$\begin{aligned} \frac{\partial D}{\partial \alpha_0} &= -2 \int_{-\pi}^{\pi} \left[f(x) - \alpha_0 - \sum_{n=1}^N (\alpha_n \cos nx + \beta_n \sin nx) \right] dx = 0 \\ \int_{-\pi}^{\pi} f(x) dx &= \int_{-\pi}^{\pi} \left[\alpha_0 + \sum_{n=1}^N (\alpha_n \cos nx + \beta_n \sin nx) \right] dx = 2\pi\alpha_0 \implies \alpha_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = a_0 \\ \frac{\partial D}{\partial \alpha_m} &= 0 \implies \alpha_m = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos mx dx = a_m \\ \frac{\partial D}{\partial \beta_m} &= 0 \implies \beta_m = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin mx dx = b_m \end{aligned}$$

The best finite trigonometric approximation of a periodic function is the finite Fourier series. ■

9.2 Differentiation and Integration of Fourier Series

Theorem 9.117 ■ Differentiating a Fourier series

If $f(x)$ is periodic with period T , continuous and has a Fourier series, then its derivative is also periodic, and its Fourier series is obtained by differentiating the Fourier series of the function.

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi}{T} nx + b_n \sin \frac{2\pi}{T} nx$$

$$\Rightarrow f'(x) = \sum_{n=1}^{\infty} \left(\frac{2\pi}{T} n b_n \right) \cos \frac{2\pi}{T} nx + \left(-\frac{2\pi}{T} n a_n \right) \sin \frac{2\pi}{T} nx$$

Theorem 9.118 ■ Integrating a Fourier series

If $f(x)$ is periodic with period T and has a Fourier series, then the definite integral of this function can be computed by integrating the Fourier series of the function.

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi}{T} nx + b_n \sin \frac{2\pi}{T} nx$$

$$\Rightarrow \int_0^x f(t) dt = a_0 x + \sum_{n=1}^{\infty} \left(\frac{T}{2\pi n} a_n \sin \frac{2\pi}{T} nx - \frac{T}{2\pi n} b_n \cos \frac{2\pi}{T} nx + \frac{T}{2\pi n} b_n \right)$$

9.3 Rayleigh's Theorem

Theorem 9.119 ■ Rayleigh's theorem

If the functions $f(x)$ and $g(x)$ are periodic with period T and their Fourier series exist as

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi}{T} nx + b_n \sin \frac{2\pi}{T} nx$$

$$g(x) = \alpha_0 + \sum_{n=1}^{\infty} \alpha_n \cos \frac{2\pi}{T} nx + \beta_n \sin \frac{2\pi}{T} nx$$

then

$$\frac{2}{T} \int_T f(x) \overline{g(x)} dx = 2a_0 \overline{\alpha_0} + \sum_{n=1}^{\infty} a_n \overline{\alpha_n} + b_n \overline{\beta_n}$$

Proof.

Proof, assuming $g(x)$ is real:

$$\begin{aligned}
 \frac{2}{T} \int_T f(x)g(x) dx &= \frac{2}{T} \int_T f(x) \left(\alpha_0 + \sum_{n=1}^{\infty} \alpha_n \cos \frac{2\pi}{T} nx + \beta_n \sin \frac{2\pi}{T} nx \right) dx \\
 &= \frac{2}{T} \int_T f(x) \alpha_0 dx + \sum_{n=1}^{\infty} \frac{2}{T} \int_T f(x) \left(\alpha_n \cos \frac{2\pi}{T} nx + \beta_n \sin \frac{2\pi}{T} nx \right) dx \\
 &= \underbrace{\frac{2}{T} \int_T f(x) dx}_{2a_0} \alpha_0 + \sum_{n=1}^{\infty} \underbrace{\frac{2}{T} \int_T f(x) \cos \frac{2\pi}{T} nx dx}_{a_n} \alpha_n + \underbrace{\frac{2}{T} \int_T f(x) \sin \frac{2\pi}{T} nx dx}_{b_n} \beta_n \\
 &= 2a_0 \alpha_0 + \sum_{n=1}^{\infty} a_n \alpha_n + b_n \beta_n
 \end{aligned}$$

■

9.4 Parseval's Theorem

Theorem 9.120 ■ Parseval's theorem

If the function $f(x)$ is periodic with period T and its Fourier series exists as

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi}{T} nx + b_n \sin \frac{2\pi}{T} nx$$

then

$$\frac{2}{T} \int_T |f(x)|^2 dx = 2|a_0|^2 + \sum_{n=1}^{\infty} (|a_n|^2 + |b_n|^2)$$

Proof.

In Rayleigh's identity, replacing $g(x)$ by $f(x)$ gives Parseval's identity.

■

By Parseval's identity, the power of the function $f(x)$ can be obtained from the Fourier coefficients. If the function is expressed by an N -term approximation of sines and cosines, then by the relation above the power of the approximating function is less than the power of $f(x)$, and we have:

Key result 9.121 ■ Bessel's inequality

$$\frac{2}{T} \int_T |f(x)|^2 dx \geq 2|a_0|^2 + \sum_{n=1}^N (|a_n|^2 + |b_n|^2)$$

Example 9.122

For the function $f(x)$ below, how many terms of the Fourier series are needed for at least 94 percent of the power to remain in the approximating function?

$$f(x) = \sum_{n=1}^{\infty} \frac{2k}{n\pi} (1 - \cos n\pi) \sin nx$$

$$\frac{2}{T} \int_T |f(x)|^2 dx = \frac{1}{\pi} \int_{-\pi}^{\pi} k^2 dx = \frac{2}{\pi} [k^2 x]_0^{\pi} = 2k^2$$

$$\frac{\sum_{n=1}^N \left(\frac{2k}{n\pi} (1 - \cos n\pi) \right)^2}{2k^2} \geq 0.94$$

$$\frac{\sum_{n=1}^7 \left(\frac{2k}{n\pi} (1 - \cos n\pi) \right)^2}{2k^2} \cong 0.9496, \quad \frac{\sum_{n=1}^5 \left(\frac{2k}{n\pi} (1 - \cos n\pi) \right)^2}{2k^2} \cong 0.9331 \implies N = 7$$

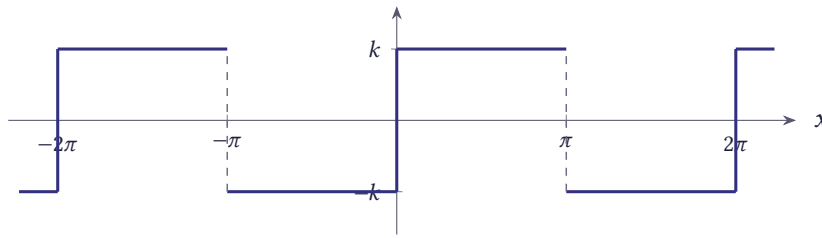


Figure 9.56 The square wave $\pm k$ with period 2π used in the power example

9.5 The Complex Form of the Fourier Series

Suppose $f(x)$ is periodic and, merely for simplicity, has period 2π . Then

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

Using Euler's formula $e^{ix} = \cos x + i \sin x$:

$$\cos nx = \frac{e^{inx} + e^{-inx}}{2}, \quad \sin nx = \frac{e^{inx} - e^{-inx}}{2i}, \quad \frac{1}{i} = -i$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \left(\frac{e^{inx} + e^{-inx}}{2} \right) + b_n \left(\frac{e^{inx} - e^{-inx}}{2i} \right) = a_0 + \sum_{n=1}^{\infty} \underbrace{\left(\frac{a_n - ib_n}{2} \right)}_{c_n} e^{inx} + \underbrace{\left(\frac{a_n + ib_n}{2} \right)}_{c_{-n}} e^{-inx}$$

$$c_n = \frac{a_n - ib_n}{2} = \frac{1}{2} \left\{ \frac{1}{\pi} \int_{2\pi} f(x) \cos nx dx - i \frac{1}{\pi} \int_{2\pi} f(x) \sin nx dx \right\}$$

$$= \frac{1}{2\pi} \int_{2\pi} f(x) (\cos nx - i \sin nx) dx = \frac{1}{2\pi} \int_{2\pi} f(x) e^{-inx} dx$$

$$c_{-n} = \frac{a_n + ib_n}{2} = \frac{1}{2\pi} \int_{2\pi} f(x)(\cos nx + i \sin nx) dx = \frac{1}{2\pi} \int_{2\pi} f(x)e^{inx} dx$$

Key result 9.123 ■ The complex form of the Fourier series

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}, \quad c_n = \frac{1}{2\pi} \int_{2\pi} f(x)e^{-inx} dx$$

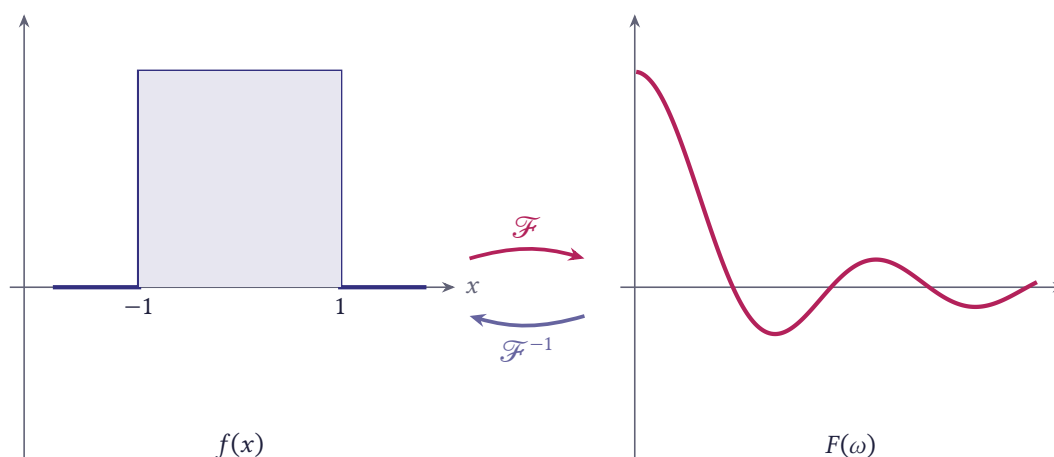
For an arbitrary period:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i\frac{2\pi}{T}nx}, \quad c_n = \frac{1}{T} \int_T f(x)e^{-i\frac{2\pi}{T}nx} dx$$

$$\frac{1}{T} \int_T |f(x)|^2 dx = \sum_{n=-\infty}^{\infty} |c_n|^2$$

The Fourier Integral

انتگرال فوريه



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10.1 Fourier Analysis of Non-Periodic Functions

To be able to use Fourier analysis for non-periodic functions, we first regard the function as periodic and then let the period tend to infinity.

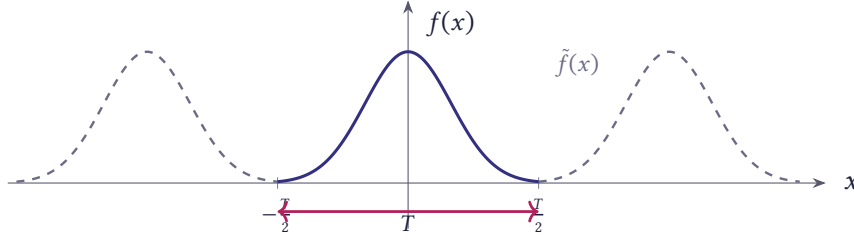


Figure 10.57 A non-periodic function $f(x)$ and its periodic extension $\tilde{f}(x)$ with period T

$$\tilde{f}(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi}{T} nx + b_n \sin \frac{2\pi}{T} nx$$

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} f(x) dx, \quad a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(x) \cos \frac{2\pi}{T} nx dx, \quad b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(x) \sin \frac{2\pi}{T} nx dx$$

$$\begin{aligned} \tilde{f}(x) &= \left(\frac{1}{T} \int_{-T/2}^{T/2} f(t) dt \right) + \sum_{n=1}^{\infty} \left\{ \left(\frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos \frac{2\pi}{T} nt dt \right) \cos \frac{2\pi}{T} nx \right. \\ &\quad \left. + \left(\frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin \frac{2\pi}{T} nt dt \right) \sin \frac{2\pi}{T} nx \right\} \end{aligned}$$

$$\tilde{f}(x) = \left(\frac{1}{T} \int_{-T/2}^{T/2} f(t) dt \right) + \frac{2}{T} \sum_{n=1}^{\infty} \left\{ \int_{-T/2}^{T/2} f(t) \left(\cos \frac{2\pi}{T} nt \cos \frac{2\pi}{T} nx + \sin \frac{2\pi}{T} nt \sin \frac{2\pi}{T} nx \right) dt \right\}$$

$$\tilde{f}(x) = \left(\frac{1}{T} \int_{-T/2}^{T/2} f(t) dt \right) + \frac{2}{T} \sum_{n=1}^{\infty} \left\{ \int_{-T/2}^{T/2} f(t) \cos \frac{2\pi}{T} n(t-x) dt \right\}$$

- To obtain the Fourier analysis of the function $f(x)$ we must let T tend to **infinity**.
- Suppose $f(x)$ is an integrable function, so that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f(t) dt = 0$$

Then

$$f(x) = \lim_{T \rightarrow \infty} \frac{2}{T} \sum_{n=1}^{\infty} \left\{ \int_{-T/2}^{T/2} f(t) \cos \frac{2\pi}{T} n(t-x) dt \right\}$$

$$\omega_n \triangleq \frac{2\pi}{T} n \implies \Delta\omega \triangleq \omega_n - \omega_{n-1} = \frac{2\pi}{T} n - \frac{2\pi}{T} (n-1) = \frac{2\pi}{T} \implies \frac{2}{T} = \frac{\Delta\omega}{\pi}$$

$$f(x) = \lim_{T \rightarrow \infty} \frac{1}{\pi} \sum_{n=1}^{\infty} \left\{ \int_{-T/2}^{T/2} f(t) \cos \omega_n(t-x) dt \right\} \Delta\omega$$

If $T \rightarrow \infty$, then $\sum_{n=1}^{\infty} \rightarrow \int_0^{\infty}$, $\Delta\omega \rightarrow d\omega$ and $\omega_n \rightarrow \omega$:

$$\begin{aligned}
 f(x) &= \frac{1}{\pi} \int_0^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) \cos \omega(t-x) dt \right\} d\omega \\
 &= \frac{1}{\pi} \int_0^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) (\cos \omega t \cos \omega x + \sin \omega t \sin \omega x) dt \right\} d\omega \\
 &= \frac{1}{\pi} \int_0^{\infty} \left\{ \cos \omega x \int_{-\infty}^{\infty} (f(t) \cos \omega t) dt + \sin \omega x \int_{-\infty}^{\infty} (f(t) \sin \omega t) dt \right\} d\omega \\
 A(\omega) &\triangleq \int_{-\infty}^{\infty} f(t) \cos \omega t dt, \quad B(\omega) \triangleq \int_{-\infty}^{\infty} f(t) \sin \omega t dt \quad \Rightarrow \quad f(x) \\
 &= \frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} d\omega
 \end{aligned}$$

10.2 The Fourier Integral

Key result 10.124 ■ The Fourier integral and the sine and cosine transforms

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} d\omega \quad (\text{the Fourier integral; synthesis equation})$$

$$A(\omega) \triangleq \int_{-\infty}^{\infty} f(t) \cos \omega t dt \quad (\text{the Fourier cosine transform})$$

$$B(\omega) \triangleq \int_{-\infty}^{\infty} f(t) \sin \omega t dt \quad (\text{the Fourier sine transform})$$

(the analysis equations)

Theorem 10.125

A sufficient condition for the existence of the Fourier integral of a function $f(x)$ is that the function be piecewise continuous, have right and left derivatives at every point, and be absolutely integrable, i.e.

$$\int_{-\infty}^{\infty} |f(x)| dx < \infty$$

At points of discontinuity, the value of the Fourier integral equals the average of the left and right limits of $f(x)$ at those points.

10.3 The Complex Form of the Fourier Integral (the Fourier Transform)

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) \cos \omega(t-x) dt \right\} d\omega$$

The expression $\int_{-\infty}^{\infty} f(t) \cos \omega(t-x) dt$ is an even function of ω , hence

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) \cos \omega(t-x) dt \right\} d\omega$$

and the expression $\int_{-\infty}^{\infty} f(t) \sin \omega(t-x) dt$ is an odd function of ω , hence

$$\frac{-i}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) \sin \omega(t-x) dt \right\} d\omega = 0$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) \cos \omega(t-x) dt \right\} d\omega - \frac{i}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) \sin \omega(t-x) dt \right\} d\omega$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) (\cos \omega(t-x) - i \sin \omega(t-x)) dt \right\} d\omega$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) e^{-i\omega(t-x)} dt \right\} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \right\} e^{i\omega x} d\omega$$

Key result 10.126 ■ The Fourier transform

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega x} d\omega \quad (\text{the Fourier integral; synthesis equation})$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \quad (\text{the Fourier transform; analysis equation})$$

The complex form of the Fourier series (review):

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i\frac{2\pi}{T}nx} \quad (\text{Fourier series; synthesis equation}), \quad c_n = \frac{1}{T} \int_T f(x) e^{-i\frac{2\pi}{T}nx} dx$$

(Fourier coefficients; analysis equation)

10.4 The Function Si(u)

$$\text{Si}(x) \triangleq \int_0^x \frac{\sin w}{w} dw$$

- odd;
- has finite asymptotic values at infinity.

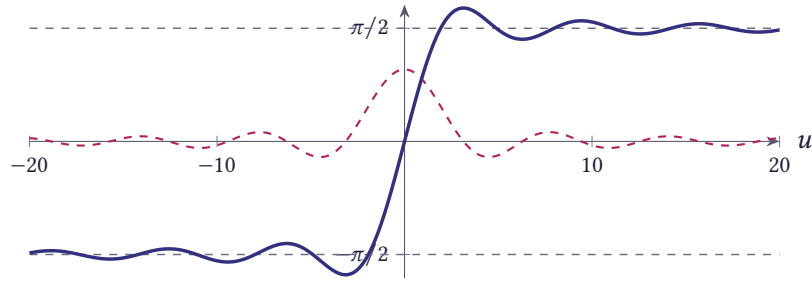


Figure 10.58 The function $\text{Si}(u)$ (solid) and the function $\sin u/u$ (dashed)

Theorem 10.127 ■ Lemma

$$\int_0^{\infty} \frac{\sin \alpha w}{w} dw = \begin{cases} \frac{\pi}{2} & \alpha > 0 \\ 0 & \alpha = 0 \\ -\frac{\pi}{2} & \alpha < 0 \end{cases}$$

Proof.

For $\alpha = 0$ the lemma is obvious. With the substitution $\tau = \alpha w$:

$$\int_0^x \frac{\sin \alpha w}{w} dw = \int_0^{\alpha x} \frac{\sin \tau}{\tau/\alpha} \frac{d\tau}{\alpha} = \int_0^{\alpha x} \frac{\sin \tau}{\tau} d\tau = \text{Si}(\alpha x), \quad \text{Si}(\alpha x) = -\text{Si}(-\alpha x)$$

So it suffices to prove the lemma for positive values of α .

$$F(\alpha) \triangleq \int_0^{\infty} e^{-\alpha t} \frac{\sin t}{t} dt, \quad F(0) = \text{Si}(+\infty) = ?$$

$$F'(\alpha) = \int_0^{\infty} -t e^{-\alpha t} \frac{\sin t}{t} dt = - \int_0^{\infty} e^{-\alpha t} \sin t dt = - \left[\frac{e^{-\alpha t}}{1 + \alpha^2} (-\alpha \sin t - \cos t) \right]_0^{\infty} = \frac{-1}{1 + \alpha^2}$$

$$\alpha > 0 : \quad F(\alpha) = -\tan^{-1} \alpha + C, \quad F(\alpha \rightarrow \infty) = 0 \implies C = F(\infty) + \tan^{-1} \infty = \frac{\pi}{2}$$

$$F(\alpha) = -\tan^{-1} \alpha + \frac{\pi}{2} \implies F(0) = \text{Si}(+\infty) = \frac{\pi}{2}$$

■

10.5 Examples of the Fourier Integral

Example 10.128

Express the rectangular pulse below as a Fourier integral.

$$f(x) = \begin{cases} 1 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$$

Since $f(x)$ is even, the Fourier sine transform vanishes: $B(\omega) = 0$.

$$A(\omega) = \int_{-\infty}^{\infty} f(t) \cos \omega t \, dt = \int_{-1}^1 \cos \omega t \, dt = \frac{2}{\omega} \sin \omega$$

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \frac{2}{\omega} \sin \omega \cos \omega x \, d\omega = \frac{1}{\pi} \int_0^{\infty} \frac{1}{\omega} [\sin(1-x)\omega + \sin(1+x)\omega] \, d\omega$$

$$= \frac{1}{\pi} \left[\int_0^{\infty} \frac{\sin(1-x)\omega}{\omega} \, d\omega + \int_0^{\infty} \frac{\sin(1+x)\omega}{\omega} \, d\omega \right] = \begin{cases} 0 & x < -1 \\ 1/2 & x = -1 \\ 1 & -1 < x < 1 \\ 1/2 & x = 1 \\ 0 & x > 1 \end{cases}$$

Note: the value of the Fourier integral at the points of discontinuity (using the lemma above).

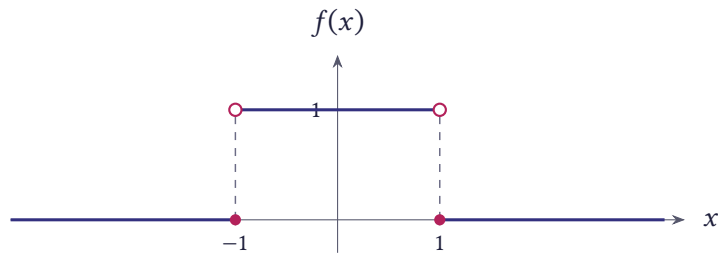


Figure 10.59 The rectangular pulse $f(x) = 1$ for $|x| < 1$

Example 10.129

Express the exponential function below as a Fourier integral.

$$f(x) = \begin{cases} 0 & x < 0 \\ e^{-x} & x > 0 \end{cases}$$

$$A(\omega) = \int_{-\infty}^{\infty} f(x) \cos \omega x \, dx = \int_0^{\infty} e^{-x} \cos \omega x \, dx = \frac{e^{-x}}{1 + \omega^2} (-\cos \omega x + \omega \sin \omega x) \Big|_0^{\infty} = \frac{1}{1 + \omega^2}$$

$$B(\omega) = \int_{-\infty}^{\infty} f(x) \sin \omega x \, dx = \int_0^{\infty} e^{-x} \sin \omega x \, dx = \frac{e^{-x}}{1 + \omega^2} (-\sin \omega x - \omega \cos \omega x) \Big|_0^{\infty} = \frac{\omega}{1 + \omega^2}$$

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} d\omega = \frac{1}{\pi} \int_0^{\infty} \frac{\cos \omega x + \omega \sin \omega x}{1 + \omega^2} d\omega$$

$$f(x=0) = \frac{1}{\pi} \int_0^{\infty} \frac{1}{1 + \omega^2} d\omega = \frac{1}{\pi} [\tan^{-1} \omega]_0^{\infty} = \frac{1}{\pi} \times \frac{\pi}{2} = \frac{1}{2}$$

Example 10.130

Express the function below as a Fourier integral.

$$f(x) = e^{-|x|}$$

Since $f(x)$ is even, the Fourier sine transform vanishes.

$$A(\omega) = \int_{-\infty}^{\infty} f(x) \cos \omega x \, dx = 2 \int_0^{\infty} e^{-x} \cos \omega x \, dx = \frac{2e^{-x}}{1 + \omega^2} (-\cos \omega x + \omega \sin \omega x) \Big|_0^{\infty} = \frac{2}{1 + \omega^2}$$

$$B(\omega) = \int_{-\infty}^{\infty} e^{-|x|} \sin \omega x \, dx = 0$$

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} d\omega = \frac{2}{\pi} \int_0^{\infty} \frac{\cos \omega x}{1 + \omega^2} d\omega$$

$$f(x=0) = \frac{2}{\pi} \int_0^{\infty} \frac{1}{1 + \omega^2} d\omega = \frac{2}{\pi} [\tan^{-1} \omega]_0^{\infty} = \frac{2}{\pi} \times \frac{\pi}{2} = 1$$

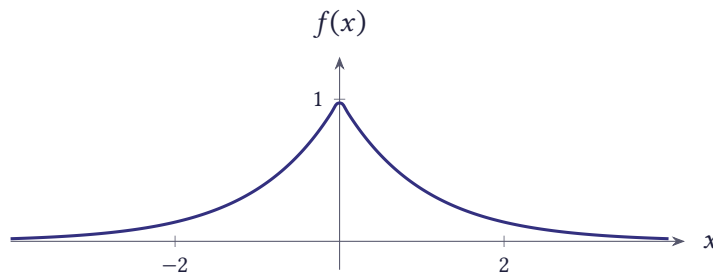


Figure 10.60 The function $f(x) = e^{-|x|}$

Example 10.131

Express the function below as a Fourier integral.

$$f(x) = \begin{cases} \sin x & |x| \leq \pi \\ 0 & |x| > \pi \end{cases}$$

Since $f(x)$ is odd, the Fourier cosine transform vanishes.

$$A(\omega) = \int_{-\pi}^{\pi} \sin x \cos \omega x \, dx = 0$$

$$\begin{aligned} B(\omega) &= \int_{-\pi}^{\pi} \sin x \sin \omega x \, dx = 2 \int_0^{\pi} \sin x \sin \omega x \, dx = \int_0^{\pi} [\cos(1-\omega)x - \cos(1+\omega)x] \, dx \\ &= \left[\frac{\sin(1-\omega)x}{1-\omega} - \frac{\sin(1+\omega)x}{1+\omega} \right]_0^{\pi} = \frac{2 \sin(\omega\pi)}{1-\omega^2} \end{aligned}$$

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} \, d\omega = \frac{2}{\pi} \int_0^{\infty} \frac{\sin(\omega\pi) \sin \omega x}{1-\omega^2} \, d\omega$$

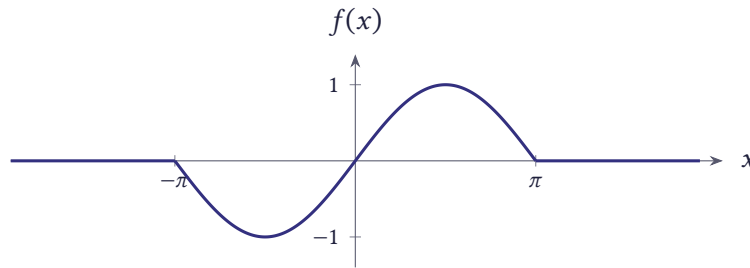


Figure 10.61 The function $f(x) = \sin x$ for $|x| \leq \pi$ and zero outside

Example 10.132

Express the function below as a Fourier integral.

$$f(x) = \begin{cases} x^2 & |x| \leq a \\ 0 & |x| > a \end{cases}$$

Since $f(x)$ is even, the Fourier sine transform vanishes.

$$B(\omega) = \int_{-a}^a x^2 \sin \omega x \, dx = 0$$

$$\begin{aligned} A(\omega) &= \int_{-\infty}^{\infty} f(x) \cos \omega x \, dx = 2 \int_0^a x^2 \cos \omega x \, dx = 2 \left[\frac{x^2}{\omega} \sin \omega x + \frac{2x}{\omega^2} \cos \omega x - \frac{2}{\omega^3} \sin \omega x \right]_0^a \\ &= 2 \left[\left(\frac{a^2}{\omega} - \frac{2}{\omega^3} \right) \sin \omega a + \frac{2a}{\omega^2} \cos \omega a \right] \end{aligned}$$

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \left[\left(\frac{a^2}{\omega} - \frac{2}{\omega^3} \right) \sin \omega a + \frac{2a}{\omega^2} \cos \omega a \right] \cos \omega x \, d\omega$$

10.6 Some Properties of the Fourier Integral

All the properties follow from the synthesis and analysis equations:

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} d\omega, \quad A(\omega) \triangleq \int_{-\infty}^{\infty} f(x) \cos \omega x dx,$$

$$B(\omega) \triangleq \int_{-\infty}^{\infty} f(x) \sin \omega x dx$$

■ Time Scaling

$$f(x) \rightarrow f(ax), \quad a \in \mathbb{R} \quad \Rightarrow \quad \begin{cases} A(\omega) \rightarrow \frac{1}{|a|} A\left(\frac{\omega}{a}\right) \\ B(\omega) \rightarrow \frac{1}{|a|} B\left(\frac{\omega}{a}\right) \end{cases}$$

Proof.

$$\begin{aligned} A_1(\omega) &= \int_{-\infty}^{\infty} f(ax) \cos \omega x dx \\ &= \int_{-\infty}^{\infty} f(ax) \cos \omega x dx \begin{cases} \stackrel{a>0}{=} \int_{u=ax}^{\infty} f(u) \cos \frac{\omega}{a} u \frac{du}{a} = \frac{1}{a} A\left(\frac{\omega}{a}\right) \\ \stackrel{a<0}{=} \int_{u=ax}^{-\infty} f(u) \cos \frac{\omega}{a} u \frac{du}{a} = -\frac{1}{a} \int_{-\infty}^{\infty} f(u) \cos \frac{\omega}{a} u du = -\frac{1}{a} A\left(\frac{\omega}{a}\right) \end{cases} \\ \Rightarrow A_1(\omega) &= \frac{1}{|a|} A\left(\frac{\omega}{a}\right) \end{aligned}$$

■

■ Parseval's Identity

$$\int_{-\infty}^{\infty} f^2(x) dx = \frac{1}{\pi} \int_0^{\infty} \{A^2(\omega) + B^2(\omega)\} d\omega$$

Proof.

$$\begin{aligned} \int_{-\infty}^{\infty} f^2(x) dx &= \int_{-\infty}^{\infty} f(x) \left(\frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} d\omega \right) dx \\ &= \frac{1}{\pi} \int_0^{\infty} \left\{ A(\omega) \left[\int_{-\infty}^{\infty} f(x) \cos \omega x dx \right] + B(\omega) \left[\int_{-\infty}^{\infty} f(x) \sin \omega x dx \right] \right\} d\omega \\ &= \frac{1}{\pi} \int_0^{\infty} \{A^2(\omega) + B^2(\omega)\} d\omega \end{aligned}$$

■

■ The Time Derivative

$$\begin{aligned} f(x) &= \frac{1}{\pi} \int_0^{\infty} \{A(\omega) \cos \omega x + B(\omega) \sin \omega x\} d\omega \quad \Rightarrow \quad f'(x) \\ &= \frac{1}{\pi} \int_0^{\infty} \{\omega B(\omega) \cos \omega x - \omega A(\omega) \sin \omega x\} d\omega \end{aligned}$$

(using the rule for differentiating an integral:)

$$\frac{\partial}{\partial z} \int_{a(z)}^{b(z)} f(x, z) dx = \int_{a(z)}^{b(z)} \frac{\partial f(x, z)}{\partial z} dx + f(b(z), z) \frac{db(z)}{dz} - f(a(z), z) \frac{da(z)}{dz}$$

■ The Frequency Derivative

$$\frac{dA(\omega)}{d\omega} = A'(\omega) = \int_{-\infty}^{\infty} -xf(x) \sin \omega x \, dx \implies \int_{-\infty}^{\infty} (xf(x)) \sin \omega x \, dx = -A'(\omega)$$

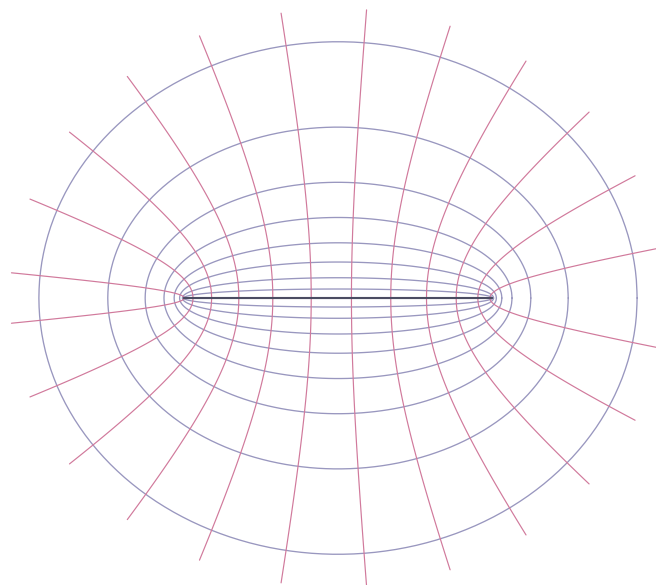
$$\frac{dB(\omega)}{d\omega} = B'(\omega) = \int_{-\infty}^{\infty} xf(x) \cos \omega x \, dx \implies \int_{-\infty}^{\infty} (xf(x)) \cos \omega x \, dx = B'(\omega)$$

$$xf(x) = \frac{1}{\pi} \int_0^{\infty} \{B'(\omega) \cos \omega x - A'(\omega) \sin \omega x\} d\omega$$

Exercise 10.133

Prove the following identities:

1. $\int_0^{\infty} \frac{\cos xw + w \sin xw}{1 + w^2} dw = \begin{cases} 0 & x < 0 \\ \pi/2 & x = 0 \\ \pi e^{-x} & x > 0 \end{cases}$
2. $\int_0^{\infty} \frac{\sin \pi w \sin xw}{1 - w^2} dw = \begin{cases} \frac{\pi}{2} \sin x & 0 \leq x \leq \pi \\ 0 & x > \pi \end{cases}$
3. $\int_0^{\infty} \frac{1 - \cos \pi w}{w} \sin xw \, dw = \begin{cases} \frac{1}{2}\pi & 0 < x < \pi \\ 0 & x > \pi \end{cases}$
4. $\int_0^{\infty} \frac{\cos \frac{1}{2}\pi w}{1 - w^2} \cos xw \, dw = \begin{cases} \frac{1}{2}\pi \cos x & 0 < |x| < \frac{1}{2}\pi \\ 0 & |x| \geq \frac{1}{2}\pi \end{cases}$
5. $\int_0^{\infty} \frac{\sin w - w \cos w}{w^2} \sin xw \, dw = \begin{cases} \frac{1}{2}\pi x & 0 < x < 1 \\ \frac{1}{4}\pi & x = 1 \\ 0 & x > 1 \end{cases}$



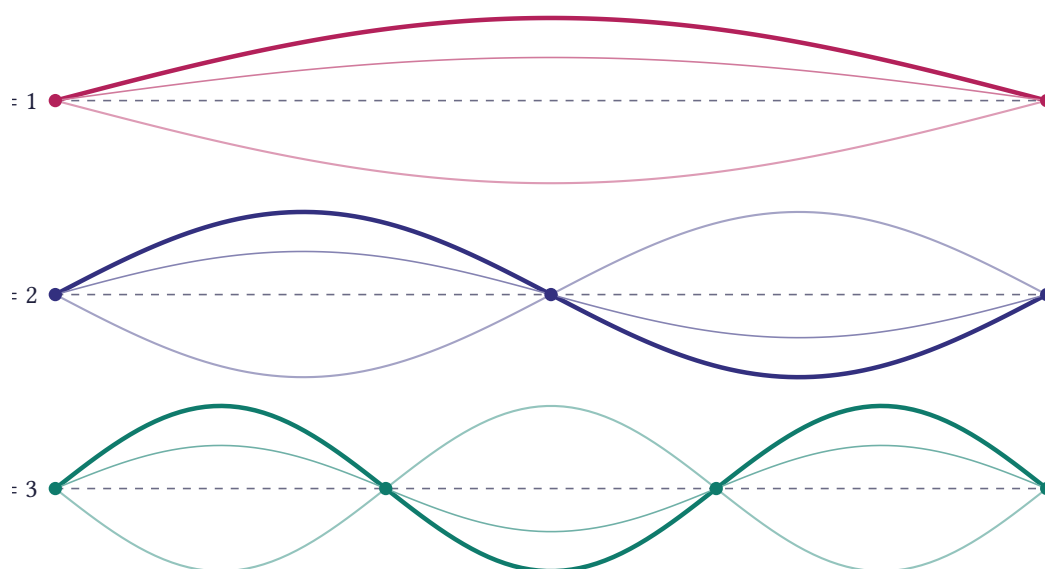
Partial Differential Equations and the Laplace Transform

معادلات دیفرانسیل با مشتقات جزئی و تبدیل لاپلاس

Chapters 11–14

The One-Dimensional Wave Equation and Separation of Variables

معادله موجی یک بعدی و روش جداسازی متغیرها



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11.1 Introduction

Definition 11.134 ■ Differential equation

An equation in which a function and its derivatives appear.

Definition 11.135 ■ Ordinary differential equation

An equation written in terms of only one independent variable, with the derivatives of the function taken only with respect to that variable.

Definition 11.136 ■ Partial differential equation

A differential equation in which the function depends on several independent variables and the partial derivatives of the function with respect to the different variables appear in the equation.

The wave equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad \frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \quad \frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

The heat equation:

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad \frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \quad \frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

Laplace's equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0, \quad \nabla^2 u = 0$$

11.2 The One-Dimensional Wave Equation

Modelling a vibrating string (deriving the equation governing $u(x, t)$).

Physical assumptions:

- The mass is distributed uniformly along the string (a homogeneous string), and the string is perfectly flexible.
- The string is stretched so tightly that the force of gravity on it is negligible.
- The deflection of the string from equilibrium is very small.

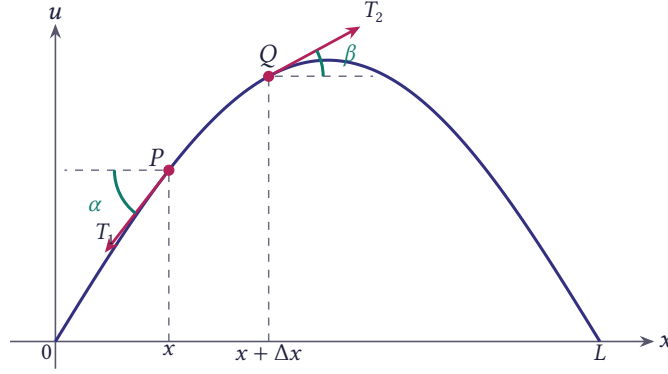


Figure 11.62 A segment PQ of the vibrating string: the tensions T_1 and T_2 at its ends, at angles α and β

$$T_1 \cos \alpha = T_2 \cos \beta = T = \text{const.}$$

$$T_2 \sin \beta - T_1 \sin \alpha = \rho \Delta s \frac{\partial^2 u}{\partial t^2}$$

where ρ is the mass density of the string and Δs the differential of the string length. Dividing by T :

$$T(\tan \beta - \tan \alpha) = \rho \Delta s \frac{\partial^2 u}{\partial t^2} \implies T \left(\frac{\partial u}{\partial x} \Big|_{(x+\Delta x, t)} - \frac{\partial u}{\partial x} \Big|_{(x, t)} \right) = \rho \Delta s \frac{\partial^2 u}{\partial t^2}$$

By the mean value theorem ($f(b) - f(a) = f'(l)(b - a)$, $l \in [a, b]$):

$$T \left(\frac{\partial u}{\partial x} \Big|_{(x+\Delta x, t)} - \frac{\partial u}{\partial x} \Big|_{(x, t)} \right) = T \frac{\partial^2 u}{\partial x^2} \Big|_{(l, t)} \Delta x = \rho \Delta s \frac{\partial^2 u}{\partial t^2}, \quad x < l < x + \Delta x$$

By assumption $\lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta s} = 1$, so

$$T \frac{\partial^2 u}{\partial x^2} = \rho \frac{\partial^2 u}{\partial t^2}, \quad c^2 \triangleq \frac{T}{\rho} \implies \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

For a unique solution of this equation we need boundary conditions and initial conditions:

- Boundary conditions: $u(0, t) = u(L, t) = 0$
- Initial conditions: $u(x, 0) = f(x)$ and $u_t(x, 0) = g(x)$, where $u_t \triangleq \frac{\partial u}{\partial t}$

11.3 Solving the One-Dimensional Wave Equation

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \\ u(0, t) = u(L, t) = 0 \\ u(x, 0) = f(x) \\ u_t(x, 0) = g(x) \end{cases}$$

Here we solve the equation by **separation of variables**. Assume

$$u(x, t) = X(x) \times T(t)$$

$$\frac{\partial^2 u}{\partial x^2} = X''(x)T(t), \quad \frac{\partial^2 u}{\partial t^2} = X(x)T''(t) \implies X(x)T''(t) = c^2 X''(x)T(t) \implies \frac{T''}{c^2 T} = \frac{X''}{X} = k$$

k must necessarily be a negative number. Examining the cases:

If $k = 0$:

$$\frac{X''}{X} = 0 \Rightarrow X'' = 0 \Rightarrow X(x) = Ax + B, \quad X(0) = 0 \Rightarrow B = 0, \quad X(L) = 0 \Rightarrow A = 0 \Rightarrow X(x) = 0$$

If $k > 0$ ($k = \lambda^2$):

$$\frac{X''}{X} = \lambda^2 \Rightarrow X'' - \lambda^2 X = 0 \Rightarrow X(x) = Ae^{\lambda x} + Be^{-\lambda x}, \quad X(0) = 0 \Rightarrow A + B = 0, \quad X(L) = 0 \Rightarrow Ae^{\lambda L} + Be^{-\lambda L} = 0$$

$$\Rightarrow A = B = 0 \Rightarrow X(x) = 0$$

Hence k must be negative ($k = -\lambda^2$):

$$\frac{T''}{c^2 T} = \frac{X''}{X} = -\lambda^2 \Rightarrow \begin{cases} X'' + \lambda^2 X = 0 \\ T'' + c^2 \lambda^2 T = 0 \end{cases}$$

$$X(x) = A \cos \lambda x + B \sin \lambda x, \quad X(0) = 0 \Rightarrow A = 0$$

$$X(L) = 0 \Rightarrow B \sin \lambda L = 0 \Rightarrow \lambda L = n\pi \quad (n \in \mathbb{Z}) \Rightarrow \lambda = \frac{n\pi}{L} = \lambda_n$$

$$X_n(x) = B_n \sin \lambda_n x, \quad \lambda_n = \frac{n\pi}{L} \quad (n \in \mathbb{Z})$$

$$T'' + c^2 \lambda_n^2 T = 0 \Rightarrow T_n(t) = C_n \cos(c\lambda_n t) + D_n \sin(c\lambda_n t)$$

$$u_n(x, t) = X_n(x) \times T_n(t) = B_n \sin \lambda_n x \times (C_n \cos(c\lambda_n t) + D_n \sin(c\lambda_n t))$$

Superposing the solutions:

$$u(x, t) = \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{cn\pi}{L}t\right) + b_n \sin\left(\frac{cn\pi}{L}t\right) \right] \sin \frac{n\pi}{L}x$$

Applying the initial conditions. The condition $u(x, 0) = f(x)$:

$$u(x, 0) = \sum_{n=1}^{\infty} a_n \sin \frac{n\pi}{L}x = f(x)$$

$f(x)$ has been set equal to a periodic odd function, so to compute the unknown Fourier coefficients we must regard $f(x)$ as a periodic odd function (with period $2L$). Then

$$a_n = \frac{2}{2L} \int_{-L}^L f(x) \sin\left(\frac{2\pi}{2L}nx\right) dx = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

The condition $u_t(x, 0) = g(x)$:

$$u_t(x, t) = \sum_{n=1}^{\infty} \left[-a_n \frac{cn\pi}{L} \sin\left(\frac{cn\pi}{L}t\right) + b_n \frac{cn\pi}{L} \cos\left(\frac{cn\pi}{L}t\right) \right] \sin \frac{n\pi}{L}x$$

$$u_t(x, 0) = \sum_{n=1}^{\infty} b_n \frac{cn\pi}{L} \sin \frac{n\pi}{L}x = g(x) \Rightarrow b_n = \frac{2}{cn\pi} \int_0^L g(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

Key result 11.137 ■ Solution of the one-dimensional wave equation (separation of variables)

$$u(x, t) = \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{cn\pi}{L}t\right) + b_n \sin\left(\frac{cn\pi}{L}t\right) \right] \sin \frac{n\pi}{L}x$$

$$a_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx, \quad b_n = \frac{2}{cn\pi} \int_0^L g(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

Example 11.138 ■ Example 1

Find the solution of the following one-dimensional wave equation.

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = u(\pi, t) = 0, \quad u(x, 0) = k \sin 2x, \quad u_t(x, 0) = 0$$

Since the initial velocity of the vibrating string is zero, $g(x) = 0 \Rightarrow b_n = 0$. Hence ($c = 1$, $L = \pi$):

$$u(x, t) = \sum_{n=1}^{\infty} a_n \cos\left(\frac{cn\pi}{L}t\right) \sin \frac{n\pi}{L}x = \sum_{n=1}^{\infty} a_n \cos nt \sin nx$$

$$a_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx = \frac{2}{\pi} \int_0^{\pi} k \sin 2x \sin nx dx = \begin{cases} 0 & n \neq 2 \\ k & n = 2 \end{cases}$$

$$u(x, t) = k \sin 2x \cos 2t$$

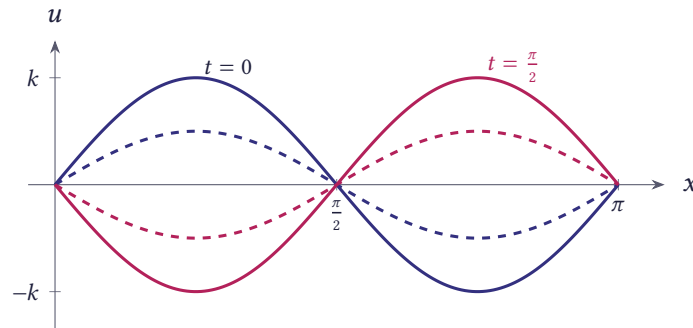


Figure 11.63 The second mode of the string: $u(x, t) = k \sin 2x \cos 2t$ at different times

Example 11.139 ■ Example 2

Find the solution of the following one-dimensional wave equation.

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = u(\pi, t) = 0, \quad u(x, 0) = f(x), \quad u_t(x, 0) = 0$$

where $f(x)$ is the triangular function: $f(x) = \frac{k}{a}x$ for $0 \leq x \leq a$ and $f(x) = \frac{k}{a-\pi}(x-\pi)$ for $a \leq x \leq \pi$.

$u_t(x, 0) = g(x) = 0 \Rightarrow b_n = 0$ and

$$u(x, t) = \sum_{n=1}^{\infty} a_n \cos nt \sin nx$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx = \frac{2}{\pi} \int_0^a \frac{k}{a} x \sin nx \, dx + \frac{2}{\pi} \int_a^{\pi} \frac{k}{a-\pi} (x-\pi) \sin nx \, dx \\ &= \frac{2k}{a\pi} \left[x \left(\frac{-1}{n} \cos nx \right) - (1) \left(\frac{-1}{n^2} \sin nx \right) \right]_0^a + \frac{2k}{(a-\pi)\pi} \left[(x-\pi) \left(\frac{-1}{n} \cos nx \right) - (1) \left(\frac{-1}{n^2} \sin nx \right) \right]_a^{\pi} \\ &= \dots = \frac{2k \sin na}{an^2(\pi-a)} \end{aligned}$$

$$u(x, t) = \frac{2k}{a(\pi-a)} \sum_{n=1}^{\infty} \frac{\sin na}{n^2} \cos nt \sin nx$$

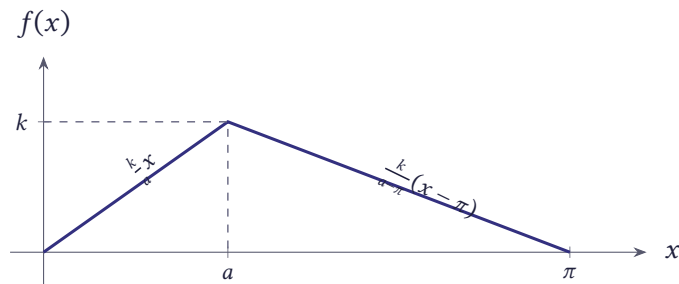


Figure 11.64 The triangular initial profile $f(x)$ with peak value k at $x = a$

Example 11.140 ■ Example 3

Find the solution of the following one-dimensional wave equation.

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = u(\pi, t) = 0, \quad u(x, 0) = kx(\pi - x), \quad u_t(x, 0) = 0$$

$g(x) = 0 \Rightarrow b_n = 0$ and

$$u(x, t) = \sum_{n=1}^{\infty} a_n \cos nt \sin nx$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} kx(\pi - x) \sin nx \, dx \\ &= \frac{2k}{\pi} \left[(-x^2 + \pi x) \left(\frac{-1}{n} \cos nx \right) - (-2x + \pi) \left(\frac{-1}{n^2} \sin nx \right) + (-2) \left(\frac{1}{n^3} \cos nx \right) \right]_0^{\pi} = \dots \\ &= \frac{4k(1 - (-1)^n)}{\pi n^3} \end{aligned}$$

$$u(x, t) = \frac{4k}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^3} \cos nt \sin nx$$

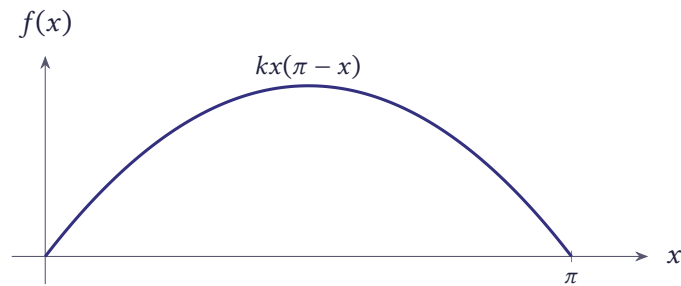


Figure 11.65 The parabolic initial profile $f(x) = kx(\pi - x)$

Exercise 11.141

- Find the ratio $\frac{a_1^2}{a_1^2 + a_2^2 + \dots}$.
- Plot the ratio a_{2n-1}/a_{2n+1} as a function of $n \in \mathbb{N}$.

11.4 The Method of Separation of Variables: Examples

Example 11.142 ■ Example 1

Solve the following differential equation by separation of variables: $xu_x = yu_y$.

$$x \frac{\partial u}{\partial x} = y \frac{\partial u}{\partial y}, \quad u(x, y) = X(x) \cdot Y(y)$$

$$xX'Y = yXY' \Rightarrow \frac{xX'}{X} = \frac{yY'}{Y} = k$$

$$\frac{X'}{X} = \frac{k}{x} \Rightarrow \ln X = k \ln x + c_1 \Rightarrow X(x) = C_1 x^k, \quad \frac{Y'}{Y} = \frac{k}{y} \Rightarrow \ln Y = k \ln y + c_2 \Rightarrow Y(y) = C_2 y^k$$

$$u(x, y) = X(x)Y(y) = C_1 x^k C_2 y^k = C(xy)^k$$

Example 11.143 ■ Example 2

Solve the following differential equation by separation of variables: $u_{xx} + u_{yy} = 0$.

$$u(x, y) = X(x) \cdot Y(y) \Rightarrow X''Y + XY'' = 0 \Rightarrow \frac{X''}{X} = -\frac{Y''}{Y} = k$$

$$\begin{cases} X'' - kX = 0 \Rightarrow X(x) = Ae^{\sqrt{k}x} + Be^{-\sqrt{k}x} \\ Y'' + kY = 0 \Rightarrow Y(y) = Ce^{\sqrt{-k}y} + De^{-\sqrt{-k}y} \end{cases}$$

$$u(x, y) = X(x)Y(y) = (Ae^{\sqrt{k}x} + Be^{-\sqrt{k}x})(Ce^{\sqrt{-k}y} + De^{-\sqrt{-k}y})$$

Example 11.144 ■ Example 3

Solve the following differential equation by separation of variables: $x^2 u_{xy} = -3y^2 u$.

$$x^2 \frac{\partial^2 u}{\partial x \partial y} + 3y^2 u = 0, \quad u(x, y) = X(x) \cdot Y(y) \Rightarrow x^2 X'Y' + 3y^2 XY = 0 \Rightarrow \frac{x^2 X'}{X} = \frac{-3y^2 Y'}{Y} = k$$

$$\frac{X'}{X} = \frac{k}{x^2} \Rightarrow \ln X = \frac{-k}{x} + c_1 \Rightarrow X(x) = C_1 e^{-k/x}$$

$$\frac{Y'}{Y} = \frac{-3y^2}{k} \Rightarrow \ln Y = -\frac{y^3}{k} + c_2 \Rightarrow Y(y) = C_2 e^{-y^3/k}$$

$$u(x, y) = X(x)Y(y) = C_1 e^{-k/x} \times C_2 e^{-y^3/k} = C e^{-(k/x + y^3/k)}$$

11.5 The One-Dimensional Wave Equation with Nonzero Boundary Values

Example 11.145 ■ Problem

Find the solution of the following one-dimensional wave equation.

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = P_0, \quad u(L, t) = P_1, \quad u(x, 0) = f(x), \quad u_t(x, 0) = g(x)$$

$$u(x, t) = w(x, t) + v(x) \implies \begin{cases} w(0, t) + v(0) = P_0 \\ w(L, t) + v(L) = P_1 \end{cases}$$

$$\frac{\partial^2 w}{\partial t^2} = c^2 \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 v}{\partial x^2} \right)$$

We choose $v(x)$ so that the problem for w has homogeneous boundary conditions:

$$\begin{cases} \frac{\partial^2 v}{\partial x^2} = 0 \\ v(0) = P_0 \\ v(L) = P_1 \end{cases} \implies v(x) = \left(\frac{P_1 - P_0}{L} \right) x + P_0$$

$$\begin{cases} \frac{\partial^2 w}{\partial t^2} = c^2 \frac{\partial^2 w}{\partial x^2} \\ w(0, t) = w(L, t) = 0 \end{cases} \implies w(x, t) = \sum_{n=1}^{\infty} \left[A_n \cos\left(\frac{cn\pi}{L}t\right) + B_n \sin\left(\frac{cn\pi}{L}t\right) \right] \sin \frac{n\pi}{L}x$$

$$u(x, t) = \underbrace{\sum_{n=1}^{\infty} \left[A_n \cos\left(\frac{cn\pi}{L}t\right) + B_n \sin\left(\frac{cn\pi}{L}t\right) \right] \sin \frac{n\pi}{L}x}_{w(x, t)} + \underbrace{\left(\frac{P_1 - P_0}{L} \right) x + P_0}_{v(x)}$$

The initial condition $u(x, 0) = f(x)$:

$$\sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L}x = f(x) - \left(\frac{P_1 - P_0}{L} \right) x - P_0 \triangleq f_1(x)$$

$$A_n = \frac{2}{L} \int_0^L f_1(x) \sin \frac{n\pi}{L}x \, dx = \frac{2}{L} \int_0^L \left[f(x) - \left(\frac{P_1 - P_0}{L} \right) x - P_0 \right] \sin \frac{n\pi}{L}x \, dx$$

Exercise 11.146

Find B_n in the cases (a) $g(x) = 0$ and (b) $g(x)$ arbitrary.

11.6 The One-Dimensional Wave Equation with Zero Initial Velocity (Travelling Waves)

If the initial velocity is zero ($g(x) = 0$), then

$$u(x, t) = \sum_{n=1}^{\infty} a_n \cos\left(\frac{cn\pi}{L}t\right) \sin \frac{n\pi}{L}x, \quad u(x, 0) = \sum_{n=1}^{\infty} a_n \sin \frac{n\pi}{L}x = f(x) \equiv \tilde{f}(x)$$

Using $\sin \alpha \cos \beta = \frac{1}{2}(\sin(\alpha + \beta) + \sin(\alpha - \beta))$:

$$\begin{aligned} u(x, t) &= \sum_{n=1}^{\infty} a_n \frac{1}{2} \left[\sin\left(\frac{n\pi}{L}x + \frac{cn\pi}{L}t\right) + \sin\left(\frac{n\pi}{L}x - \frac{cn\pi}{L}t\right) \right] \\ &= \frac{1}{2} \sum_{n=1}^{\infty} a_n \left[\sin \frac{n\pi}{L}(x + ct) + \sin \frac{n\pi}{L}(x - ct) \right] \\ &= \frac{1}{2} \sum_{n=1}^{\infty} a_n \sin \frac{n\pi}{L}(x + ct) + \frac{1}{2} \sum_{n=1}^{\infty} a_n \sin \frac{n\pi}{L}(x - ct) = \frac{1}{2} [\tilde{f}(x + ct) + \tilde{f}(x - ct)] \end{aligned}$$

Key result 11.147 ■ Travelling waves

$$u(x, t) = \frac{1}{2} [\tilde{f}(x + ct) + \tilde{f}(x - ct)]$$

$\tilde{f}$ is the odd periodic extension (with period $2L$) of the initial function f ; the solution is the superposition of two waves of half the initial amplitude travelling at speed c in opposite directions.

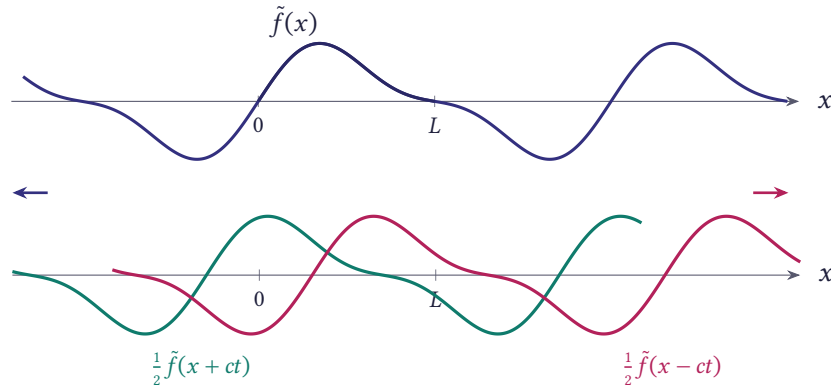


Figure 11.66 The odd periodic extension $\tilde{f}(x)$ and the two waves $\frac{1}{2}\tilde{f}(x + ct)$ and $\frac{1}{2}\tilde{f}(x - ct)$ travelling in opposite directions

Example 11.148

Plot the solution of the following one-dimensional wave equation, assuming zero initial velocity, at different times.

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = u(L, t) = 0, \quad u(x, 0) = f(x), \quad u_t(x, 0) = 0$$

$$f(x) = \begin{cases} \frac{2k}{L}x & 0 < x < \frac{L}{2} \\ \frac{2k}{L}(L-x) & \frac{L}{2} < x < L \end{cases}$$

Plotting $\frac{1}{2}\tilde{f}(x+ct)$ and $\frac{1}{2}\tilde{f}(x-ct)$ and adding them at the times $t = 0, t = L/5c, t = 2L/5c, t = L/2c$ and $t = 3L/5c$ gives the shape of the solution.

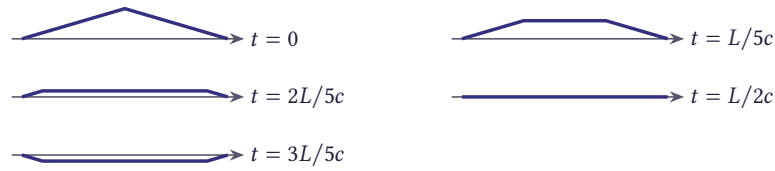
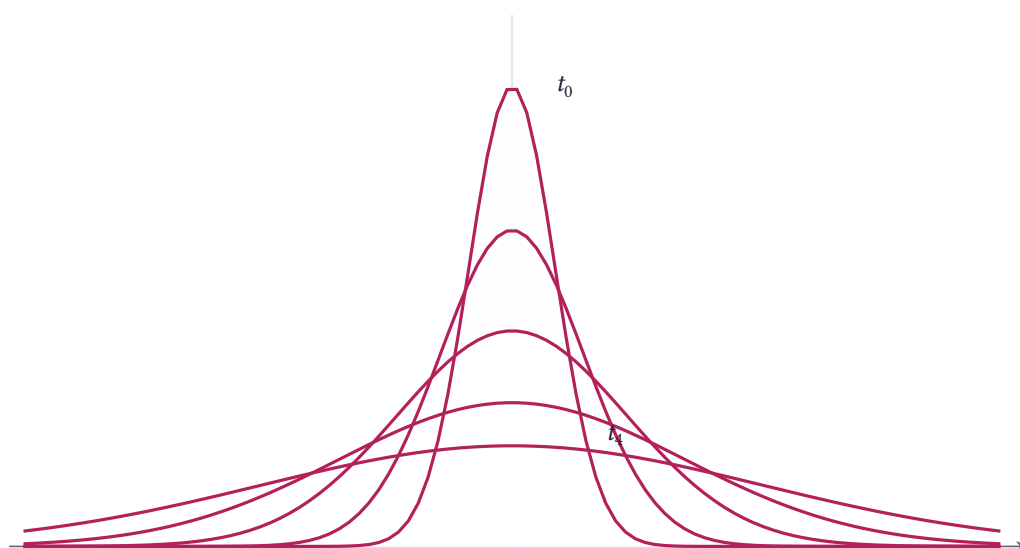


Figure 11.67 The solution of the example at $t = 0, L/5c, 2L/5c, L/2c$ and $3L/5c$; at $t = L/2c$ the string is flat

d'Alembert's Method and the One-Dimensional Heat Equation

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12.1 d'Alembert's Method for the One-Dimensional Wave Equation

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \\ u(0, t) = u(L, t) = 0 \\ u(x, 0) = f(x) \\ u_t(x, 0) = g(x) \end{cases}$$

We rewrite and solve the equation using the following change of variables.

$$u(x, t) \Rightarrow u(\varphi, \psi), \quad \begin{cases} \varphi = x + ct \\ \psi = x - ct \end{cases}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \varphi} \frac{\partial \varphi}{\partial x} + \frac{\partial u}{\partial \psi} \frac{\partial \psi}{\partial x} = \frac{\partial u}{\partial \varphi} + \frac{\partial u}{\partial \psi}$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial \varphi} \frac{\partial \varphi}{\partial t} + \frac{\partial u}{\partial \psi} \frac{\partial \psi}{\partial t} = c \frac{\partial u}{\partial \varphi} - c \frac{\partial u}{\partial \psi}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial \varphi} + \frac{\partial u}{\partial \psi} \right) = \left(\frac{\partial^2 u}{\partial \varphi^2} + \frac{\partial^2 u}{\partial \varphi \partial \psi} \right) \cdot 1 + \left(\frac{\partial^2 u}{\partial \psi \partial \varphi} + \frac{\partial^2 u}{\partial \psi^2} \right) \cdot 1 = \frac{\partial^2 u}{\partial \varphi^2} + 2 \frac{\partial^2 u}{\partial \varphi \partial \psi} + \frac{\partial^2 u}{\partial \psi^2}$$

(using $\frac{\partial^2 u}{\partial \varphi \partial \psi} = \frac{\partial^2 u}{\partial \psi \partial \varphi}$.)

Exercise 12.149

In the same way, prove the identity

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial \varphi^2} - 2 \frac{\partial^2 u}{\partial \varphi \partial \psi} + \frac{\partial^2 u}{\partial \psi^2} \right)$$

Substituting the two relations above into the one-dimensional wave equation:

$$c^2 \left(\frac{\partial^2 u}{\partial \varphi^2} - 2 \frac{\partial^2 u}{\partial \varphi \partial \psi} + \frac{\partial^2 u}{\partial \psi^2} \right) = c^2 \left(\frac{\partial^2 u}{\partial \varphi^2} + 2 \frac{\partial^2 u}{\partial \varphi \partial \psi} + \frac{\partial^2 u}{\partial \psi^2} \right) \implies \frac{\partial^2 u}{\partial \varphi \partial \psi} = 0$$

The general solution:

$$\frac{\partial u}{\partial \varphi} = \phi_1(\varphi) \implies u(\varphi, \psi) = \int \phi_1(\varphi) d\varphi + \Psi(\psi) = \Phi(\varphi) + \Psi(\psi) \implies u(x, t) = \Phi(x + ct) + \Psi(x - ct)$$

Applying the initial conditions:

$$u(x, 0) = \Phi(x) + \Psi(x) = f(x)$$

$$u_t(x, t) = \frac{\partial \Phi}{\partial \varphi} \frac{\partial \varphi}{\partial t} + \frac{\partial \Psi}{\partial \psi} \frac{\partial \psi}{\partial t} = c\Phi'(\varphi) - c\Psi'(\psi) = c\Phi'(x + ct) - c\Psi'(x - ct)$$

$$u_t(x, 0) = c\Phi'(x) - c\Psi'(x) = c[\Phi'(x) - \Psi'(x)] = g(x)$$

$$\begin{cases} \Phi(x) + \Psi(x) = f(x) \\ \Phi'(x) - \Psi'(x) = \frac{1}{c}g(x) \end{cases} \implies \begin{cases} \Phi'(x) + \Psi'(x) = f'(x) \\ \Phi'(x) - \Psi'(x) = \frac{1}{c}g(x) \end{cases} \implies \begin{cases} \Phi'(x) = \frac{1}{2}f'(x) + \frac{1}{2c}g(x) \\ \Psi'(x) = \frac{1}{2}f'(x) - \frac{1}{2c}g(x) \end{cases}$$

$$\begin{cases} \Phi(x) = \frac{1}{2}f(x) + \frac{1}{2c} \int_0^x g(s) ds + k_1 \\ \Psi(x) = \frac{1}{2}f(x) - \frac{1}{2c} \int_0^x g(s) ds + k_2 \end{cases}$$

$$u(x, t) = \frac{1}{2} \left(f(x+ct) + \frac{1}{c} \int_0^{x+ct} g(s) ds \right) + \frac{1}{2} \left(f(x-ct) - \frac{1}{c} \int_0^{x-ct} g(s) ds \right) + k_1 + k_2$$

Key result 12.150 ■ d'Alembert's formula

$$u(x, t) = \frac{1}{2} \left[f(x+ct) + f(x-ct) \right] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds + K, \quad u(0, 0) = 0 \Rightarrow K = 0$$

$$u(x, t) = \frac{1}{2} \left[f(x+ct) + f(x-ct) \right] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$$

Exercise 12.151

Prove that if the boundary conditions are to hold, the functions $f(x)$ and $g(x)$ in the formula above must both be periodic with period $2L$ and odd.

12.2 The Heat Equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad \frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \quad \frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

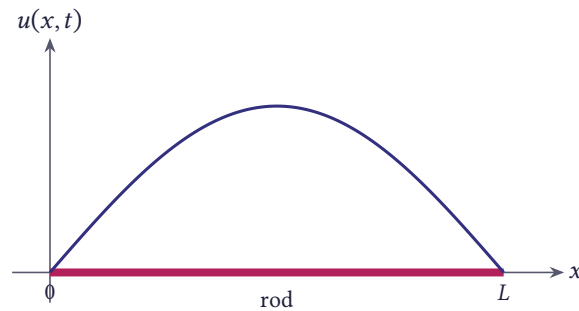
■ The One-Dimensional Heat Equation


Figure 12.68 A one-dimensional rod $0 \leq x \leq L$ with temperature distribution $u(x, t)$

For a unique solution we need boundary conditions and initial conditions:

- Boundary conditions: $u(0, t) = u(L, t) = 0$
- Initial conditions: $u(x, 0) = f(x)$

12.3 Solving the One-Dimensional Heat Equation

$$\begin{cases} \frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \\ u(0, t) = u(L, t) = 0 \\ u(x, 0) = f(x) \end{cases}$$

Here we solve the equation by **separation of variables**. Assume $u(x, t) = X(x) \times T(t)$.

$$\frac{\partial^2 u}{\partial x^2} = X''(x)T(t), \quad \frac{\partial u}{\partial t} = X(x)T'(t) \implies X(x)T'(t) = c^2 X''(x)T(t) \implies \frac{T'}{c^2 T} = \frac{X''}{X} = k$$

k must necessarily be negative (for $k = 0$ and $k > 0$ one gets $X(x) = 0$ exactly as in the wave problem).

With $k = -\lambda^2$:

$$\frac{T'}{c^2 T} = \frac{X''}{X} = -\lambda^2 \implies \begin{cases} X'' + \lambda^2 X = 0 \\ T' + c^2 \lambda^2 T = 0 \end{cases}$$

$$X(x) = A \cos \lambda x + B \sin \lambda x, \quad X(0) = 0 \implies A = 0, \quad X(L) = 0 \implies B \sin \lambda L = 0 \implies \lambda = \frac{n\pi}{L} = \lambda_n$$

$$X_n(x) = B_n \sin \lambda_n x, \quad T' + c^2 \lambda_n^2 T = 0 \implies T_n(t) = C_n e^{-(c\lambda_n)^2 t}$$

$$u_n(x, t) = X_n(x) \times T_n(t) = B_n \sin\left(\frac{n\pi}{L}x\right) \times C_n e^{-\left(\frac{cn\pi}{L}\right)^2 t}$$

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-(c\lambda_n)^2 t} \sin(\lambda_n x), \quad \lambda_n = \frac{n\pi}{L}$$

The initial condition $u(x, 0) = f(x)$:

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin \lambda_n x = f(x)$$

$f(x)$ has been set equal to a periodic odd function, so to compute the unknown Fourier coefficients we must regard $f(x)$ as a periodic odd function (with period $2L$):

$$A_n = \frac{2}{2L} \int_{-L}^L f(x) \sin\left(\frac{2\pi}{2L}nx\right) dx = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

Key result 12.152 ■ Solution of the one-dimensional heat equation (separation of variables)

$$u(x, t) = \sum_{n=1}^{\infty} A_n e^{-(c\lambda_n)^2 t} \sin(\lambda_n x), \quad \lambda_n = \frac{n\pi}{L}, \quad A_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

Example 12.153 ■ Example 1

Find the solution of the following one-dimensional heat equation.

$$\begin{aligned}\frac{\partial u}{\partial t} &= \frac{\partial^2 u}{\partial x^2}, & u(0, t) &= u(10, t) = 0, & u(x, 0) &= f(x) = \begin{cases} x & 0 \leq x < 5 \\ 10 - x & 5 \leq x \leq 10 \end{cases} \\ A_n &= \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx = 0.2 \int_0^5 x \sin\left(\frac{n\pi}{10}x\right) dx + 0.2 \int_5^{10} (10 - x) \sin\left(\frac{n\pi}{10}x\right) dx \\ &= 0.2 \left[(x) \left(\frac{-10}{n\pi} \cos \frac{n\pi}{10}x \right) - (1) \left(\frac{-100}{n^2\pi^2} \sin \frac{n\pi}{10}x \right) \right]_0^5 \\ &\quad + 0.2 \left[(10 - x) \left(\frac{-10}{n\pi} \cos \frac{n\pi}{10}x \right) - (-1) \left(\frac{-100}{n^2\pi^2} \sin \frac{n\pi}{10}x \right) \right]_5^{10} = \frac{40}{n^2\pi^2} \sin \frac{n\pi}{2} \\ u(x, t) &= \sum_{n=1}^{\infty} \left(\frac{40}{n^2\pi^2} \sin \frac{n\pi}{2} \right) e^{-\left(\frac{cn\pi}{10}\right)^2 t} \sin\left(\frac{n\pi}{10}x\right)\end{aligned}$$

(with $c = 1$.)

Example 12.154 ■ Example 2 (nonzero boundary values)

Find the solution of the following one-dimensional heat equation.

$$\begin{aligned}\frac{\partial u}{\partial t} &= c^2 \frac{\partial^2 u}{\partial x^2}, & u(0, t) &= T_0, & u(L, t) &= T_1, & u(x, 0) &= f(x) \\ u(x, t) &= w(x, t) + v(x) \implies \begin{cases} w(0, t) + v(0) = T_0 \\ w(L, t) + v(L) = T_1 \end{cases} \\ \frac{\partial w}{\partial t} &= c^2 \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 v}{\partial x^2} \right) \\ \begin{cases} \frac{\partial^2 v}{\partial x^2} = 0 \\ v(0) = T_0 \\ v(L) = T_1 \end{cases} &\implies v(x) = \left(\frac{T_1 - T_0}{L} \right) x + T_0, & \begin{cases} \frac{\partial w}{\partial t} = c^2 \frac{\partial^2 w}{\partial x^2} \\ w(0, t) = w(L, t) = 0 \end{cases} &\implies w(x, t) \\ &= \sum_{n=1}^{\infty} A_n e^{-(c\lambda_n)^2 t} \sin(\lambda_n x) \\ u(x, t) &= \underbrace{\sum_{n=1}^{\infty} A_n e^{-(c\lambda_n)^2 t} \sin(\lambda_n x)}_{w(x, t)} + \underbrace{\left(\frac{T_1 - T_0}{L} \right) x + T_0}_{v(x)}, & \lambda_n &= \frac{n\pi}{L}\end{aligned}$$

The initial condition:

$$\begin{aligned}\sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x &= f(x) - \left(\frac{T_1 - T_0}{L} \right) x - T_0 \triangleq f_1(x) \implies A_n \\ &= \frac{2}{L} \int_0^L \left[f(x) - \left(\frac{T_1 - T_0}{L} \right) x - T_0 \right] \sin \frac{n\pi}{L} x dx\end{aligned}$$

Example 12.155 ■ Example 3 (insulated ends)

Find the solution of the following one-dimensional heat equation.

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad u_x(0, t) = u_x(L, t) = 0, \quad u(x, 0) = f(x)$$

We solve this problem by **separation of variables**: $u(x, t) = X(x) \times T(t)$.

$$\frac{T'}{c^2 T} = \frac{X''}{X} = -\lambda^2 \Rightarrow \begin{cases} X'' + \lambda^2 X = 0 \Rightarrow X(x) = A \cos \lambda x + B \sin \lambda x \\ T' + c^2 \lambda^2 T = 0 \end{cases}$$

$$X'(x) = -A\lambda \sin \lambda x + B\lambda \cos \lambda x, \quad X'(0) = 0 \Rightarrow B = 0, \quad X'(L) = 0 \Rightarrow -A\lambda \sin \lambda L = 0 \\ \Rightarrow \lambda L = n\pi \Rightarrow \lambda_n = \frac{n\pi}{L}$$

$$X_n(x) = A_n \cos \lambda_n x, \quad T_n(t) = C_n e^{-(c\lambda_n)^2 t}$$

$$u(x, t) = \sum_{n=1}^{\infty} B_n e^{-(c\lambda_n)^2 t} \cos(\lambda_n x), \quad \lambda_n = \frac{n\pi}{L}$$

The initial condition $u(x, 0) = \sum_{n=1}^{\infty} B_n \cos\left(\frac{n\pi}{L}x\right) = f(x)$ gives

$$B_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi}{L}x\right) dx$$

12.4 The One-Dimensional Heat Equation on an Infinite Rod

$$\begin{cases} \frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \\ u(\infty, t) = u(-\infty, t) = 0 \\ u(x, 0) = f(x), \quad -\infty < x < \infty \end{cases}$$

Assuming $u(x, t) = X(x) \times T(t)$:

$$X(x)T'(t) = c^2 X''(x)T(t) \Rightarrow \frac{T'}{c^2 T} = \frac{X''}{X} = -\lambda^2 \\ \Rightarrow \begin{cases} X(x) = A_1(\lambda) \cos \lambda x + B_1(\lambda) \sin \lambda x = X_\lambda(x) \\ T(t) = C_1(\lambda) e^{-c^2 \lambda^2 t} = T_\lambda(t) \end{cases}$$

$$u_\lambda(x, t) = [A(\lambda) \cos \lambda x + B(\lambda) \sin \lambda x] e^{-c^2 \lambda^2 t} \Rightarrow u(x, t) \\ = \frac{1}{\pi} \int_0^\infty [A(\lambda) \cos \lambda x + B(\lambda) \sin \lambda x] e^{-c^2 \lambda^2 t} d\lambda$$

Checking the equation:

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{\pi} \int_0^\infty -\lambda^2 [A \cos \lambda x + B \sin \lambda x] e^{-c^2 \lambda^2 t} d\lambda, \quad \frac{\partial u}{\partial t} \\ = \frac{1}{\pi} \int_0^\infty -c^2 \lambda^2 [A \cos \lambda x + B \sin \lambda x] e^{-c^2 \lambda^2 t} d\lambda \Rightarrow \frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \checkmark$$

Applying the initial condition: $u(x, 0) = \frac{1}{\pi} \int_0^\infty [A(\lambda) \cos \lambda x + B(\lambda) \sin \lambda x] d\lambda = f(x)$, i.e. the Fourier integral (Chapter 10):

$$\begin{aligned} A(\lambda) &= \mathfrak{F}_C\{f(x)\} = \int_{-\infty}^\infty f(x) \cos \lambda x dx, & B(\lambda) &= \mathfrak{F}_S\{f(x)\} = \int_{-\infty}^\infty f(x) \sin \lambda x dx \\ u(x, t) &= \frac{1}{\pi} \int_0^\infty \left[\left(\int_{-\infty}^\infty f(v) \cos \lambda v dv \right) \cos \lambda x + \left(\int_{-\infty}^\infty f(v) \sin \lambda v dv \right) \sin \lambda x \right] e^{-c^2 \lambda^2 t} d\lambda \\ &= \frac{1}{\pi} \int_0^\infty \left\{ \int_{-\infty}^\infty f(v) [\cos \lambda v \cos \lambda x + \sin \lambda v \sin \lambda x] dv \right\} e^{-c^2 \lambda^2 t} d\lambda \\ &= \frac{1}{\pi} \int_0^\infty \left\{ \int_{-\infty}^\infty f(v) \cos \lambda(x-v) dv \right\} e^{-c^2 \lambda^2 t} d\lambda \end{aligned}$$

Key result 12.156 ■ First form of the solution

$$u(x, t) = \frac{1}{\pi} \int_{-\infty}^\infty f(v) \left\{ \int_0^\infty e^{-c^2 \lambda^2 t} \cos \lambda(x-v) d\lambda \right\} dv$$

■ Mathematical Lemmas

Theorem 12.157 ■ Lemma (a)

$$I = \int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

Proof.

$$\begin{aligned} I^2 &= \int_0^\infty e^{-x^2} dx \times \int_0^\infty e^{-y^2} dy = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy \stackrel{\rho^2=x^2+y^2}{=} \int_0^{\pi/2} \int_0^\infty \rho e^{-\rho^2} d\rho d\theta \\ &= \int_0^{\pi/2} \left[\frac{-1}{2} e^{-\rho^2} \right]_0^\infty d\theta = \frac{\pi}{4} \end{aligned}$$

■

Theorem 12.158 ■ Lemma (b)

$$\int_0^\infty e^{-s^2} \cos(2bs) ds = \frac{\sqrt{\pi}}{2} e^{-b^2}$$

Proof.

$$I(b) = \int_0^\infty e^{-s^2} \cos(2bs) ds \Rightarrow \frac{dI(b)}{db} = - \int_0^\infty 2se^{-s^2} \sin(2bs) ds$$

Integrating by parts ($du = -2se^{-s^2} ds$, $v = \sin 2bs$):

$$\begin{aligned} \frac{dI(b)}{db} &= e^{-s^2} \sin 2bs \Big|_0^\infty - 2b \int_0^\infty e^{-s^2} \cos 2bs ds = -2b I(b) \Rightarrow I(b) = ke^{-b^2}, \quad I(0) = k = \int_0^\infty e^{-s^2} ds \\ &= \frac{\sqrt{\pi}}{2} \end{aligned}$$

With the substitution $c\lambda\sqrt{t} = s$:

$$\int_0^\infty e^{-c^2\lambda^2 t} \cos \lambda(x-v) d\lambda = \frac{1}{c\sqrt{t}} \int_0^\infty e^{-s^2} \cos \left[2 \frac{(x-v)}{2c\sqrt{t}} s \right] ds = \frac{1}{c\sqrt{t}} \frac{\sqrt{\pi}}{2} e^{-\left(\frac{x-v}{2c\sqrt{t}}\right)^2}$$

Key result 12.159 ■ Second form of the solution

$$u(x, t) = \frac{1}{2c\sqrt{\pi t}} \int_{-\infty}^\infty f(v) e^{-\frac{(x-v)^2}{4c^2 t}} dv$$

With the substitution $z = \frac{v-x}{2c\sqrt{t}}$ (i.e. $v = x + 2c\sqrt{t}z$ and $dv = 2c\sqrt{t}dz$):

Key result 12.160 ■ Third form of the solution

$$u(x, t) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^\infty f(x + 2c\sqrt{t}z) e^{-z^2} dz$$

12.5 The Error Function

In mathematics the error function is most often defined as

$$\operatorname{erf}(x) \triangleq \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Properties of the error function:

- odd symmetry: $\operatorname{erf}(-x) = -\operatorname{erf}(x)$
- asymptotic property: $\operatorname{erf}(\infty) = 1$
- its Maclaurin expansion is given below (start from the expansion of e^x and prove this identity):

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \left[x - \frac{x^3}{1! \times 3} + \frac{x^5}{2! \times 5} - \frac{x^7}{3! \times 7} + \dots \right] = \frac{2}{\sqrt{\pi}} \sum_{n=1}^\infty (-1)^{n+1} \frac{x^{2n-1}}{(n-1)!(2n-1)}$$

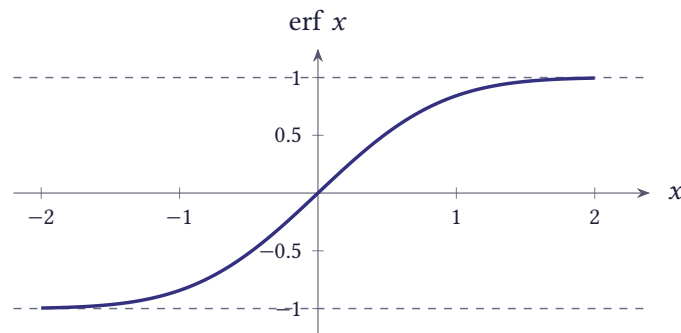


Figure 12.69 The graph of the error function $\operatorname{erf}(x)$ for $-2 \leq x \leq 2$

Example 12.161 ■ Example 1

Find the solution of the equation above for the following initial condition.

$$f(x) = \begin{cases} U_0 & |x| < 1 \\ 0 & |x| \geq 1 \end{cases}$$

We use the second form of the solution:

$$\begin{aligned} u(x, t) &= \frac{U_0}{2c\sqrt{\pi t}} \int_{-1}^1 e^{-\frac{(x-v)^2}{4c^2 t}} dv \stackrel{z=\frac{v-x}{2c\sqrt{t}}}{=} \frac{U_0}{\sqrt{\pi}} \int_{-\frac{1+x}{2c\sqrt{t}}}^{\frac{1-x}{2c\sqrt{t}}} e^{-z^2} dz \\ &= \frac{U_0}{\sqrt{\pi}} \left\{ \int_{-\frac{1+x}{2c\sqrt{t}}}^0 e^{-z^2} dz + \int_0^{\frac{1-x}{2c\sqrt{t}}} e^{-z^2} dz \right\} = \frac{U_0}{\sqrt{\pi}} \frac{\sqrt{\pi}}{2} \left\{ \frac{2}{\sqrt{\pi}} \int_0^{\frac{1+x}{2c\sqrt{t}}} e^{-z^2} dz + \frac{2}{\sqrt{\pi}} \int_0^{\frac{1-x}{2c\sqrt{t}}} e^{-z^2} dz \right\} \\ u(x, t) &= \frac{U_0}{2} \left[\operatorname{erf}\left(\frac{1+x}{2c\sqrt{t}}\right) + \operatorname{erf}\left(\frac{1-x}{2c\sqrt{t}}\right) \right] \end{aligned}$$

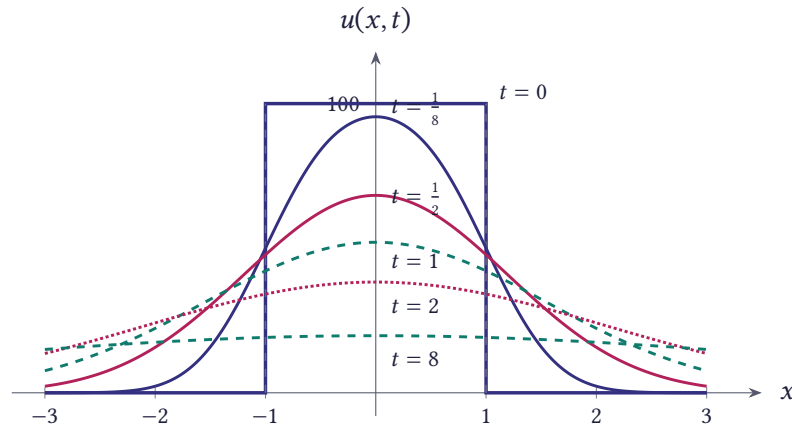


Figure 12.70 The temperature distribution $u(x, t)$ at $t = 0, \frac{1}{8}, \frac{1}{2}, 1, 2$ and 8 for the initial pulse $U_0 = 100$ and $c = 1$

Example 12.162 ■ Example 2

Find the solution of the equation above for the following initial condition.

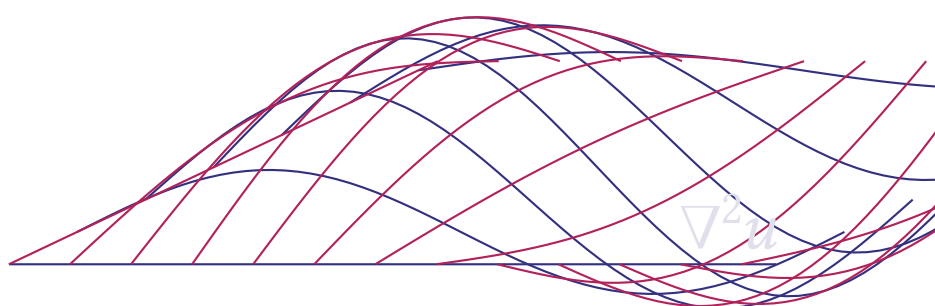
$$f(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

We use the second form of the solution:

$$\begin{aligned} u(x, t) &= \frac{1}{2c\sqrt{\pi t}} \int_0^\infty e^{-\frac{(x-v)^2}{4c^2 t}} dv \stackrel{z=\frac{v-x}{2c\sqrt{t}}}{=} \frac{1}{\sqrt{\pi}} \int_{-\frac{x}{2c\sqrt{t}}}^\infty e^{-z^2} dz = \frac{1}{\sqrt{\pi}} \left\{ \int_{-\frac{x}{2c\sqrt{t}}}^0 e^{-z^2} dz + \int_0^\infty e^{-z^2} dz \right\} \\ u(x, t) &= \frac{1}{2} \left[\operatorname{erf}\left(\frac{x}{2c\sqrt{t}}\right) + 1 \right] \end{aligned}$$

Two-Dimensional Wave and Heat Equations: Rectangle and Disc

معادلات دوبعدی موج و گرما: مستطیل و دایره



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13.1 The Two-Dimensional Wave Equation

Modelling a vibrating membrane (deriving the equation governing $u(x, y, t)$).

Physical assumptions:

- The mass is distributed uniformly over the membrane (a homogeneous membrane), and the membrane is perfectly flexible.
- The membrane is stretched so tightly that the force of gravity on it is negligible.
- The deflection of the membrane from equilibrium is very small.

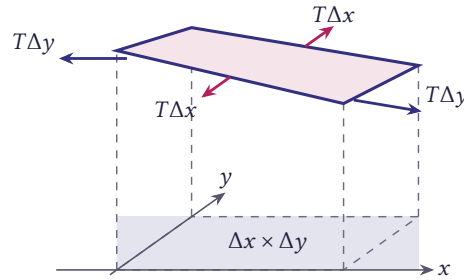


Figure 13.71 An element $\Delta x \times \Delta y$ of the vibrating membrane and the tensile forces $T\Delta x$ and $T\Delta y$ acting on it

By a modelling process analogous to the one-dimensional case, the equation of the vibrating two-dimensional membrane is obtained:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = c^2 \nabla^2 u$$

For a unique solution we need boundary conditions and initial conditions:

- Boundary conditions: $u(x, y, t) = 0$ on the boundary of the membrane
- Initial conditions: $u(x, y, 0) = f(x, y)$ and $u_t(x, y, 0) = g(x, y)$, where $u_t \triangleq \frac{\partial u}{\partial t}$

13.2 The Two-Dimensional Wave Equation with a Rectangular Boundary

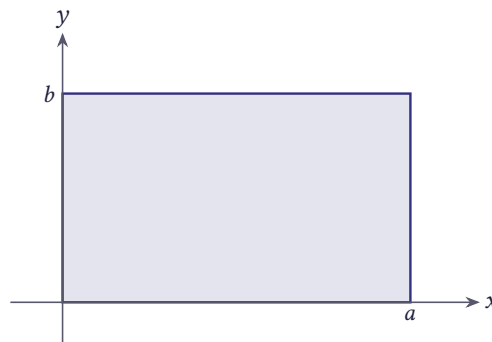


Figure 13.72 The rectangular region $0 \leq x \leq a, 0 \leq y \leq b$

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

- Boundary conditions: $u(0, y, t) = u(a, y, t) = 0$ and $u(x, 0, t) = u(x, b, t) = 0$
- Initial conditions: $u(x, y, 0) = f(x, y)$ and $u_t(x, y, 0) = g(x, y)$

■ Solving the Equation

Here we solve the equation by **separation of variables**. Assume $u(x, y, t) = X(x) \times Y(y) \times T(t)$.

$$\frac{\partial^2 u}{\partial x^2} = X''YT, \quad \frac{\partial^2 u}{\partial y^2} = XY''T, \quad \frac{\partial^2 u}{\partial t^2} = XYT'' \implies XYT'' = c^2(X''YT + XY''T)$$

$$\frac{T''}{c^2T} = \frac{X''}{X} + \frac{Y''}{Y} = -\lambda^2 \implies \frac{Y''}{Y} = -\lambda^2 - \frac{X''}{X} = -\rho^2 \implies \frac{X''}{X} = -\lambda^2 + \rho^2 = -\mu^2$$

$$\begin{cases} X'' + \mu^2 X = 0 \\ Y'' + \rho^2 Y = 0 \\ T'' + c^2 \lambda^2 T = 0 \end{cases} \quad \lambda^2 = \mu^2 + \rho^2$$

The solution in the x direction:

$$X(x) = A_1 \cos \mu x + A_2 \sin \mu x, \quad X(0) = 0 \implies A_1 = 0, \quad X(a) = 0 \implies A_2 \sin \mu a = 0 \implies \mu a = n\pi \implies \mu_n = \frac{n\pi}{a}$$

$$X_n(x) = A_n \sin \mu_n x, \quad \mu_n = \frac{n\pi}{a}$$

The solution in the y direction:

$$Y(y) = B_1 \cos \rho y + B_2 \sin \rho y, \quad Y(0) = 0 \implies B_1 = 0, \quad Y(b) = 0 \implies B_2 \sin \rho b = 0 \implies \rho b = m\pi \implies \rho_m = \frac{m\pi}{b}$$

$$Y_m(y) = B_m \sin \rho_m y, \quad \rho_m = \frac{m\pi}{b}$$

The solution in time:

$$\lambda_{nm}^2 = \mu_n^2 + \rho_m^2 = \left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2, \quad T_{nm}(t) = C_1 \cos(c\lambda_{nm}t) + C_2 \sin(c\lambda_{nm}t)$$

$$u_{nm}(x, y, t) = [A_{nm} \cos(c\lambda_{nm}t) + B_{nm} \sin(c\lambda_{nm}t)] \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{m\pi}{b}y\right), \quad n, m = 1, 2, \dots$$

$$u(x, y, t) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} [A_{nm} \cos(c\lambda_{nm}t) + B_{nm} \sin(c\lambda_{nm}t)] \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{m\pi}{b}y\right)$$

Applying the initial condition $u(x, y, 0) = f(x, y)$:

$$f(x, y) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} A_{nm} \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{m\pi}{b}y\right) = \sum_{n=1}^{\infty} K_n(y) \sin \frac{n\pi}{a}x, \quad K_n(y) \triangleq \sum_{m=1}^{\infty} A_{nm} \sin \frac{m\pi}{b}y$$

$$K_n(y) = \frac{2}{a} \int_0^a f(x, y) \sin \frac{n\pi}{a}x \, dx, \quad A_{nm} = \frac{2}{b} \int_0^b K_n(y) \sin \frac{m\pi}{b}y \, dy$$

$$A_{nm} = \frac{4}{ab} \int_0^b \left\{ \int_0^a f(x, y) \sin \frac{n\pi}{a}x \, dx \right\} \sin \frac{m\pi}{b}y \, dy$$

Exercise 13.163

Prove that

$$B_{nm} = \frac{4}{abc\lambda_{nm}} \int_0^b \left\{ \int_0^a g(x, y) \sin \frac{n\pi}{a} x dx \right\} \sin \frac{m\pi}{b} y dy$$

Key result 13.164 ■ Solution of the two-dimensional wave equation on a rectangle

$$u(x, y, t) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} [A_{nm} \cos(c\lambda_{nm}t) + B_{nm} \sin(c\lambda_{nm}t)] \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{m\pi}{b}y\right), \quad \lambda_{nm}^2 = \left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2$$

$$\begin{aligned} A_{nm} &= \frac{4}{ab} \int_0^b \left\{ \int_0^a f(x, y) \sin \frac{n\pi}{a} x dx \right\} \sin \frac{m\pi}{b} y dy, \quad B_{nm} \\ &= \frac{4}{abc\lambda_{nm}} \int_0^b \left\{ \int_0^a g(x, y) \sin \frac{n\pi}{a} x dx \right\} \sin \frac{m\pi}{b} y dy \end{aligned}$$

Example 13.165

Find the solution of the two-dimensional motion equation above for $g(x, y) = 0$ and $f(x, y) = xy$.

$$g(x, y) = 0 \Rightarrow B_{nm} = 0$$

$$\begin{aligned} A_{nm} &= \frac{4}{ab} \int_0^b \left\{ \int_0^a xy \sin \frac{n\pi}{a} x dx \right\} \sin \frac{m\pi}{b} y dy = \frac{4}{ab} \left\{ \left[\int_0^b y \sin \frac{m\pi}{b} y dy \right] \times \left[\int_0^a x \sin \frac{n\pi}{a} x dx \right] \right\} \\ &= \frac{4}{ab} \left[y \left(\frac{-b}{m\pi} \cos \frac{m\pi}{b} y \right) - (1) \left(\frac{-b^2}{m^2\pi^2} \sin \frac{m\pi}{b} y \right) \right]_0^b \times \left[x \left(\frac{-a}{n\pi} \cos \frac{n\pi}{a} x \right) - (1) \left(\frac{-a^2}{n^2\pi^2} \sin \frac{n\pi}{a} x \right) \right]_0^a \\ &= \frac{4}{ab} \cdot \frac{-b^2}{m\pi} (-1)^m \cdot \frac{-a^2}{n\pi} (-1)^n = \frac{4ab}{nm\pi^2} (-1)^{n+m} \end{aligned}$$

$$u(x, y, t) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{4ab}{nm\pi^2} (-1)^{n+m} \cos(c\lambda_{nm}t) \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{m\pi}{b}y\right)$$

13.3 The Two-Dimensional Heat Equation with a Rectangular Boundary

$$\frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = c^2 \nabla^2 u$$

- Boundary condition: $u(x, y, t) = 0$ on the boundary of the plate
- Initial condition: $u(x, y, 0) = f(x, y)$

■ Solving the Equation

Assume $u(x, y, t) = X(x) \times Y(y) \times T(t)$.

$$\frac{\partial^2 u}{\partial x^2} = X''YT, \quad \frac{\partial^2 u}{\partial y^2} = XY''T, \quad \frac{\partial u}{\partial t} = XYT' \implies XYT' = c^2(X''YT + XY''T)$$

$$\frac{T'}{c^2T} = \frac{X''}{X} + \frac{Y''}{Y} = -\lambda^2, \quad \frac{Y''}{Y} = -\lambda^2 - \frac{X''}{X} = -\rho^2, \quad \frac{X''}{X} = -\lambda^2 + \rho^2 = -\mu^2$$

$$\begin{cases} X'' + \mu^2 X = 0 \\ Y'' + \rho^2 Y = 0 \\ T' + c^2 \lambda^2 T = 0 \end{cases} \quad \lambda^2 = \mu^2 + \rho^2$$

As in the wave case ($X_n(x) = A_n \sin \mu_n x$ with $\mu_n = n\pi/a$ and $Y_m(y) = B_m \sin \rho_m y$ with $\rho_m = m\pi/b$), and with

$$T_{nm}(t) = C_1 e^{-c^2 \lambda_{nm}^2 t}, \quad \lambda_{nm}^2 = \mu_n^2 + \rho_m^2 = \left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2$$

$$u_{nm}(x, y, t) = A_{nm} \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{m\pi}{b}y\right) e^{-c^2 \lambda_{nm}^2 t}, \quad n = 1, 2, \dots, m = 1, 2, \dots$$

$$u(x, y, t) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} A_{nm} \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{m\pi}{b}y\right) e^{-c^2 \lambda_{nm}^2 t}$$

Applying the initial condition: exactly as in the wave case,

$$A_{nm} = \frac{4}{ab} \int_0^b \left\{ \int_0^a f(x, y) \sin \frac{n\pi}{a} x dx \right\} \sin \frac{m\pi}{b} y dy$$

13.4 The Laplacian in Polar Coordinates

The aim is to write $\nabla^2 u = u_{xx} + u_{yy}$ in polar coordinates in terms of r, θ .

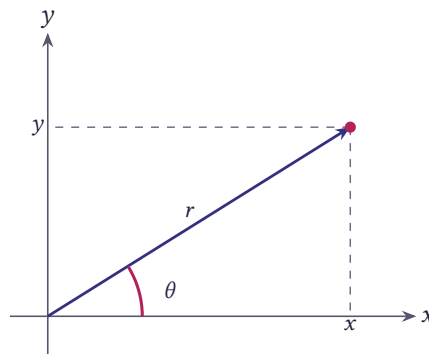


Figure 13.73 Polar coordinates: $x = r \cos \theta$, $y = r \sin \theta$

$$r = \sqrt{x^2 + y^2}, \quad x = r \cos \theta, \quad \theta = \tan^{-1}\left(\frac{y}{x}\right), \quad y = r \sin \theta$$

$$u_x = u_r r_x + u_\theta \theta_x \implies u_{xx} = (u_r r_x + u_\theta \theta_x)_x = u_{rr} r_x^2 + u_{\theta\theta} \theta_x^2 + 2r_x \theta_x u_{r\theta} + r_{xx} u_r + \theta_{xx} u_\theta$$

$$u_y = u_r r_y + u_\theta \theta_y \implies u_{yy} = u_{rr} r_y^2 + u_{\theta\theta} \theta_y^2 + 2r_y \theta_y u_{r\theta} + r_{yy} u_r + \theta_{yy} u_\theta$$

$$r_x = \frac{x}{r}, \quad r_{xx} = \frac{y^2}{r^3}, \quad \theta_x = \frac{-y}{r^2}, \quad \theta_{xx} = \frac{2xy}{r^4}, \quad r_y = \frac{y}{r}, \quad r_{yy} = \frac{x^2}{r^3}, \quad \theta_y = \frac{x}{r^2}, \quad \theta_{yy} = \frac{-2xy}{r^4}$$

$$u_{xx} = \frac{x^2}{r^2} u_{rr} + \frac{y^2}{r^4} u_{\theta\theta} + \frac{-2xy}{r^3} u_{r\theta} + \frac{y^2}{r^3} u_r + \frac{2xy}{r^4} u_\theta$$

$$u_{yy} = \frac{y^2}{r^2} u_{rr} + \frac{x^2}{r^4} u_{\theta\theta} + \frac{2xy}{r^3} u_{r\theta} + \frac{x^2}{r^3} u_r + \frac{-2xy}{r^4} u_\theta$$

Adding these two relations:

Key result 13.166 ■ The Laplacian in polar coordinates

$$\nabla^2 u = u_{rr} + \frac{1}{r^2} u_{\theta\theta} + \frac{1}{r} u_r$$

13.5 The Two-Dimensional Wave Equation with a Circular Boundary

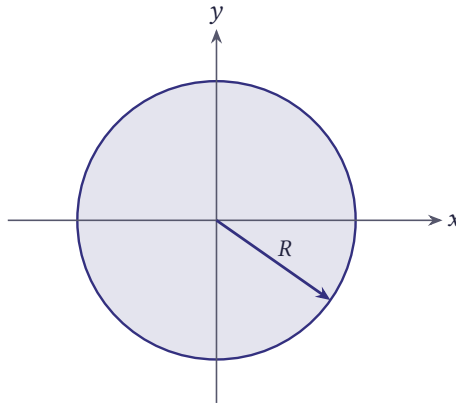


Figure 13.74 A circular membrane of radius R

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} &= c^2 \nabla^2 u, & u(R, t) &= 0 \text{ (boundary condition),} & u(r, 0) &= f(r), & u_t(r, 0) &= g(r) \\ & & & & & & & \text{(initial conditions)} \end{aligned}$$

■ Solving the Equation

Because of the circular symmetry of this problem it is better to solve it in polar coordinates. On a circular membrane the vibration is not a function of θ , so

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = c^2 \left(u_{rr} + \frac{1}{r} u_r \right) \\ u(R, t) = 0, \quad u(r, 0) = f(r), \quad u_t(r, 0) = g(r) \end{cases}$$

Now, assuming separable variables $u(r, t) = P(r) \times T(t)$:

$$PT'' = c^2 \left(P''T + \frac{1}{r} P'T \right) \Rightarrow \frac{T''}{c^2 T} = \frac{P''}{P} + \frac{1}{r} \frac{P'}{P} = -\lambda^2 \Rightarrow \begin{cases} P'' + \frac{1}{r} P' + \lambda^2 P = 0 \\ T'' + c^2 \lambda^2 T = 0 \end{cases}$$

Bessel's Differential Equation of Order n

$$y'' + \frac{1}{x}y' + \left(1 - \frac{n^2}{x^2}\right)y = 0 \implies y(x) = AJ_n(x) + BY_n(x)$$

where J_n are the Bessel functions of the first kind of order n and Y_n the Bessel functions of the second kind of order n .

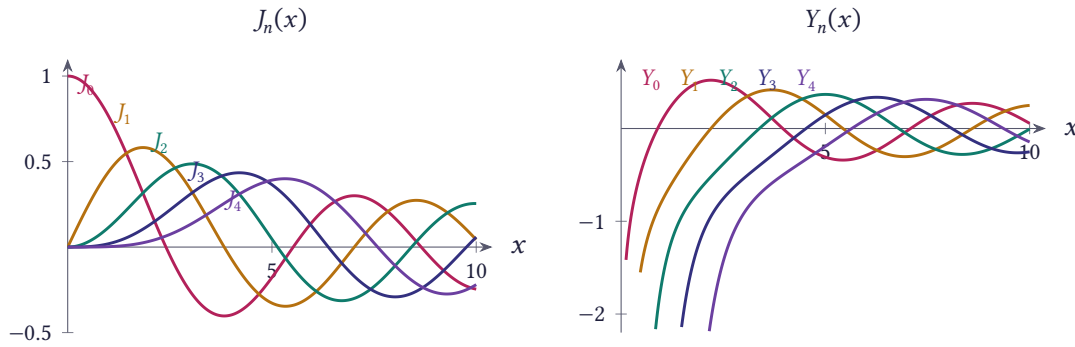


Figure 13.75 The Bessel functions of the first kind $J_0, \dots, J_4$ and of the second kind $Y_0, \dots, Y_4$ for $0 < x \leq 10$

With the substitution $s = \lambda r$:

$$P' = \frac{dP}{dr} = \frac{dP}{ds} \frac{ds}{dr} = \lambda \frac{dP}{ds}, \quad P'' = \lambda^2 \frac{d^2P}{ds^2}, \quad \frac{1}{r} \lambda = \frac{\lambda^2}{s}$$

$$\lambda^2 \frac{d^2P}{ds^2} + \frac{1}{r} \lambda \frac{dP}{ds} + \lambda^2 P = 0 \implies \frac{d^2P}{ds^2} + \frac{1}{s} \frac{dP}{ds} + P = 0$$

This is Bessel's equation of order zero ($n = 0$), so

$$P(r) = AJ_0(\lambda r) + BY_0(\lambda r)$$

Applying the boundary conditions:

- Since $P(0) \neq \infty$ and Y_0 is unbounded at the origin, the coefficient of Y_0 is zero ($D = 0$) and $P(r) = CJ_0(\lambda r)$.
- $P(R) = 0$ gives $J_0(\lambda R) = 0$, i.e. $\lambda R = \alpha_m$, where α_m are the roots of J_0 :

$$\lambda = \lambda_m = \frac{\alpha_m}{R}, \quad m = 1, 2, \dots \implies P_m(r) = C_m J_0\left(\frac{\alpha_m}{R} r\right)$$

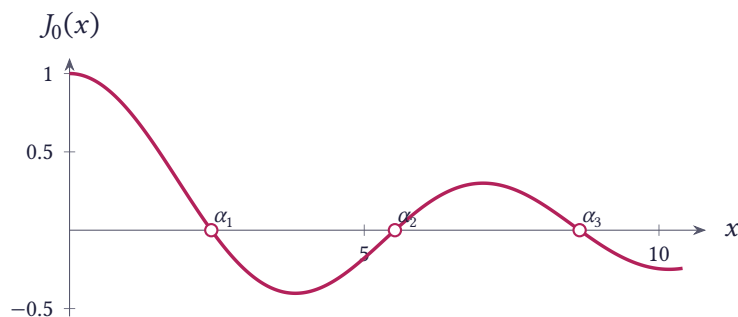


Figure 13.76 The function $J_0(x)$ and its roots $\alpha_1, \alpha_2, \alpha_3$

$$T'' + c^2 \lambda_m^2 T = 0 \implies T_m(t) = A_m \cos(c \lambda_m t) + B_m \sin(c \lambda_m t)$$

Key result 13.167 ■ Solution of the two-dimensional wave equation on a disc

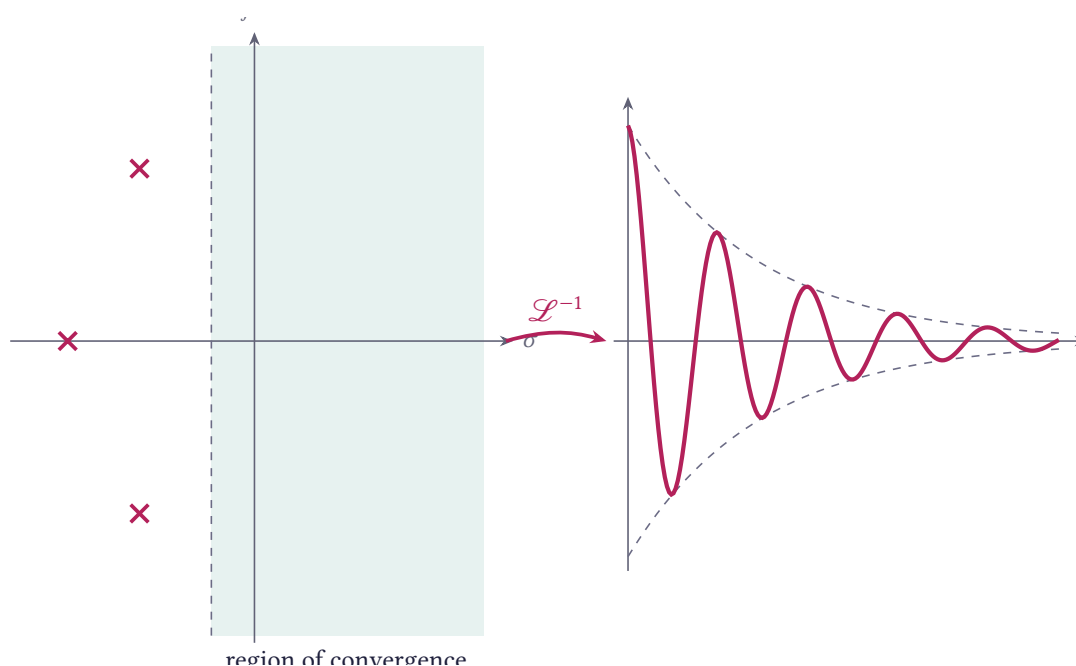
$$u(r, t) = \sum_{m=1}^{\infty} (A_m \cos(c\lambda_m t) + B_m \sin(c\lambda_m t)) J_0\left(\frac{\alpha_m}{R} r\right), \quad \lambda_m = \frac{\alpha_m}{R}$$

Research task (up to 1 bonus point)

Prepare a complete report on Bessel's differential equation of order n , the Bessel functions of the first and second kind and their properties, as well as the complete solution of the two-dimensional wave equation with a circular boundary (the report must be typed, in Word format).

The Laplace Transform

تبدیل لاپلاس



In this chapter

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14.1 The One-Sided Laplace Transform

Definition 14.168 ■ The one-sided Laplace transform

$$F(s) = \mathcal{L}\{f(t)\} \triangleq \int_0^{\infty} f(t) e^{-st} dt \quad \Longleftrightarrow \quad f(t) = \mathcal{L}^{-1}\{F(s)\}$$

The Laplace transform is one of the important tools in various engineering and mathematical applications. One of its applications is solving differential equations (read Chapter 6 of the course textbook).

The Laplace transform is a linear operator:

$$\mathcal{L}\{af(t) + bg(t)\} = a \mathcal{L}\{f(t)\} + b \mathcal{L}\{g(t)\}$$

Example 14.169 ■ Example 1

$$f(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases} \implies \mathcal{L}\{f(t)\} = F(s) = ?$$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} 1 \times e^{-st} dt = -\frac{1}{s} e^{-st} \Big|_0^{\infty} = \frac{1}{s}$$

Example 14.170 ■ Example 2

$$f(t) = \begin{cases} e^{at} & t \geq 0 \\ 0 & t < 0 \end{cases} \implies \mathcal{L}\{f(t)\} = F(s) = ?$$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{at} \times e^{-st} dt = \int_0^{\infty} e^{-(s-a)t} dt = -\frac{1}{s-a} e^{-(s-a)t} \Big|_0^{\infty} = \frac{1}{s-a}$$

Example 14.171 ■ Example 3

$$\mathcal{L}\{\cosh(at)\} = \mathcal{L}\left\{\frac{1}{2}e^{at} + \frac{1}{2}e^{-at}\right\} = \frac{1}{2}\mathcal{L}\{e^{at}\} + \frac{1}{2}\mathcal{L}\{e^{-at}\} = \frac{1}{2} \frac{1}{s-a} + \frac{1}{2} \frac{1}{s+a} = \frac{s}{s^2 - a^2}$$

Example 14.172 ■ Example 4

$$\mathcal{L}\{\sinh(at)\} = \mathcal{L}\left\{\frac{1}{2}e^{at} - \frac{1}{2}e^{-at}\right\} = \frac{1}{2}\mathcal{L}\{e^{at}\} - \frac{1}{2}\mathcal{L}\{e^{-at}\} = \frac{1}{2} \frac{1}{s-a} - \frac{1}{2} \frac{1}{s+a} = \frac{a}{s^2 - a^2}$$

Example 14.173 ■ Example 5

Using $\int e^{ax} \cos nx \, dx = \frac{e^{ax}}{a^2 + n^2} (a \cos nx + n \sin nx)$:

$$\mathcal{L}\{\cos \omega t\} = \int_0^{\infty} \cos \omega t e^{-st} \, dt = \frac{e^{-st}}{s^2 + \omega^2} (-s \cos \omega t + \omega \sin \omega t) \Big|_0^{\infty} = \frac{s}{s^2 + \omega^2}$$

Example 14.174 ■ Example 6

Using $\int e^{ax} \sin nx \, dx = \frac{e^{ax}}{a^2 + n^2} (a \sin nx - n \cos nx)$:

$$\mathcal{L}\{\sin \omega t\} = \int_0^{\infty} \sin \omega t e^{-st} \, dt = \frac{e^{-st}}{s^2 + \omega^2} (-s \sin \omega t - \omega \cos \omega t) \Big|_0^{\infty} = \frac{\omega}{s^2 + \omega^2}$$

■ **Table of Laplace Transforms (Table 6.1 of the textbook; a longer table is in Section 6.9)**

	$f(t)$	$\mathcal{L}(f)$		$f(t)$	$\mathcal{L}(f)$
1	1	$1/s$	7	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
2	t	$1/s^2$	8	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
3	t^2	$2!/s^3$	9	$\cosh at$	$\frac{s}{s^2 - a^2}$
4	t^n ($n = 0, 1, \dots$)	$\frac{n!}{s^{n+1}}$	10	$\sinh at$	$\frac{a}{s^2 - a^2}$
5	t^a (a positive)	$\frac{\Gamma(a+1)}{s^{a+1}}$	11	$e^{at} \cos \omega t$	$\frac{s-a}{(s-a)^2 + \omega^2}$
6	e^{at}	$\frac{1}{s-a}$	12	$e^{at} \sin \omega t$	$\frac{\omega}{(s-a)^2 + \omega^2}$

$$\Gamma(x) = \int_0^{\infty} e^{-t} t^{x-1} \, dt$$

■ **Properties of the One-Sided Laplace Transform (the table of Section 6.8 of the textbook)**

With the unit step function

$$u(t) \triangleq \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

Property	Formula
Definition of the Laplace transform	$F(s) = \mathcal{L}\{f(t)\} \triangleq \int_0^{\infty} f(t)e^{-st} dt$
Inverse Laplace transform	$f(t) = \mathcal{L}^{-1}\{F(s)\}$
Linearity	$\mathcal{L}\{af(t) + bg(t)\} = a\mathcal{L}\{f(t)\} + b\mathcal{L}\{g(t)\}$
Time shift	$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$
Shift in the s domain	$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$
Time scaling	$\mathcal{L}\{f(at)\} = \frac{1}{a}F(s/a)$
Time derivative	$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{(n-1)}(0)$
Time integral	$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{1}{s}F(s)$
Derivative in the s domain	$\mathcal{L}\{tf(t)\} = -F'(s)$
Integral in the s domain	$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^{\infty} F(v) dv$

14.2 Using the Laplace Transform to Solve Differential Equations

The steps for solving a differential equation with initial conditions using the Laplace transform:

1. convert the initial-value problem into an algebraic problem;
2. compute the solution of the algebraic problem;
3. solve the initial-value problem (return the solution to the time domain).



Figure 14.77 The steps of solving an initial-value problem with the Laplace transform: initial-value problem $\xrightarrow{1}$ algebraic problem $\xrightarrow{2}$ solution of the algebraic problem $\xrightarrow{3}$ solution of the initial-value problem

■ Ordinary Differential Equations

Example 14.175

Solve the differential equation $y'' - y = t$, $y(0) = 1$, $y'(0) = 1$ using the Laplace transform.

$$s^2 Y - sy(0) - y'(0) - Y = \frac{1}{s^2} \implies (s^2 - 1)Y(s) = s + 1 + \frac{1}{s^2}$$

$$Y(s) = \frac{s+1}{s^2-1} + \frac{1}{s^2(s^2-1)} = \frac{1}{s-1} + \left(\frac{1}{s^2-1} - \frac{1}{s^2}\right)$$

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s^2-1}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = e^t u(t) + \sinh t u(t) - t u(t)$$

Partial Differential Equations

Example 14.176

Solve the following partial differential equation using the Laplace transform.

$$\frac{\partial w}{\partial x} + x \frac{\partial w}{\partial t} = 0, \quad w(x, 0) = 0, \quad w(0, t) = t u(t)$$

We take the Laplace transform of both sides with respect to t :

$$\mathcal{L}_t \left\{ \frac{\partial w}{\partial x} + x \frac{\partial w}{\partial t} \right\} = \mathcal{L}_t \left\{ \frac{\partial w}{\partial x} \right\} + x \mathcal{L}_t \left\{ \frac{\partial w}{\partial t} \right\} = 0$$

$$\frac{\partial}{\partial x} w(x, s) + x[s w(x, s) - w(x, 0)] = 0, \quad w(x, 0) = 0$$

$$\frac{\partial}{\partial x} w(x, s) + xs w(x, s) = 0 \Rightarrow \frac{dw}{w} = -sx dx \Rightarrow \ln w = -s \frac{x^2}{2} + \ln C \Rightarrow w(x, s) = C e^{-\frac{x^2}{2}s}$$

From $w(0, t) = t u(t)$ we have $w(0, s) = \mathcal{L}\{w(0, t)\} = 1/s^2 = C$:

$$w(x, s) = \frac{1}{s^2} e^{-\frac{x^2}{2}s}$$

Using the time-shift property $\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$:

$$w(x, t) = \left(t - \frac{x^2}{2}\right) \times u\left(t - \frac{x^2}{2}\right)$$